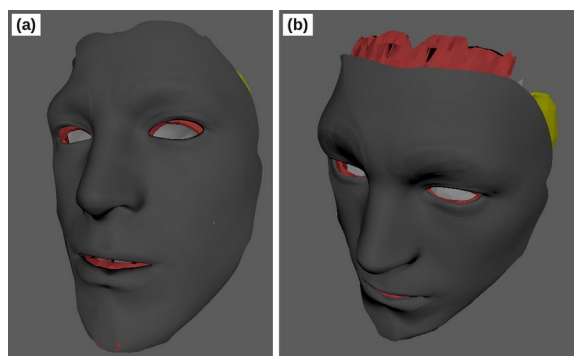


Automated Detection, Correction, and Surface Fairing for Skin-to-Muscle Registration Artifacts

A post-registration study of artifact detection, safety-constrained correction and surface fairing for a layered biomechanical facial model in Autodesk Maya

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Pre-manual-cleanup stage (a) and saved final stage (b) of the skin, Maya viewport screenshots from the research journal (different camera angles).

Abstract

Registering a generic facial skin mesh to personalised internal anatomy can leave two kinds of defect on the skin: sharp or broad geometric artifacts (bumps, ridges, faceted patches) and closed-skin regions through which fat, muscle or bone protrudes. This report documents the post-registration research that detected, corrected and faired those defects on a fixed-topology skin mesh of 15,905 vertices and 15,636 faces registered to 43 anatomy meshes in Autodesk Maya. Five detector milestones (M1–M5) progressed from a single-scale umbrella-Laplacian score, through target-referenced and multi-scale variants, to an anatomy-derived reference obtained by projecting the skin onto an offset iso-surface of an unsigned distance field, fused with the geometric evidence by set union. A triangle–triangle skin–anatomy intersection test then replaced vertex distance as the safety criterion; the pre-cleanup skin contained 93 forbidden intersecting faces (497 crossing pairs). Repair and safety-constrained smoothing (normal push, harmonic patch repair, global-step bisection, a unified per-vertex trust-region solver) reached zero intersections but plateaued in smoothness. A deliberately unsafe pure-Laplacian run then showed that the surface could be smoothed immediately (rough-core mean roughness $0.276 - 0.114$) at the price of 143 forbidden faces, and a blocked Laplacian that freezes only the locally offending vertices retained 92% of that reduction with no forbidden faces in an offline reproduction. The saved final mesh (pure Laplacian followed by about 130 documented manual vertex edits) lowers skin-wide mean roughness by 28% relative to the pre-cleanup skin and the 99th-percentile roughness by 41%, and reduces forbidden faces from 93 to 3; those residual faces and every unsupported claim are reported explicitly. The central finding is methodological: the decisive result came from simplifying the problem, so that anatomical constraints are localised instead of the smoothing being weakened globally.

Keywords: mesh fairing; umbrella Laplacian; Taubin smoothing; signed/unsigned distance field; skin–anatomy intersection; facial registration; Autodesk Maya

Executive Summary

Problem. After physical registration of a template skin onto personalised muscles, fat and bone, the 15,905-vertex skin still showed local bumps and ridges and, in 93 closed-skin faces, internal anatomy crossing the surface. Eye and mouth openings legitimately expose anatomy and had to be preserved.

Approach. Detection was developed in five milestones (umbrella Laplacian, reference-Laplacian, multi-scale/hybrid, anatomy-SDF reference, M3UM4 fusion) and frozen at M5. A triangle–triangle intersection detector became the safety metric. Repair and constrained-smoothing solvers were then studied until a pure Laplacian diagnostic showed that the constraints, not the operator, limited progress; a blocked Laplacian localised the constraints, and a small documented manual pass finished the mesh.

Key results (all recomputed offline from the saved Maya scene).

Table 1. Headline before/after values (repository intersection detector; roughness = umbrella-Laplacian norm in world units).

Metric	Pre-cleanup skin	After unified solver (_solver200)	Saved final scene
Forbidden intersecting faces (pairs)	93 (497)	0 (0)	3 (15)
All-vertex mean roughness	0.1345	0.1133	0.0965
All-vertex p99 roughness	0.681	0.666	0.402
Rough-core mean roughness (n=3,564)	0.252	0.276	0.191

Conclusion. The research question is answered: bumps are removable by simple Laplacian smoothing, and the anatomical constraint belongs in a local freeze rather than in a global weakening of the smoother. The saved final mesh is a hybrid result (automated detection and smoothing plus a documented manual pass). It still contains 3 forbidden faces by the offline detector, and no fully automatic production system is claimed; both points are treated in Section 14 and Section 15.

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1. Introduction and Background

1.1 Context: registering a skin to personalised anatomy

Physically based facial animation has long paired an anatomical model (muscle, fat and bone) with a deformable skin layer [25, 21]. The framework this project belongs to, described in the supplied manuscript [19], drives an epidermal skin surface by propagating internal muscle dynamics outward. A topologically reduced musculature proxy, augmented with fat compartments, is used to register a generic template skin to the personalised internal tissues; the manuscript characterises this registration as a one-time, offline preprocessing step per character. In the project’s Maya implementation the template skin is shrink-wrapped toward the anatomy with cloth-style forces (compare [3, 6]), using an unsigned distance query against the nearest anatomy meshes blended by an exponential soft minimum, and a collision response that keeps the skin at a small offset from the tissue.

The registration stage is therefore a geometric optimisation with many competing terms (landmark constraints, cloth-like regularisation, anatomy collision and offset). Its output is good enough to animate but, as the starting-state screenshots in Section 2 show, it is not consistently clean: the gray skin can carry local bumps and ridges, and yellow (fat) or red (muscle and other tissue) geometry can be visible through areas that should be closed skin. Which of the registration terms produces each defect was not separately isolated by experiment in this work; the report does not claim a root-cause attribution beyond the mechanisms it tested.

1.2 What existed before this project, and what this project contributes

Table 2. Scope split between the existing framework and this project.

Existing framework (Beneath the Skin and the project registration script)	Contribution of this research (post-registration cleanup)
Template skin registered to muscles, fat and bone as a one-time offline step; cloth-style forces, soft-minimum unsigned distance to the $K = 5$ nearest anatomy meshes, closest-point collision response with an offset (project defaults <code>skin_offset = 1.5</code> , <code>sdf_blend_k = 1.0</code> , 6 SDF smoothing passes).	Automatic <i>detection</i> of residual artifacts on the registered skin (milestones M1–M5), including an anatomy-derived reference built from the same kind of unsigned distance field.
Muscle-driven propagation of deformation to the skin; expression controllers; the internal solver.	An exact triangle–triangle skin–anatomy intersection detector used as the safety metric, kept separate from vertex clearance.
No post-registration quality check of the registered surface is described.	Repair and constrained-smoothing experiments, a unified per-vertex trust-region solver, a pure-Laplacian diagnostic, a blocked Laplacian, and a documented manual finish.
(unchanged by this work)	A quantitative before/after evaluation and an offline reproduction pipeline for the saved Maya states.

This report does not modify the internal muscle solver, the expression controllers, or the registration algorithm that produced the initial skin. Everything reported here happens after registration, on a mesh whose topology never changes.

1.3 Contributions

The research makes six contributions. First, a family of five detectors (M1–M5) with an explicit account of which artifact class each one finds and why each was not sufficient alone. Second, a triangle-level skin–anatomy intersection detector that separates true surface crossing from proximity and from legitimate eye and mouth exposure. Third, a set of correction experiments (normal push, harmonic patch repair, global-step

constrained smoothing, a unified solver) together with a technical account of why each plateaued. Fourth, a pure-Laplacian isolation experiment that separated an operator problem from a constraint problem. Fifth, a blocked Laplacian that localises the anatomical constraint and, in offline reproduction, keeps most of the smoothing effect with no forbidden faces. Sixth, an evidence base: every number in this report was recomputed from the saved Maya scene with a stated method, and every claim that could not be supported is listed in Appendix E.

1.4 Organisation of this report

Section 2 states the problem and success criteria. Section 3 describes the mesh, scene, software and evidence conventions. Section 4 collects the equations used. Section 5 walks through milestones M1, M2, M3 (V1 and V2), M4 and M5, each with the same ten-part structure. Section 6 to Section 9 cover the intersection detector, repair experiments, constrained smoothing and the unified solver. Section 10 to Section 13 cover fairing, the pure and blocked Laplacians and the manual finish. Section 14 gives the quantitative evaluation, Section 15 argues why the research is complete (and where it is not), and later sections cover figures, literature, limitations, reproducibility and a workshop extension. Appendices hold formulas, architecture, state catalogue, commands and the list of missing evidence.

1.5 Terminology and visual conventions

The following terms are used consistently. They matter because several of the early misunderstandings in the project came from treating distinct phenomena as one.

Table 3. Terminology and screenshot conventions.

Term / convention	Meaning in this report
Closed-skin anatomy protrusion	Internal anatomy (fat, muscle, bone) that crosses the skin surface in a region that should remain covered. This is the forbidden condition.
Valid exposure through openings	Anatomy seen through the eye and mouth openings (and other true boundaries). Legitimate; not an intersection.
Sharp local artifact	A small bump, dent, ridge or faceted patch that differs from its neighbourhood; visible to local shape measures such as the Laplacian.
Broad smooth artifact	A wide region that is wrong (for example drifted) yet locally smooth; largely invisible to local measures.
Surface intersection	A skin triangle geometrically crossing an anatomy triangle (exact triangle-triangle test).
Low anatomy clearance	A small unsigned distance from a skin vertex to the anatomy. Proximity, not penetration.
Vertex penetration vs face intersection	A vertex distance can be positive while a skin face is pierced by an anatomy triangle; the two are measured separately.
Local roughness	Norm of the umbrella-Laplacian vector at a vertex, Eq. (3).
Yellow dots (screenshots)	Vertices <i>selected</i> by a detector (Maya component selection display).
Solid yellow / red patches (screenshots)	Fat (yellow) and muscle or other anatomy (red) geometry visible through the skin or through openings. Not selected vertices.
M3 vs M4	M3 measures local geometric irregularity; M4 measures disagreement with an anatomy-derived reference. M4 distance is unsigned.

2. Problem Statement

2.1 The starting state

The work started from the converged registration result (scene t34.mb, derived from the t32 result of 1,600 registration iterations, which itself includes manual smoothing by the supervisor according to the repository notes). Figure 1 shows the registered skin before any cleanup. The face is anatomically plausible, but three things are visible: fine bumps and ridges on the chin, jaw line and cheeks; small solid yellow and red patches near the jaw and chin where fat and muscle come through the gray surface; and, at the top and back, larger yellow and red areas that belong to the open boundary of the skin cap and are not defects.

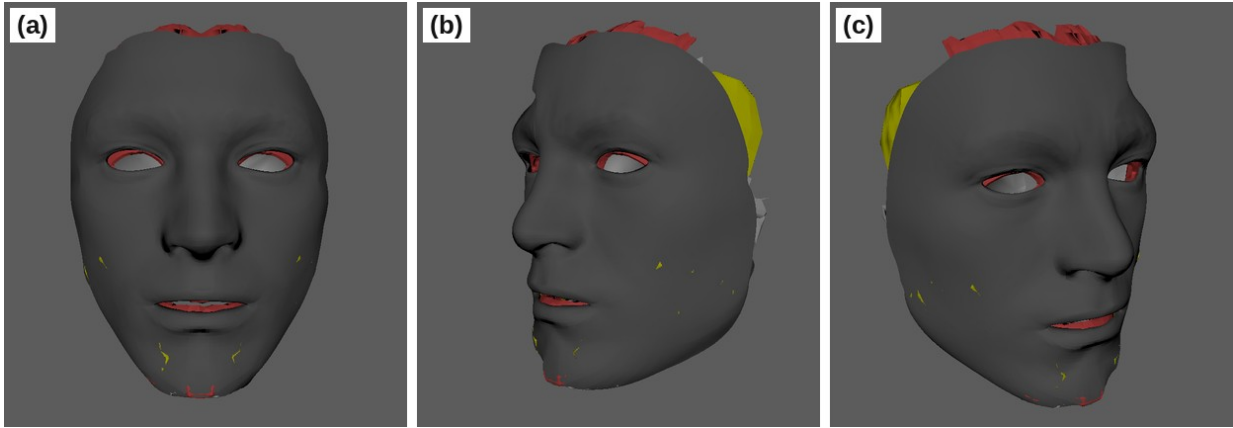


Figure 1. Baseline (M0) skin as registered: (a) front, (b) left, (c) right. Small yellow specks on the jaw and chin are fat showing through closed skin; the large yellow and red areas at the back and top lie behind the open cap boundary and are not defects. Journal screenshots M0_front/left/right.

2.2 Failure modes

Five phenomena had to be told apart. A *sharp local artifact* is a bump or ridge that differs from its one-ring neighbours. A *broad smooth artifact* is a wide region whose shape is wrong but whose local Laplacian is small, for example a cheek that has drifted off the underlying fat; it is the hardest to detect by local geometry. A *closed-skin anatomy protrusion* is a skin face crossed by an anatomy triangle where the skin should be closed. *Valid exposure* is anatomy seen through the eyes, the mouth and the outer cap boundary; it is expected and must survive cleanup. Finally *low clearance* means the skin is close to the anatomy without crossing it; it is a risk indicator, not a violation. An early and persistent source of confusion was that the project's distance backend is unsigned, so a small distance cannot distinguish a skin vertex just outside a muscle from one just inside it.

2.3 Formal statement

Let the skin be a triangle/quad mesh with fixed connectivity, vertex positions $X = (x_1, \dots, x_n)$, one-ring neighbourhoods $N(i)$ and a set B of boundary vertices; let $A_1, \dots, A_m$ be the anatomy triangle meshes. Write $R(X)$ for a roughness functional (the mean umbrella-Laplacian norm, Eq. (3)) and $F(X)$ for the set of forbidden skin faces, defined in Eq. (12). The cleanup task is

$$\begin{aligned} & \min_{X'} R(X'|_{\mathcal{A}}) \\ \text{s.t. } & \mathcal{F}(X') = \emptyset, \quad x'_b = x_b \quad \forall b \in B, \quad \|x'_i - x_i\| \text{ small, } \quad \text{connectivity unchanged} \end{aligned} \quad (1)$$

where $\mathcal{A}$ is the region judged to contain artifacts and the third condition expresses that the facial form and identity must be preserved. This is a conceptual statement, not an optimisation that was solved: the project used heuristic operators and evaluated them against these criteria. Two of the conditions are hard (no forbidden faces; boundary vertices fixed) and two are soft (roughness reduction; small displacement). The rest of the report can be read as an exploration of how to satisfy the hard conditions without spoiling the soft ones.

2.4 Research questions

The work was organised around six questions. Can problematic skin regions be found automatically from geometry rather than by hand? Can local shape measures separate true artifacts from legitimate anatomical curvature? Can an anatomy-derived reference expose broad smooth misregistration? How can smoothing be made strong enough to remove visible bumps without exposing anatomy? What is the smallest safety mechanism that preserves most of the smoothing effect? And what evidence is sufficient to declare the mesh complete rather than continuing to iterate on a solver?

2.5 Success criteria and non-goals

A result was accepted when (i) closed skin visually covers the anatomy while eye and mouth exposure remains, (ii) the major bumps, ridges and faceted patches are removed or reduced below visual distraction at normal inspection distance, (iii) correction adds no large spikes or repair mounds, (iv) the broad facial form is preserved, (v) the result can be reproduced from documented code and saved Maya stages with any manual finishing stated explicitly, and (vi) further automated iteration shows diminishing returns. The non-goals are equally explicit: the work does not redesign the facial simulation, does not test generalisation to other characters, and does not claim a fully automatic production pipeline.

3. Technical Setup

3.1 Mesh, scene and coordinates

The skin is a fixed-topology, quad-dominant mesh (Figure 2). Its connectivity never changes during the research; every experiment moves existing vertices only, which makes vertex-by-vertex comparison between saved states exact. Table 4 lists the mesh statistics measured from the scene.

Table 4. Skin mesh statistics measured from t34.mb.

Property	Value
Vertices / edges / faces	15,905 / 31,543 / 15,636
Face types	15,589 quads + 47 triangles
Boundary loops / boundary vertices	4 (eyes, mouth, outer cap) / 589
Euler characteristic $V - E + F$	-2 (genus 0 with four boundary loops)
Mean edge length (final state)	2.505 world units (CV = 0.475)
Anatomy meshes	43 (bone: skull and jaw; muscle: middleres_*, fat: fat_*, fat_element_*; cartilage and other)
Live skin node; scene	skin_cloth_copy_v5_pull_back; t34.mb
Working-skin world transform	uniform scale 1.18498945 about pivot (-0.1216, -29.0359, 45.8617), then translation (0, 0, 20.5503)
Non-boundary vertices used for skin-wide statistics	15,316

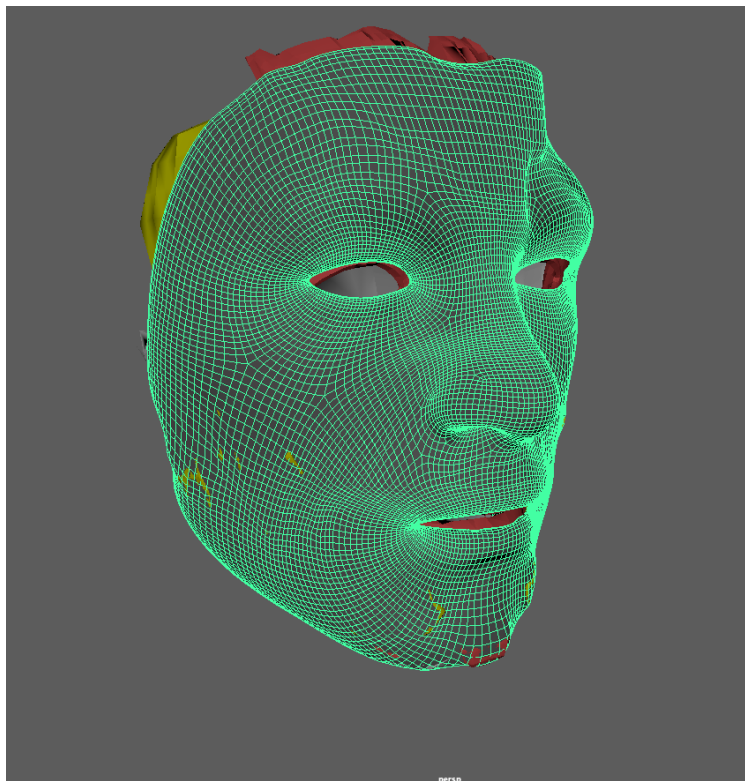


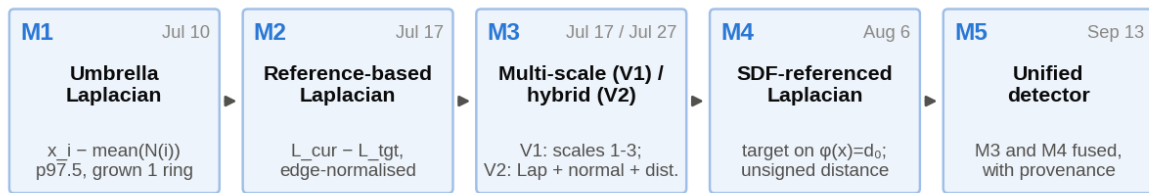
Figure 2. The skin mesh in Maya (wireframe, node skin_cloth_copy_v5_pull_back selected): quad-dominant topology with denser rings around the eyes, nostrils and mouth. Cropped from the journal screenshot “Maya Screenshot.png”.

All positions in the logs and in this report are in world space of the working skin, that is, after the uniform scale and translation listed above; lengths are therefore in the scene’s world units, not millimetres. The mean edge length is about 2.5 units, so a displacement of 1 unit is roughly 40% of an edge.

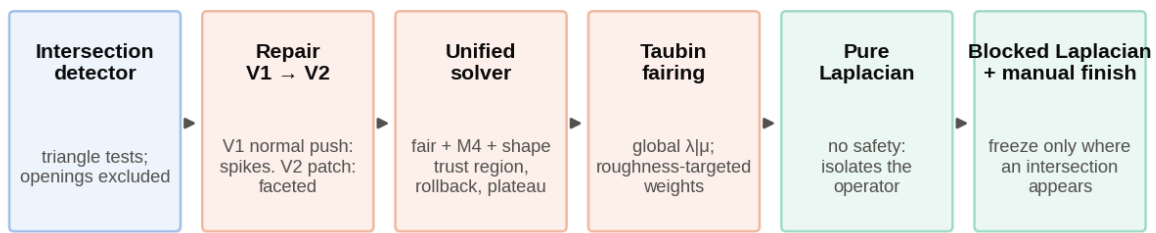
3.2 Software architecture

The code lives in the repository `skin-to-muscle-registration` (`src/`), with a Maya-facing driver that keeps detection and correction modules independent. Figure 3 sketches the pipeline; Table 5 lists the modules with their sizes. The important design choice for research was that milestone branches were kept rather than overwritten, so every detector and every correction policy can be re-run, and that detection was frozen at M5 so later correction experiments could not silently change the region being evaluated.

A. Artifact detection lineage (what to fix)



B. Correction and fairing lineage (how to fix it safely)



M5 mask defines the solver scopes

Orange: constrained-correction machinery that plateaued. Aqua: the simplification — smooth first, then localise the anatomy constraint.

Dates: git commit dates on branch `feature/m5-final-artifact-detection`.

Figure 3. Research pipeline: registered skin → detection (M1–M5, frozen) → true-intersection safety metric → correction and fairing experiments → pure/blocked Laplacian → manual finish → evaluation. Generated for this report from the repository structure.

Table 5. Source modules of the research code base (line counts measured from the repository).

Module (<code>src/</code>)	Lines	Role
<code>d98-target_shrinkwrap_registration.py</code>	4,367	Maya driver: registration and thin wrappers that run every detector and cleanup experiment.
<code>artifact_detection.py</code>	3,443	M1–M5 detectors, anatomy SDF query and iso-surface projection, hybrid scores, unified M5 fusion.
<code>anatomy_constraint.py</code>	3,415	Closest-anatomy queries, clearance, exact triangle–triangle intersection analysis, repair helpers (V1, V2).
<code>final_cleanup_solver.py</code>	3,268	Unified cleanup solver, Taubin final fairing, roughness-targeted weights, feasibility phase.
<code>smoothing_utils.py</code>	1,574	Laplacian, Taubin and constrained smoothers; region-aware smoothing.
<code>cleanup_pipeline.py</code>	860	Cleanup orchestration, reports and logging.
<code>mesh_utils.py</code>	535	Mesh I/O, adjacency, boundary, vector helpers, closest-point queries.

Module (src/)	Lines	Role
pure_laplacian_smoothing.py	427	Independent, deliberately unsafe pure Jacobi Laplacian diagnostic.
blocked_laplacian_smoothing.py	500	Smooth freely; roll back and freeze only vertices of faces that create a forbidden intersection.
micro_clearance_push.py	400	One-shot local outward push for residual forbidden faces (implemented; not shown to have been applied to the final scene).
region_selection.py, maya_io.py, metrics_utils.py	162 / 150 / 101	Region growth, Maya I/O, displacement and diagnostic metrics.

3.3 Evidence classes and the offline evaluation pipeline

Three classes of evidence are used and are labelled wherever they matter. *Logged* values were printed by the Maya session during development and are quoted from the research notes. *Screenshot* evidence is a journal image of the Maya viewport. *Offline* values were recomputed for this report: the binary Maya scene t34.mb was parsed outside Maya to recover the vertex positions of every saved skin state and of the 43 anatomy meshes, node transforms were applied, and the repository modules for Laplacian statistics, pure and blocked Laplacian smoothing and the triangle-triangle intersection detector were imported and executed with Maya's input/output replaced by in-memory stand-ins that operate on those positions. The clearance, Hausdorff-distance and mesh-quality measurements are separate Python implementations written for this report (analysis scripts in the supplementary folder).

The offline pipeline was validated against what Maya logged. The Jacobi smoother reproduces the saved states to about 10^{-5} world units. The pure-Laplacian run (percentile 75, growth 2, $\alpha = 0.35$, 10 iterations) reproduces the logged roughness statistics exactly (mean $0.27625 - 0.11366$). The median skin-to-anatomy distance of the placed anatomy is 7.1–7.2 units against 7.4 logged, and the pre-cleanup skin shows 93 forbidden faces offline against 92 logged. A one-face difference is expected from the choice of quad diagonal used to triangulate quads: with the alternative diagonal the offline count is 99, so the count is sensitive to triangulation at the level of a few faces. Maya's own triangulation is not recorded in the evidence, so absolute face counts should be read with that tolerance.

3.4 Saved states used as baselines

Table 6 lists the saved skin states in the scene, the operation that produced each, and what is known about it. Several relationships were inferred from geometry rather than recorded, and are marked.

Table 6. Saved skin states, their role, and their forbidden face counts (offline detector).

State (suffix of skin_cloth_copy_v5_pull_back)	Produced by / role	Forbidden faces / pairs
_precleanupsolver	Registered skin immediately before the unified solver (the pre-cleanup baseline).	93 / 497
_solver50	Unified solver, 50 iterations (intermediate).	0 / 0
_solver200	Unified solver, 200 iterations; the clean baseline for all later smoothing.	0 / 0
_fair_best17 / _targeted_pass1	Taubin fairing best state / roughness-targeted fairing pass (fairing branch).	1 / 12 / 0 / 0
_pre_pure_laplacian	Backup written by the pure-Laplacian module before its run.	2 / 16
_pre_blocked_laplacian	Backup before the blocked run; geometrically identical to a pure Laplacian (p75, growth 2, $\alpha = 0.10$, 8 iterations) applied to _solver200 (inferred).	21 / 94
(live skin)	Saved final scene: the state above plus about 130 manual vertex edits.	3 / 15

3.5 Measurement regions and a caveat about selection

Two vertex sets recur. The *rough core* C contains the 3,564 non-boundary vertices whose umbrella-Laplacian score in `_solver200` is at or above its 75th percentile; growing it by two rings with linear falloff gives the *active set* A of 9,035 vertices that the pure and blocked smoothers may move. Skin-wide statistics use all 15,316 non-boundary vertices. Because C is chosen by roughness in `_solver200`, that state is by construction the roughest on its own core; comparisons on C therefore favour later states and should be read together with the skin-wide statistics, which carry no such selection.

4. Mathematical Foundations

This section collects the equations that the code and the evaluation actually use. Each one is tied to the implementing function in the repository; where the working draft described a formula more loosely than the code implements it, the code is followed and the difference is noted. Notation: $x_i \in \mathbb{R}^3$ is the position of skin vertex i ; $N(i)$ its one-ring (edge-connected) neighbours; $N_r(i)$ all vertices within r edge steps (cumulative rings); $\bar{x}_i$ the centroid of $N(i)$; B the set of boundary vertices; $\varepsilon = 10^{-8}$ a guard constant.

4.1 Local shape measures

4.1.1 Umbrella Laplacian and roughness

$$L_i = x_i - \frac{1}{|N(i)|} \sum_{j \in N(i)} x_j = x_i - \bar{x}_i \quad (2)$$

L_i is the vector from the neighbour centroid to the vertex (function `compute_laplacian_scores`). It is the negative of the “uniform Laplacian displacement” used in some texts; the sign is immaterial for norms and is chosen here to match the code. The M1 score and the *local roughness* used in every evaluation table are its norm:

$$r_i = \|L_i\|_2 \quad (3)$$

Roughness is zero on any vertex lying at the centroid of its neighbours, small on gently curved regions, and large on isolated bumps and on genuinely high-curvature features such as the nose, eyelids, lips and jaw line. That dual behaviour is the reason M1 was insufficient (Section 5.1).

4.1.2 Reference-Laplacian disagreement (M2, M4)

$$\begin{aligned} L_i(T) &= t_i - \frac{1}{|N(i)|} \sum_{j \in N(i)} t_j, & s_i &= \|L_i(X) - L_i(T)\|_2 \\ \hat{s}_i &= \frac{s_i}{\bar{e}_i + \varepsilon}, & \bar{e}_i &= \frac{1}{|N(i)|} \sum_{j \in N(i)} \|x_i - x_j\| \end{aligned} \quad (4)$$

Here $T = (t_1, \dots, t_n)$ is a corresponding reference mesh (same vertex order). M2 uses the supplied target mesh (function `compute_reference_laplacian_scores`, normalisation off by default); M4 uses the anatomy-derived target of Eq. (9) (`compute_sdf_reference_laplacian_scores`, normalisation on). Comparing full vectors, not magnitudes, lets features present in both surfaces cancel. The draft called the reference term L^{ref} and normalised by “average local edge length”; the normaliser $\bar{e}_i$ above is the mean distance from x_i to its one-ring neighbours, exactly as coded.

4.1.3 Multi-scale persistence (M3 V1)

$$L_i^{(r)} = x_i - \frac{1}{|N_r(i)|} \sum_{j \in N_r(i)} x_j, \quad \rho_i^{(r)} = \frac{\|L_i^{(r)}\|}{\ell_i^{(r)} + \varepsilon}, \quad \ell_i^{(r)} = \frac{1}{|N_r(i)|} \sum_{j \in N_r(i)} \|x_i - x_j\| \quad (5)$$

$$P_i = \sum_{r \in \{1,2,3\}} \mathbf{1}[\rho_i^{(r)} \geq \tau_r], \quad \tau_r = \text{percentile}_{97.5}(\{\rho_j^{(r)}\}_j), \quad \text{retain } i \iff P_i \geq 2 \quad (6)$$

A vertex is kept only if it is an outlier (top 2.5% at that scale) at two or more of the three scales (compute_multiscale_laplacian_scores, detect_multiscale_irregular_region). Ring sets are cumulative, $N_r(i)$ = everything within r edges. The remaining steps of the detector are set operations: exclude boundary vertices and one buffer ring, drop connected components smaller than five vertices, grow by one ring, and re-exclude the boundary.

4.1.4 Hybrid score (M3 V2)

$$z(s) = \max\left(0, \frac{s - \text{median}(s)}{1.4826 \text{ MAD}(s) + \varepsilon}\right), \quad H_i = \omega_i \sum_{k \in \{\text{lap}, \text{nm}, \text{dist}\}} w_k z(s_i^k) \quad (7)$$

The three signals are the mean of scale-1 and scale-2 normalised Laplacian scores, a normal-inconsistency score $s^{\text{nm}}_i = 1 - 1/(|N(i)| \sum \text{clamp}(n_i n_j, -1, 1))$, and a robust local deviation of the distance to the anatomy, $s^{\text{dist}}_i = |d_i - \text{median}_{N_2(i)} d_j| / (\text{MAD}_{N_2(i)} + \varepsilon)$. Weights are $(w_{\text{lap}}, w_{\text{nm}}, w_{\text{dist}}) = (0.3, 0.2, 0.5)$, and ω_i is a soft boundary weight of $(0, 0.25, 0.6, 1)$ for vertices at 0, 1, 2 and ≥ 3 rings from a boundary (compute_hybrid_artifact_scores, compute_boundary_weights). The constant 1.4826 makes the median absolute deviation a consistent estimator of the standard deviation for Gaussian data.

4.2 Anatomy-derived reference (M4)

$$\varphi(x) = -\frac{1}{k} \ln \sum_{m \in K(x)} e^{-k d_m(x)} = d_{\min} - \frac{1}{k} \ln \sum_{m \in K(x)} e^{-k (d_m - d_{\min})}, \quad k = 1 \quad (8)$$

$d_m(x)$ is the unsigned distance from x to anatomy mesh m and $K(x)$ the five nearest meshes. This is the exponential smooth-minimum of the registration script (function compute_sdf_for_point), which the project's SDF parameters set with blend $k = 1$ and which the M4 query reuses. It is *unsigned*: $\varphi \geq 0$ says nothing about which side of a tissue the point lies on, and $\varphi \leq d_{\min}$ because the soft minimum is biased below the hard minimum where several meshes are near. For the target, each skin vertex is moved onto the offset iso-surface $\varphi = d_0$ by a damped Newton iteration:

$$\tau^{(q+1)} = \tau^{(q)} - \frac{\varphi(\tau^{(q)}) - d_0}{\|\nabla \varphi(\tau^{(q)})\|^2 + \varepsilon} \nabla \varphi(\tau^{(q)}), \quad \tau^{(0)} = x_i, \quad t_i = \tau^{(q^*)} \quad (9)$$

(project_point_to_iso_surface). $\nabla \varphi$ is taken to be the analytic outward unit direction returned by the distance query, so each step has length $|\varphi - d_0|$. The iteration stops when $|\varphi - d_0| \leq 10^{-4}$ or after 10 iterations; vertices that fail are excluded from scoring. The working draft summarised this target as “ $t_i = c_i + d_0 n_i$ ” (closest point plus offset); the code instead iterates to the iso-surface, which coincides with that expression only when a single surface dominates and the query is exact. The offset used in the development scene was $d_0 \approx 7.40$ world units, chosen from the scene's distance statistics (summarize_skin_anatomy_distances). The M4 score is Eq. (4) with T = these targets.

4.3 Fusion (M5)

$$\begin{aligned}\mathcal{M}_U &= \text{grow}_1\left(\text{filter}\left((M_3 \cup M_4) \setminus B^{(1)}\right) \setminus B^{(1)}\right) \\ \mathcal{M}_\cap &= M_3 \cap M_4, \quad \mathcal{M}_{3\text{-only}} = M_3 \setminus M_4, \quad \mathcal{M}_{4\text{-only}} = M_4 \setminus M_3\end{aligned}\quad (10)$$

$B^{(1)}$ is the boundary plus one ring, filter removes small components, and grow_1 adds one neighbour ring (detect_unified_artifacts). M5 fuses *sets*, not scores, so provenance survives and no arbitrary M3/M4 weight is introduced. M3 here is the V1 multi-scale detector; the V2 hybrid score is not used by default.

4.4 Surface intersection test

For a segment $P+tD$, $t \in [0,1]$, and a triangle (V_0, V_1, V_2) , the Möller–Trumbore parametrisation [17] gives

$$\begin{aligned}E_1 &= V_1 - V_0, \quad E_2 = V_2 - V_0, \quad h = D \times E_2, \quad a = E_1 \cdot h \\ u &= \frac{(P - V_0) \cdot h}{a}, \quad v = \frac{D \cdot ((P - V_0) \times E_1)}{a}, \quad t = \frac{E_2 \cdot ((P - V_0) \times E_1)}{a}\end{aligned}\quad (11)$$

and the segment crosses the triangle when $u, v \geq 0$, $u+v \leq 1$ and $0 \leq t \leq 1$ (a finite segment, not a ray). Two non-coplanar triangles intersect when an edge of one crosses the other [18]; coplanar pairs use a separating-axis overlap test. Pairs are classified crossing, touching, coplanar-overlap or uncertain. Quads are split into two triangles by a fixed diagonal. A uniform grid over axis-aligned bounding boxes is the broad phase [26]. A skin face is *forbidden* when it belongs to a crossing or coplanar-overlap pair and does not touch the one-ring-buffered boundary:

$$\mathcal{F}(X) = \left\{ f \in \text{skin faces} : f \cap B^{(1)} = \emptyset \wedge \exists a \in \bigcup_m A_m : \text{class}(f, a) \in \{\text{crossing, coplanar-overlap}\} \right\} \quad (12)$$

Touching and uncertain pairs are not counted. The boundary exclusion is what makes valid exposure through the eyes, mouth and cap edge not count as a violation: 1,171 skin faces (about 7.5% of all faces) are excluded by it in every analysed state (an offline count).

4.5 Smoothing operators

4.5.1 Weighted Jacobi Laplacian (pure and blocked)

$$x_i^{t+1} = x_i^t + \alpha w_i \left(\frac{1}{|N(i)|} \sum_{j \in N(i)} x_j^t - x_i^t \right) = x_i^t - \alpha w_i L_i^t, \quad w_i = \begin{cases} 1 & i \in \mathcal{C} \\ 1 - \frac{\varrho_i}{R+1} & \varrho_i \leq R, i \notin \mathcal{B} \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

All vertices are updated simultaneously from the previous iterate (Jacobi), ϱ_i is the ring distance from the core $\mathcal{C}$ and R the growth (2 in the logged runs, giving weights 1, 2/3, 1/3, 0). Boundary vertices are protected. This is the classic uniform Laplacian smoothing [9, 27], chosen for robustness and transparency rather than accuracy of curvature flow [8, 16].

4.5.2 Taubin ($\lambda|\mu$) smoothing

$$x' = x - \lambda L(x), \quad x'' = x' - \mu L(x'), \quad f_{\lambda\mu}(\kappa) = (1 - \lambda\kappa)(1 - \mu\kappa), \quad \kappa_{PB} = \frac{1}{\lambda} + \frac{1}{\mu} \quad (14)$$

A positive step λ followed by a negative step μ with $|\mu| > \lambda$ [23, 24] makes the per-mode transfer function $f_{\lambda\mu}$ pass low graph frequencies $\kappa < \kappa_{PB}$ nearly unchanged while attenuating high frequencies. With the project's

$(\lambda, \mu) = (0.33, -0.34)$, $\kappa_{PB} \approx 0.089$; the aggressive preset $(0.5, -0.51)$ gives $\kappa_{PB} \approx 0.039$. In the code, one iteration of `smoothing_utils.taubin_smooth` is a full $\lambda + \mu$ pair. The plain Laplacian step of Eq. (13) has transfer $(1 - \alpha\kappa)$ per iteration and therefore shrinks low frequencies as well (Figure 20).

4.5.3 Unified solver force balance

$$\Delta x_i = (1 - a_i) \left[w_i^f (\bar{x}_i - x_i) + w_i^{M4} (t_i - x_i) + w_i^{sh} (x_i^{\text{rest}} - x_i) \right] \quad (15)$$

then clamp: $\|\Delta x_i\| \leq \rho \ell_i$

$$a_i = s \left(\text{clamp} \left(\frac{q_i - R_f}{R_t}, 0, 1 \right) \right), \quad s(u) = u^2(3 - 2u), \quad w_i^{sh} = w_{sh} (\sigma + (1 - \sigma) a_i) \quad (16)$$

(`combine_step`, `shape_weight_from_anchor`, `build_active_region`). ℓ_i is the vertex's own mean one-ring edge length and $\rho = 0.5$ the trust-region ratio, so a single tight vertex limits only itself. Rings are $R_f = 6$ (fairing band) and $R_t = 4$ (transition band), a_i a smoothstep anchor that fades the update to zero at the outer edge, and the default weights are $w^f = 0.5$, $w^{M4} = 0.25$, $w_{sh} = 0.15$ with $\sigma = 0.3$ (M4-only vertices get a 0.3 fairing scale; overlap vertices a 0.6 M4 scale). The M4 pull is later decayed (patience 2, rate 0.7). The rest position x^{rest} is a low-pass filtered feasible copy (12 iterations, strength 0.3), see Section 9.

4.5.4 Global safe step (constrained smoother) and blocked acceptance

$$\alpha^* = \max \{ \alpha \in [0, 1] : \mathcal{F}(X + \alpha \Delta X) = \emptyset \}, \quad X^{t+1} = X^t + \alpha^* \Delta X^t \quad (8 \text{ bisection steps}) \quad (17)$$

$$\begin{aligned} \tilde{X} &= \text{Jacobi}(X^t; U^t), \quad V_{\text{bad}} = \text{vertices}(\mathcal{F}(\tilde{X})) \\ U^{t+1} &= U^t \setminus (V_{\text{bad}} \cup N(V_{\text{bad}})), \quad \tilde{x}_v \leftarrow x_v^t \quad \forall v \notin U^{t+1} \end{aligned} \quad (18)$$

The first rule (Eq. (17)) is what the strict constrained smoother did; the second (Eq. (18)) is the blocked Laplacian (Section 12): free vertices propose a Jacobi step, the proposal is scanned with the exact detector, the vertices of offending faces plus one ring are rolled back and frozen, and the scan is repeated once with a second ring; if validity is still not restored the proposal is discarded and the run stops with the last safe state.

4.6 Roughness-targeted weighting

The targeted fairing weight is a piecewise-linear map of a vertex's roughness through percentile knots, $r_{\min} \mapsto 0.75$, $p_{75} \mapsto 1.0$, $p_{90} \mapsto 1.25$, $p_{95} \mapsto 1.5$, $p_{99} \mapsto 1.75$, $r_{\max} \mapsto 2.0$, followed by three one-ring averaging passes so that no isolated vertex receives a spike in weight (`build_roughness_fairing_weights`).

4.7 Evaluation metrics

$$Q_p(S) = p\text{-th percentile of } \{r_i\}_{i \in S}, \quad \Delta_p \% = 100 \frac{Q_p^{\text{after}} - Q_p^{\text{before}}}{Q_p^{\text{before}}} \quad (19)$$

$$u_i = \|x_i - x_i^{\text{base}}\|, \quad \text{RMS} = \sqrt{\frac{1}{n} \sum_i u_i^2} \quad (20)$$

$$\text{CV}_e = \frac{\sigma(\ell_e)}{\mu(\ell_e)}, \quad \theta(f, g) = \arccos(\text{clamp}(n_f \cdot n_g, -1, 1))$$

$$\kappa_i = 2\pi - \sum_{c \ni i} \theta_c \text{ (interior vertices)}, \quad d(x, Y) = \min_{y \in Y} \|x - y\| \quad (21)$$

$$d_H(X, Y) = \max \left\{ \max_{x \in X} d(x, Y), \max_{y \in Y} d(y, X) \right\}$$

κ_i is the angle defect (an integral of Gaussian curvature per vertex), $\theta(f, g)$ the dihedral (normal) angle across an edge, and d_H the symmetric Hausdorff distance; point-to-surface distances use the triangulated surface. The quad diagonal used for triangulation is fixed, so Hausdorff values carry a small triangulation tolerance. Displacement and Hausdorff figures in this report are computed on the vertices that actually moved (or on a subsample where stated).

5. Milestones M1–M5: Detection Research

The detector sequence is best read as a sequence of hypotheses. Each milestone preserved the previous method, tested one specific limitation and left a reusable baseline. The point was not that each version was globally better than the last, but to learn which geometric signal corresponds to which class of artifact. Every milestone below follows the same ten-part structure: objective and hypothesis; method; parameters; screenshots; result; what worked; what failed; why it failed technically; the insight gained; and how it motivated the next milestone. In every screenshot, yellow *dots* are vertices selected by the detector; solid yellow and red patches are fat and muscle or other anatomy (see Section 1.5).

Table 7. Milestone progression: hypotheses, signals, outcomes and limits.

Milestone (commit, 2026)	Hypothesis	Signal	Main outcome	Why not the final answer
M1 (7b2627a, Jul 10)	Large one-ring Laplacian magnitude marks registration artifacts.	$r_i = \ L_i\ $, top 2.5%	Baseline detector and Maya selection/report pipeline; finds sharp local irregularity.	Also flags valid high curvature and boundaries; percentile selects 2.5% regardless of surface quality.
M2 (d207c61, Jul 17)	Subtracting a reference Laplacian cancels valid anatomy.	$\ L_i(X) - L_i(T)\ $, $T = \text{supplied target}$	Reference concept implemented and topology-validated.	The target is not an artifact-free ideal; openings and boundaries dominate.
M3 V1 (8a38590, Jul 17)	Defects persist across scales; valid features fade.	Normalised Laplacian at 1, 2, 3 rings; persistence ≥ 2	Coherent regions; strong chin detection; boundary exclusion; component filter.	Too restrictive for broad smooth errors.
M3 V2 (ae11d62, Jul 27)	Normals and distance-to-anatomy reveal broad drift.	Robust-z hybrid of Laplacian, normal, distance	Broader response over cheeks and jaw.	New false positives at folds, corners and boundaries; hand-set weights.
M4 (f2aca65, Aug 6)	An anatomy-derived reference is better than the supplied target.	$\ L_i(X) - L_i(T^A)\ $, T^A on $\varphi = \text{do}$	Captures smooth reference disagreement; complements M3.	Unsigned distance is not penetration; a signal, not a correction rule.
M5 (7a8faa2, 270bac2, Sep 13)	Fuse M3 and M4 as interpretable sets.	$M3 \cup M4$ with provenance, one growth, boundary re-exclusion	Stable, frozen detection with M3-only / M4-only / overlap / union.	Correction needs different operators and safety logic.

Commit identifiers and dates are from the repository history (branch feature/m5-final-artifact-detection). M1–M4 have journal screenshots and code but no separate written notes in the evidence; the sections below therefore rest on the code, the screenshots and the working draft.

5.1 M1: single-scale umbrella-Laplacian detection

Objective and hypothesis. M1 aimed to replace manual, visual selection of bad skin vertices with an automatic and reproducible detector. The hypothesis was that vertices whose one-ring Laplacian magnitude lies in the upper tail correspond to registration artifacts.

Method. Every vertex receives the roughness score r_i of Eq. (3); vertices at or above the 97.5th percentile are selected, the region is grown by one ring for visibility, and the vertices are selected in Maya and summarised in a report (detect_skin_artifacts, detect_irregular_region). No boundary handling, anatomy information or reference is used.

Parameters. Percentile 97.5; growth of one ring; scores unnormalised. Because the threshold is a percentile, about 2.5% of the 15,905 vertices (roughly 400) are flagged before growth *by construction*, whatever the surface looks like.

Screenshots. Figure 4 shows the M1 selection from the front and from below, with the mesh overlay in (b).

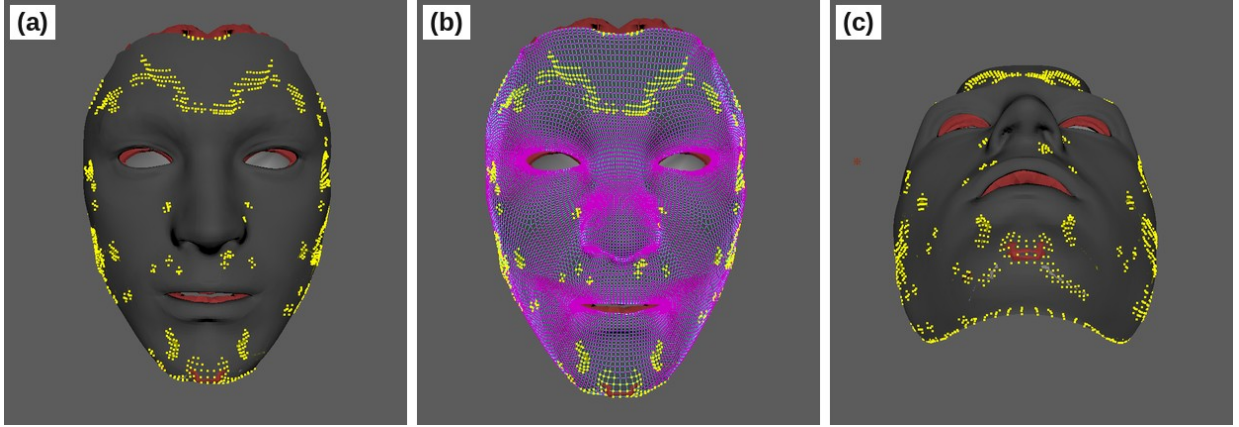


Figure 4. M1 selection (yellow dots): (a) front, (b) front with mesh overlay, (c) bottom. Selected vertices follow the forehead creases, nostrils, nasolabial and jaw folds, the chin and the outer boundary. Journal screenshots M1_front, M1_front_vertices, M1_bottom.

Result. M1 found local irregularity all over the skin and established the Maya selection and reporting pipeline. The selected dots trace the forehead ridge, the alar folds beside the nose, the lower cheek and jaw line, the chin and the outer boundary of the skin cap.

What worked. The detector is simple, fast (one pass over the adjacency) and interpretable, and it produced the first reproducible artifact mask. It also gave the metric, the roughness r_i , that all later chapters use for evaluation.

What failed. Much of what M1 selects is legitimate anatomy: the brow ridge, nostrils, lip corners and jaw line are high-curvature by nature, so they score as highly as true bumps. Boundary vertices also score high. Because selection is by percentile there is no way to say that a selection is empty, and no broad error is detected at all.

Why it failed technically. The umbrella vector L_i is a discrete curvature-like quantity: on a smooth surface it is the (scaled) mean-curvature normal, so any region of high curvature has a large $\|L_i\|$ whether or not the registration is wrong. On irregular sampling it also acquires a tangential component, and the uniform operator is known to approximate the mean-curvature normal poorly there [16, 5]. Edge lengths in this mesh vary by two orders of magnitude (0.13 to 16 units; coefficient of variation 0.48), so unnormalised scores confound sampling density with shape. Boundary vertices have one-sided rings, which inflates their scores.

Insight. “Unlike its neighbours” is not the same as “wrong”. A detector needs a notion of what the local shape *should* be.

Motivation for M2. If the expected local shape can be obtained from a reference surface, the legitimate curvature cancels and only the difference remains.

5.2 M2: reference-based Laplacian detection

Objective and hypothesis. M2 tried to preserve valid facial curvature by comparing each skin Laplacian with the Laplacian of the corresponding vertex on the supplied target mesh. If a feature exists in both surfaces, their local Laplacian coordinates agree and cancel; an artifact present only in the current skin leaves a residual.

Method. The score is $s_i = \|L_i(X) - L_i(T)\|_2$ (Eq. (4)) with T the target mesh (skin_cloth_copy_v5_pull_back_target). The full Laplacian *vectors* are compared, not magnitudes. A topology

check (`validate_corresponding_topology`) verifies equal vertex counts and identical one-ring adjacency before scoring. Selection is the same percentile rule as M1 (`detect_reference_irregular_region`).

Parameters. Percentile 97.5, growth one ring; normalisation by local edge length available but off by default; $\varepsilon = 10^{-8}$.

Screenshots. Figure 5 shows the selection; Figure 6 is the journal note that recorded the intended logic.

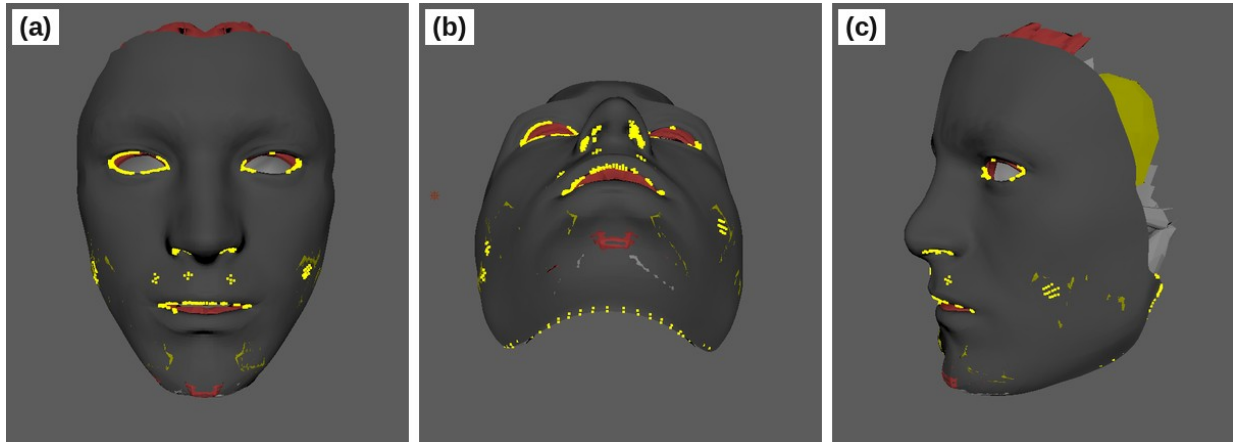


Figure 5. M2 selection (yellow dots): (a) front, (b) bottom, (c) left. Detections cluster around the eyelid rims, nostrils and lip line and along the outer edge, and are sparse on the broad cheek and forehead surfaces. Journal screenshots M2_front, M2_bottom, M2_left.

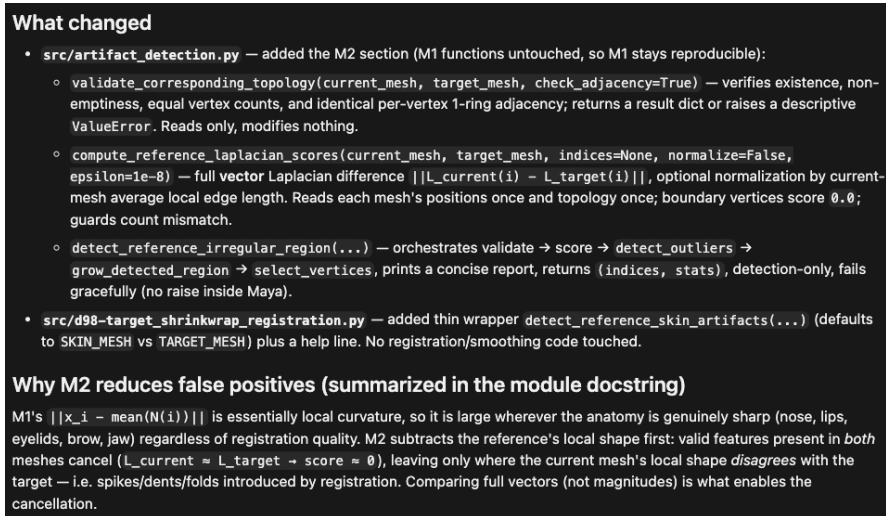


Figure 6. Journal note recording the M2 design intent (code assistant summary): full-vector Laplacian difference against the target, topology validation, and the argument that valid features present in both meshes cancel.

Result. M2 removed many of the M1 selections on the forehead and cheeks, but the surviving selections sit around the eyes, nostrils, lips and outer boundary.

What worked. The reference-difference idea is technically sound and the implementation (topology validation, vector comparison, optional normalisation) became the basis of M4. It also confirmed that comparing vectors cancels shared curvature.

What failed. The score measures disagreement with the target far better than it measures artifact probability. The registered skin was legitimately altered by anatomy, shrink-wrap behaviour, landmark

constraints and manual corrections, so it *should* differ from the target near the eyes, lips, nostrils and outer edge; those are exactly the selected places.

Why it failed technically. The target is not an artifact-free version of the registered face: it is a different surface, and $L_i(X) - L_i(T)$ is large wherever registration moved the geometry away from it, which is largest at the openings where the anatomy constrains the skin most. At boundaries and opening rims, one-sided rings make the two Laplacians differ even for small shifts. The method inherits every error of the reference.

Insight. The reference must encode what the surface should be *given the anatomy*, not what the original target was.

Motivation for M3. Before building a new reference, the local signal itself was made more selective (multi-scale persistence, explicit boundary handling, component filtering); the anatomy-derived reference followed in M4.

5.3 M3 V1: multi-scale persistence

Objective and hypothesis. A valid sharp feature (a lip corner, a nostril) loses prominence as the neighbourhood grows, whereas a genuine local defect stays unusual across scales. Requiring persistence over several ring radii, and removing boundary regions and tiny components, should raise precision compared with M1.

Method. Normalised Laplacian scores $\rho^{(r)}_i$ are computed at ring radii 1, 2 and 3 (Eq. (5)); a percentile threshold is applied at each scale; vertices exceeding it at two or more scales are kept (Eq. (6)). Topological boundary vertices and one buffer ring are excluded, connected components smaller than five vertices are dropped, and the survivors are grown by one ring (detect_multiscale_irregular_region).

Parameters. Scales (1, 2, 3); per-scale percentile 97.5; persistence count 2; boundary buffer 1 ring; minimum component size 5; final growth 1; normalisation by local mean distance on.

Screenshots. Figure 7.

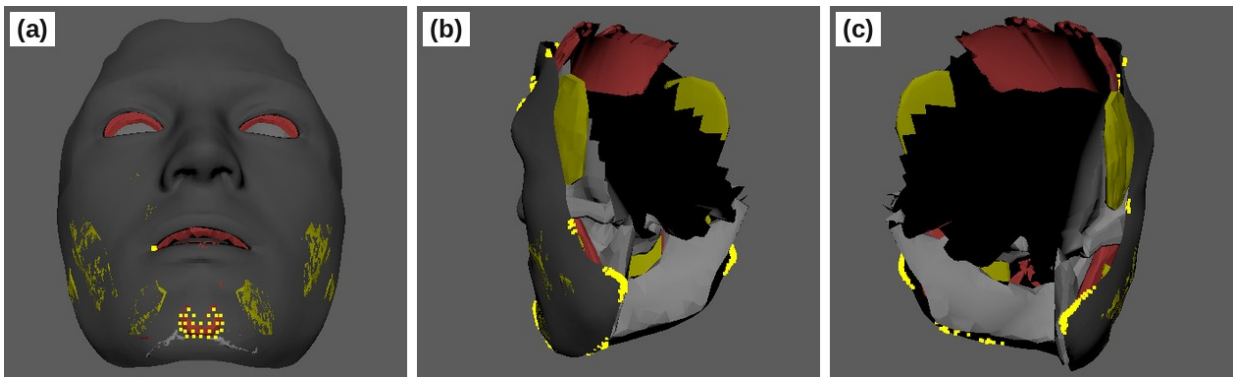


Figure 7. M3 V1 multi-scale persistence: (a) bottom view with a coherent yellow-dot cluster on the chin; (b), (c) views from behind the open skin cap, where the interior is visible through the boundary opening (valid exposure). The solid yellow speckle on the cheeks in (a) is fat visible through the skin and is not selected by the detector. Journal screenshots M3_V1_bottom, M3_V1_back_left, M3_V1_back_right.

Result. Scattered single-vertex detections largely disappeared and the surviving selections form coherent regions. The chin artifact, a rough cluster above the mentalis with red muscle showing through, was detected well.

What worked. Persistence across scales, boundary exclusion and the component filter changed the *feature being measured* rather than merely raising the M1 threshold, and produced the first mask that looked like a set of defects instead of a scatter of points.

What failed. Broad smooth errors were missed. In Figure 7(a) the cheeks visibly show solid yellow fat through the skin, yet there are no selected vertices there: a cheek can be wrong while staying locally smooth at all three scales.

Why it failed technically. The umbrella Laplacian is a second-difference (high-pass) operator. A slowly varying displacement of a region has a very small Laplacian: in graph-frequency terms the error lives at low κ where every Laplacian-based score is small (Figure 20 shows the same frequency argument for smoothing). Persistence filtering removes noise but cannot restore energy that the operator does not see, and the top-2.5% threshold per scale makes the requirement more restrictive still.

Insight. Local geometry alone is well matched to sharp defects and fundamentally weak for broad drift; information from outside the surface (the anatomy) is needed.

Motivation for M3 V2. Add signals that respond to broad drift: normal inconsistency and the local deviation of the distance to the anatomy.

5.4 M3 V2: hybrid geometry and anatomy signals

Objective and hypothesis. Combining a multi-scale Laplacian, a normal-inconsistency score and a local distance-to-anatomy deviation should detect both sharp defects and smooth drift, provided the three signals are put on a common scale.

Method. Each signal is converted to a robust z-score (median and MAD, negative values clamped to zero) and the three are combined with fixed weights and multiplied by a soft boundary weight (Eq. (7); `compute_hybrid_artifact_scores`). Selection is by percentile on the combined score, followed by the same boundary, component and growth steps as V1.

Parameters. Laplacian scales (1, 2); normal rings 1; distance rings 2; weights (0.3, 0.2, 0.5) normalised to sum 1; boundary weights (0, 0.25, 0.6, 1) at 0, 1, 2 and 3+ rings, with a stronger variant (0, 0, 0.15, 0.5, 1) tried.

Screenshots. Figure 8.

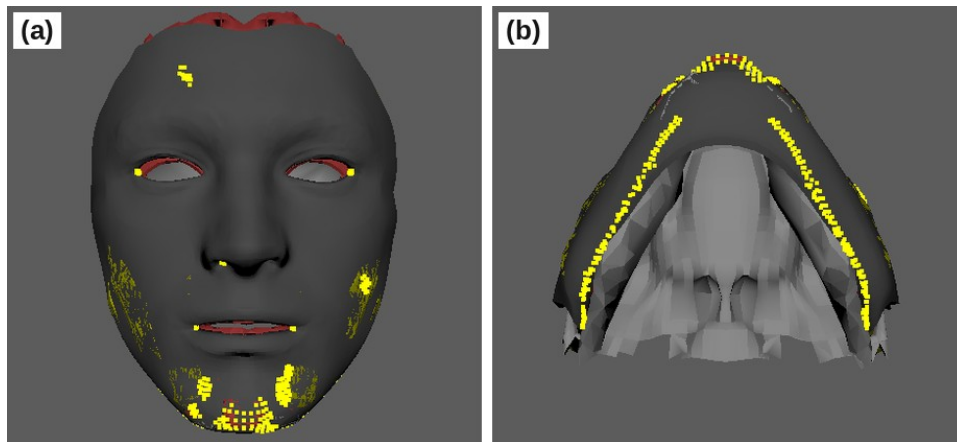


Figure 8. M3 V2 hybrid detector: (a) front view showing a chin cluster, cheek-edge selections and isolated eye- and mouth-corner points; (b) view from below in which long dotted lines follow the lower boundary of the skin, a boundary false positive. The faceted grey geometry inside (b) is the interior anatomy seen through the cap opening. Journal screenshots M3_V2_front, M3_V2_bottom.

Result. The hybrid response was broader than V1 over the cheeks and jaw, and the distance signal responded to regions that the Laplacian did not.

What worked. The distance-to-anatomy signal is the first detector input that sees broad drift, and the robust z-normalisation made three quantities with different units comparable.

What failed. New false positives appeared around folds, eye and mouth corners and along the lower boundary (Figure 8(b)), and the result depends on hand-set weights that no experiment justified.

Why it failed technically. Natural skin thickness varies, so a vertex whose distance differs from its neighbours' median is not necessarily wrong; the local comparison assumes a locally uniform offset that the anatomy does not provide. Robust z-scores are *relative*: they flag the upper tail of whatever distribution they see, so a clean patch still produces detections. Normal inconsistency responds to every legitimate fold, and boundary weighting only softens, not removes, boundary responses.

Insight. Distance to anatomy is informative, but it must be compared with an *expected* offset field, not with local statistics; and it is better to keep evidence from different sources as separate sets than to blend them with arbitrary weights.

Motivation for M4. Build the expected surface explicitly from the anatomy and compare Laplacians against it, as M2 tried to do with the target.

5.5 M4: anatomy-derived SDF reference

Objective and hypothesis. Replace the unreliable target by a reference constructed directly from the internal anatomy: the offset iso-surface $\varphi = d_0$ of the unsigned distance field, which is where the skin would sit if it followed the anatomy at a uniform offset. Disagreement between the skin's Laplacian and this reference's Laplacian should highlight broad smooth misregistration as well as local defects.

Method. The anatomy query reuses the registration script's exponential soft-minimum over the five nearest meshes (Eq. (8)). Each skin vertex is moved onto $\varphi = d_0$ by the damped Newton iteration of Eq. (9), preserving vertex index and adjacency (no new mesh, no marching cubes), and the M4 score is the reference-Laplacian disagreement of Eq. (4) with the projected positions as T . Selection is by percentile with boundary exclusion, component filtering and growth (`detect_sdf_reference_artifacts`).

Parameters. Blend $k = 1$; five nearest meshes; $d_0 \approx 7.3998$ world units, chosen from the scene's distance statistics; at most 10 Newton iterations, tolerance 10^{-4} ; percentile 97.5, threshold 2.5, normalisation on; boundary buffer 1; minimum component 3; growth 1.

Screenshots. Figure 9 shows the reference surface itself; Figure 10 shows the M4 detections.

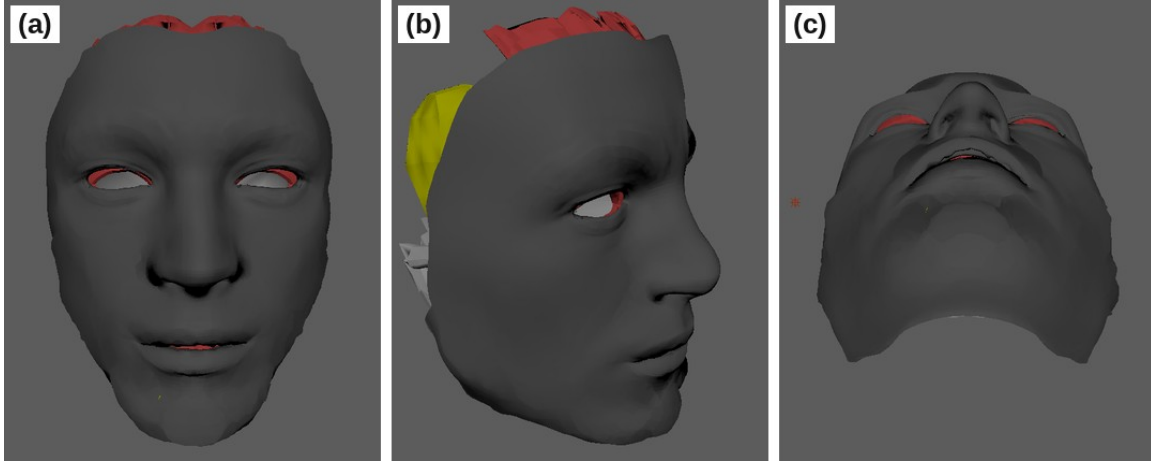


Figure 9. SDF-derived reference surface (target T , the skin projected onto $\phi = d_0$): (a) front, (b) left, (c) bottom. The reference is smoother than the registered skin and inherits large-scale facets from the anatomy. Journal screenshots SDF Skin Reference/Front, Left, Bottom.

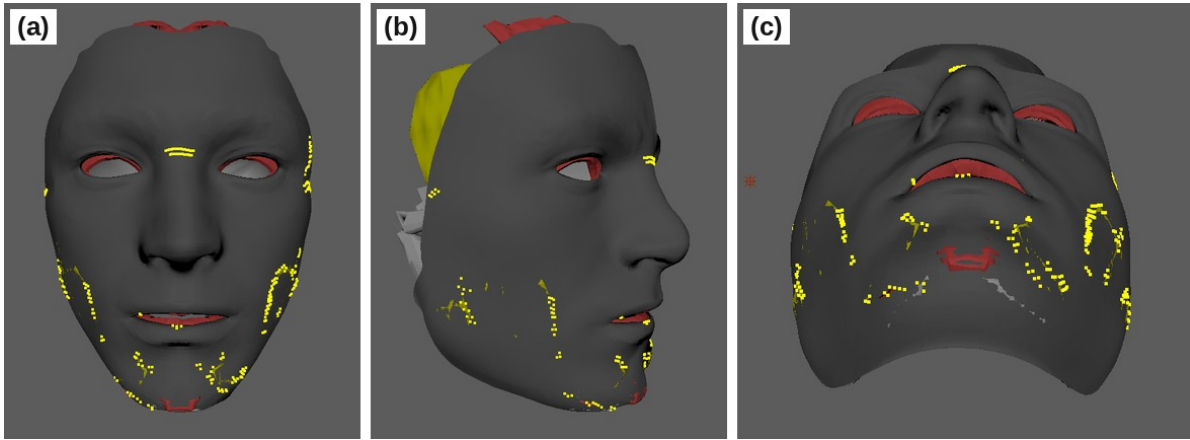


Figure 10. M4 detections (yellow dots): (a) front, (b) left, (c) bottom. Detections lie on the lower cheeks, jaw line, chin and lip corners, with a short bar between the eyebrows; the pattern differs from M3. Journal screenshots M4_V1_Front, M4_V1_Left, M4_V1_Bottom.

Result. M4 selected a different set of vertices than M3: smooth regions on the lower face whose local shape disagrees with the anatomy-derived expectation, in addition to a few sharp spots.

What worked. M4 separated two artifact types. M3 remained strongest for sharp local irregularity; M4 could highlight regions that are locally smooth but disagree with the anatomy. Using the same smooth-minimum distance as the registration keeps the reference consistent with the surface that the registration was trying to reproduce. It also established a terminology rule: the backend is unsigned, so M4 is “anatomy-aware reference disagreement”, not penetration detection.

What failed. M4 is a detection signal and not a correction rule. A single global offset d_0 ignores true variation in soft-tissue thickness, so some disagreement is legitimate, and detections still cluster near openings.

Why it failed technically. The distance is unsigned ([11] and [12] discuss signed distance fields and their sign problem), so the iteration assumes the skin lies outside the anatomy and moves along the outward gradient; if a vertex is inside, the outward direction is wrong. The smooth minimum over several meshes has kinks where the nearest mesh changes, and the reference inherits the faceting of the coarse muscle and fat

proxies (the faceted grey surface in Figure 8(b)). Finally $L_i(T)$ is only as good as the iso-surface projection, which fails or moves far for some vertices; those are excluded from scoring.

Insight. M3 and M4 answer different questions and neither is redundant; combining them should not hide which one fired.

Motivation for M5. Fuse the two candidate sets explicitly and keep their provenance.

5.6 M5: unified interpretable detector

Objective and hypothesis. Provide one detection mask that keeps the complementary M3 and M4 evidence separate and inspectable, and then freeze the detector so that later correction experiments cannot silently change the region they are evaluated on.

Method. M3 (the V1 multi-scale detector, not the V2 hybrid) and M4 are each run once without growth; their candidate sets are fused by set operations (Eq. (10)) into M3-only, M4-only, overlap and union categories; a shared boundary exclusion with buffer, a connected-component filter, a single growth ring and a boundary re-exclusion (the rim can regrow) follow (detect_unified_artifacts). Fusion of sets rather than scores means no arbitrary M3/M4 weight enters.

Parameters. M3: scales (1, 2, 3), percentile 97.5, persistence 2, buffer 1, minimum component 5. M4: percentile 97.5, buffer 1, minimum component 3, $d_0 \approx 7.3998$. Fusion mode union; final buffer 1, minimum component 3, growth 1, boundary exclusion re-applied after growth.

Screenshots. Figure 11 shows two recorded M5 runs (V1 top, V2 bottom).

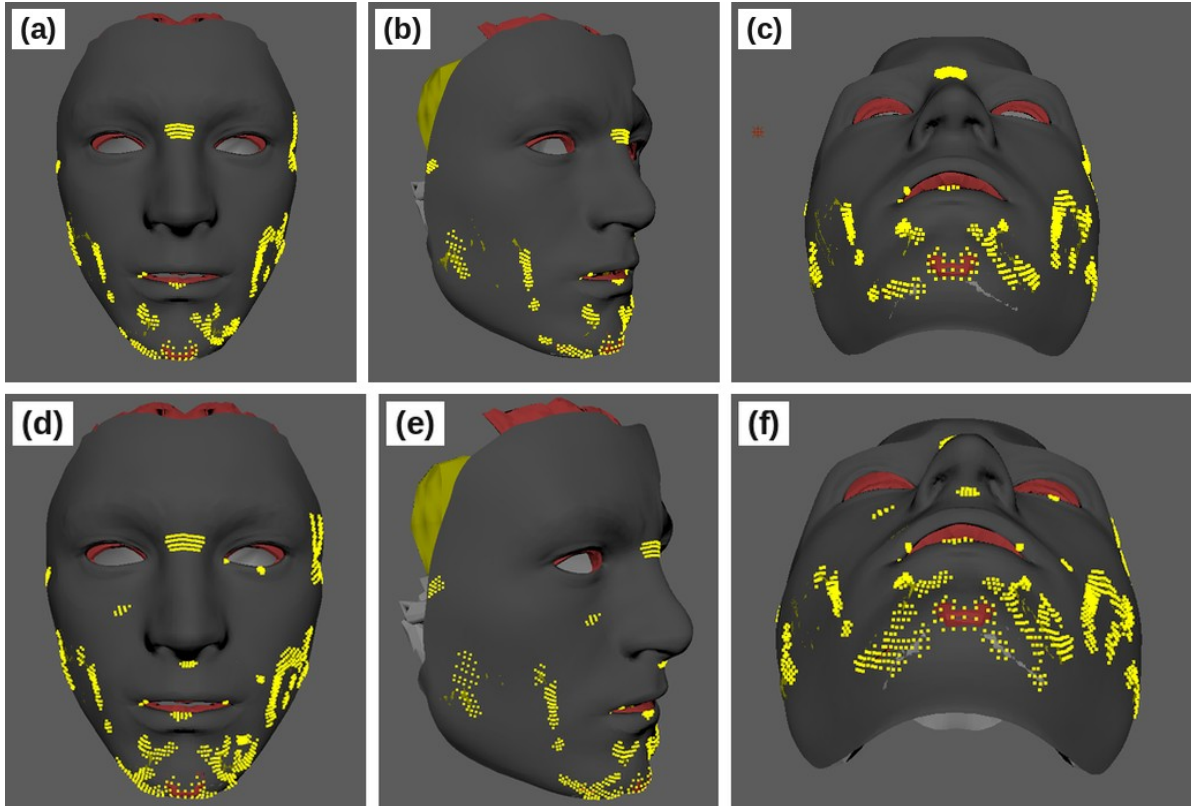


Figure 11. M5 union selections. Top row (V1): (a) front, (b) left, (c) bottom. Bottom row (V2): (d) front, (e) left, (f) bottom. V2 selects visibly more vertices on the chin, under the jaw and along the cheeks. The parameter difference between the two recorded runs is not documented in the evidence. Journal screenshots M5_V1_* and M5_V2_*.

Result. The union covers the lower cheeks, jaw line, chin, lip corners and a short forehead bar. Each selected vertex retains provenance (sharp/local evidence, anatomy-reference evidence, or both), which is what the later unified solver used to weight its forces.

What worked. Interpretability, high recall and stability: after M5 the detector was declared sufficient and frozen. Later problems could therefore be attributed to correction, not detection.

What failed. Detection is not correction. The mask also contains false positives near boundaries, and there is no labelled ground truth, so the mask can be judged visually and against the intersection detector but its precision and recall cannot be scored. Two recorded M5 runs differ visibly and the difference in their settings is unrecorded.

Why it failed technically. Percentile selection is relative, so each component always returns roughly its top 2.5% before cleaning, and a union raises recall at the cost of precision. The mask says *where* to look; nothing in it says in which direction a vertex should move or how far it may safely move relative to the anatomy.

Insight. Detection was good enough. The remaining difficulty, correcting the selected geometry without destroying skin–anatomy validity, is a different problem with a different safety metric.

Motivation for the next stage. A true safety metric was needed before any correction could be trusted; that is the triangle–triangle intersection detector of Section 6. Note that the later pure and blocked Laplacians select their own region from roughness percentiles (Section 3.5); they do not use the M5 mask.

Evidence gaps for the detection milestones

There are no written notes for M4 and M5, the difference between the two M5 screenshots is unrecorded, and no ground-truth artifact labels exist. Milestone descriptions are therefore derived from the code, the commit history, the working draft and the screenshots; precision and recall were not measured.

5.7 First correction trial: category-driven M5 cleanup

The first attempt to *correct* what M5 detected is recorded in two repository logs written on 13 September 2026 (cleanup_logs/m5_category_cleanup_*.json and *.csv; driver run_m5_iterative_cleanup). It is the transition experiment between detection and correction and is summarised here because its numbers explain several later design decisions.

Method. Each cycle re-detected M5 with provenance and then applied a different correction per category (hybrid mode). Vertices found only by M4 were pulled by 10% of the vector to their SDF target (per-step cap of half the local edge length, one extra ring at weight 0.5); vertices found only by M3 received 200 Taubin iterations at strength 0.3; a five-iteration Taubin blend at strength 0.2 followed the projection. Detection was then repeated, for at most ten cycles with a convergence patience of two. The M4 offset in the log is $d_0 = 7.3998$, which confirms the value used in Section 5.5; M3 percentile 97.5, M4 percentile 96.

Logged outcome. The longer run (ten cycles, 1,055 s in total, dominated by a detection pass of about 91 s per cycle) ended at the cycle cap rather than by convergence. The M4 target residual, the mean distance from a selected vertex to its own target, was 7.8 world units before cycle 1 and 6.9 before cycle 10, roughly three mean edge lengths, and each cycle removed only 4.0–7.1% of it. Successive selections overlapped with a Jaccard index of 0.69–0.90; per cycle between 37 and 163 vertices left the selection and between 23 and 108 new ones entered, so the mask kept churning: it began at 834 vertices and ended at 577, while 999 distinct vertices had been corrected. The largest cumulative displacement was 5.57 units at vertex 7311.

What this taught. Three things. First, a percentile detector re-run on a changing surface does not converge, because it returns its top tail whatever the surface looks like, so detection cannot serve as the stopping criterion of a correction loop. Second, the M4 target is not a nearby correction: a residual of several edge lengths means the projected iso-surface point is often far from the vertex, which is what one expects when an unsigned field jumps between anatomy meshes (see the failure analysis in Section 5.5). M4 is useful as a *direction of disagreement* but not as a place to move to. Third, none of this loop had a safety criterion; a vertex could be pulled or smoothed into anatomy without any test. Each finding is carried forward: the unified solver of Section 9 treats M4 as a decaying guidance force, and Section 6 introduces the missing safety test.

6. True Surface-Intersection Research

M5 says where the skin looks irregular; it does not say whether a region is anatomically *wrong*. Answering that needs a geometric test that is independent of both the artifact detectors and of any smoothing parameter. The repository already contained a vertex-level classifier (`analyze_skin_anatomy_penetration`) that treats an unsigned distance below a clearance floor as “below clearance” and never, by itself, as “inside”; an inside verdict is reserved for closed anatomy meshes and uses multi-ray intersection parity (seven rays, a majority of four non-grazing votes), with open meshes yielding at most a heuristic “likely penetrating”. The research then moved to the level at which the defect is actually visible: the skin *triangle*.

6.1 Four conditions that must not be confused

Table 8. The four conditions distinguished by the safety analysis.

Condition	How it is measured	What it does not imply	Evidence in this project
Low anatomy clearance	Unsigned exact distance from a skin vertex to the nearest anatomy triangle is below a floor (repository default floor 0.8 world units).	Neither penetration nor a face crossing: a vertex can be close and still outside.	The clean baseline <code>_solver200</code> has 592 non-boundary vertices closer than 1 unit yet no forbidden face.
Vertex penetration	Vertex lies inside a closed anatomy mesh (multi-ray parity). Open meshes give only a heuristic verdict.	A vertex outside every closed mesh can still belong to a pierced face.	The signed vertex test is available in the repository; its outcomes were not re-measured for this report.
Face intersection	An exact triangle–triangle test between a skin triangle and an anatomy triangle (Eq. (11), Eq. (12)).	Not proximity. Ignores exposure through true openings.	The pre-cleanup skin has 93 forbidden faces (497 crossing pairs); 78% of their vertices are under 1 unit from anatomy, but 82% of all vertices under 1 unit belong to no forbidden face (Table 10).
Valid exposure	Anatomy seen through the eye and mouth openings and the open cap boundary.	Not a defect; must survive cleanup.	Excluded by construction: 1,171 skin faces touching the one-ring-buffered boundary are never counted.

6.2 The detector

The detector is exactly the one defined in Section 4.4 and is unchanged from the moment it was introduced; every later solver, and every offline evaluation in this report, calls the same function (`anatomy_constraint.analyze_skin_anatomy_intersections`). Algorithm 1 lists the procedure. Two design decisions matter. First, the unit of failure is the skin *face*, not the vertex, because the visible protrusion is a pierced face. Second, faces touching the one-ring-buffered boundary are excluded, which is how legitimate exposure through the eyes, the mouth and the cap edge is kept out of the count without any per-region tuning.

```

Algorithm 1: forbidden skin faces
input: skin positions X, face list (quads/triangles), anatomy triangle caches {A_m}, boundary set B
1  split every skin quad into two triangles by a fixed diagonal (vertices 0-2)
2  build one axis-aligned box per skin triangle; index all anatomy triangle boxes on a uniform grid
   (broad phase)
3  for each skin triangle t that does not touch B^(1) (boundary plus one ring):
4      for each anatomy triangle a whose box overlaps t (tolerance 1e-6):
5          test the three edges of t against a and the three edges of a against t as finite segments
             (Moller-Trumbore); coplanar pairs use a 2-D separating-axis test             (narrow
phase)
6      class(t, a) <- crossing | touching | coplanar-overlap | uncertain
7  F <- { skin faces owning a triangle in a pair of class crossing or coplanar-overlap }
8  return F, the pair list, and per-anatomy-mesh counts

```

The offline evaluation reproduces the count that Maya logged for the pre-cleanup skin to within one face (93 against 92) and is sensitive, at the level of a few faces, to the diagonal used to split quads (99 with the other diagonal), so absolute counts are quoted with that tolerance (Section 3.3). Touching pairs and *uncertain* pairs (degenerate configurations such as the two zero-length edges that produce all 21 uncertain pairs in `_solver200`) are reported but not counted as violations.

6.3 What the detector found on the pre-cleanup skin

Table 9. Skin–anatomy crossing pairs by anatomy mesh, pre-cleanup skin (`_precleanup_solver`). 93 forbidden skin faces, 497 crossing pairs, 0 coplanar, 0 touching, 0 uncertain.

Intersected anatomy mesh	Tissue	Crossing pairs	Share
Mentalis (muscle)	muscle	115	23.1%
fat_8	fat	63	12.7%
Jaw (bone)	bone	60	12.1%
fat_4	fat	54	10.9%
fat_element_v2_mirrored3	fat	48	9.7%
fat_11	fat	47	9.5%
fat_12	fat	45	9.1%
fat_7	fat	35	7.0%
Depressor labii inferioris, right (muscle)	muscle	23	4.6%
Depressor labii inferioris, left (muscle)	muscle	7	1.4%
All anatomy meshes		497	100%

By tissue, 292 of the 497 crossing pairs (59%) involve fat meshes, 145 (29%) muscle meshes, and 60 (12%) the jaw bone (Figure 12). This mix matches the colours seen in the journal screenshots: yellow fat specks on the chin, jaw and lower cheeks and red muscle at the chin (compare Figure 1). The offending faces are spatially compact (Figure 13): 93 faces fall into 16 topologically connected clusters of core vertices, on the chin, along the jaw line and on the lower cheeks. The three tissue classes are examined per state in Section 14.

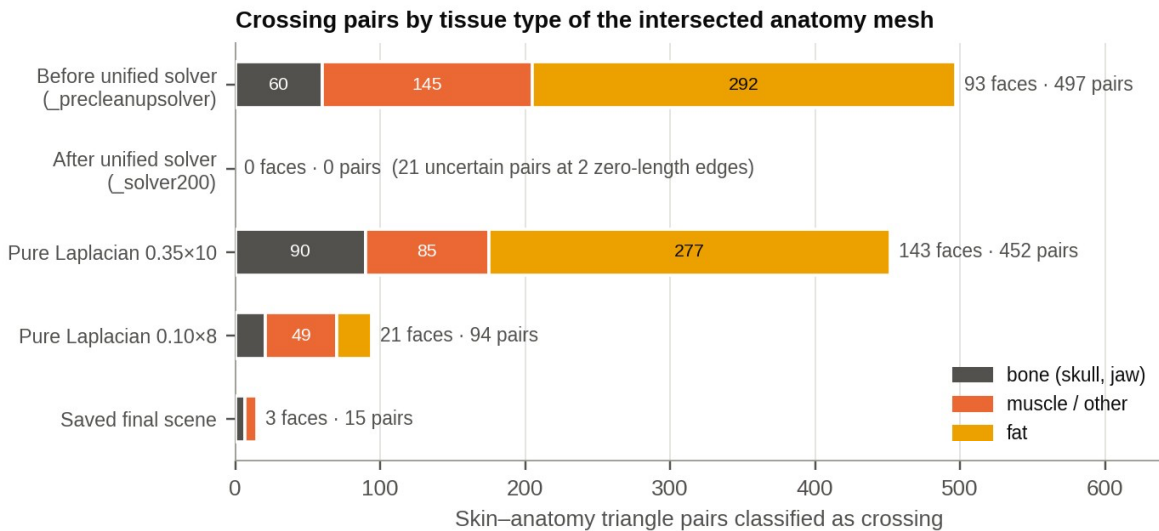


Figure 12. Skin–anatomy crossing pairs classified by tissue type of the intersected mesh, for the states analysed offline. The unified solver removes all crossings; the pure Laplacian re-creates them, mainly with fat and jaw, and the saved final scene retains 3 faces (15 pairs) on the jaw and mentalis. Offline detector run on the saved scene states.

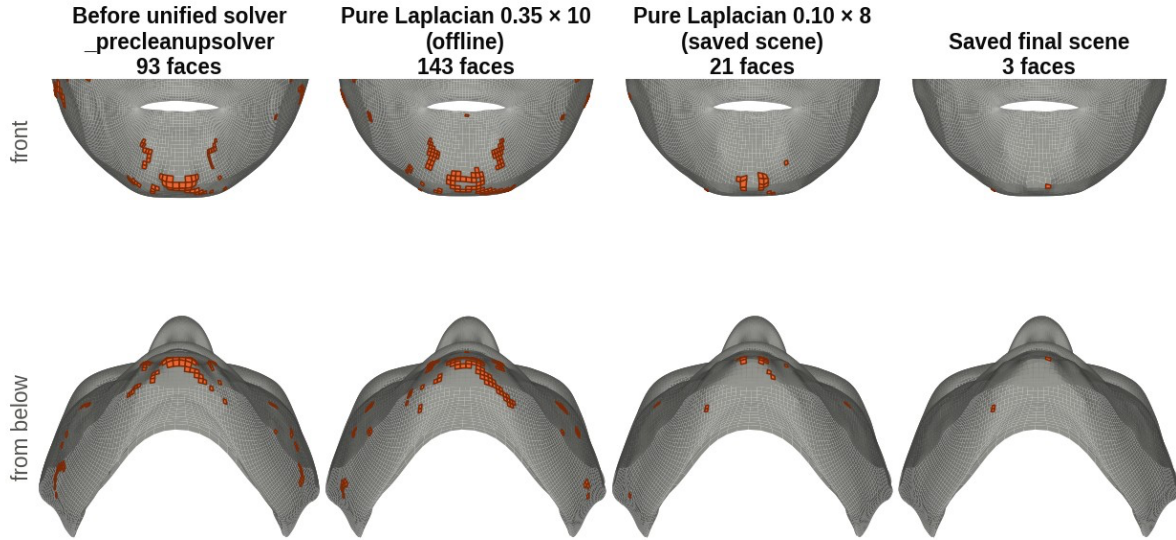


Figure 13. Forbidden skin faces (orange) drawn on the skin mesh (front view above, from below underneath) for the pre-cleanup skin (93 faces), the pure Laplacian 0.35×10 run (143), the pure 0.10×8 scene state (21) and the saved final scene (3). The faces sit on the chin, along the jaw line and on the lower cheeks; the surviving three are at the chin and jaw.

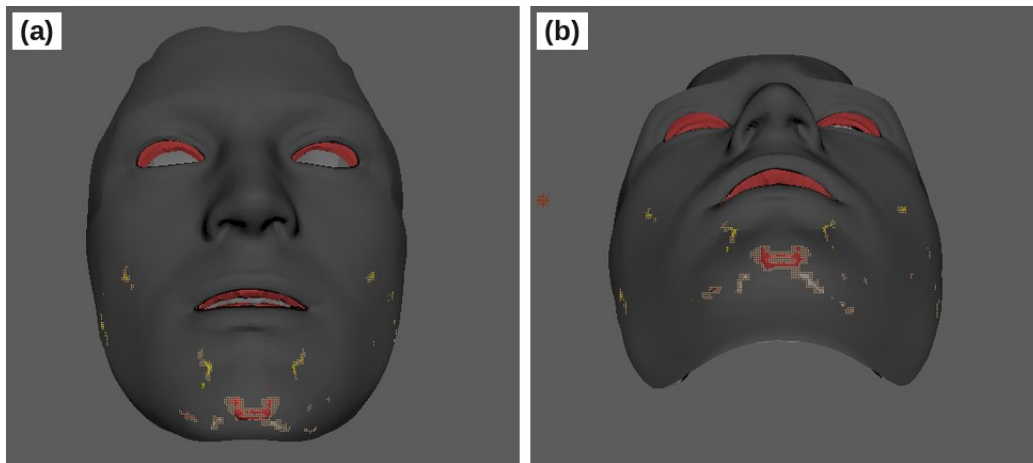


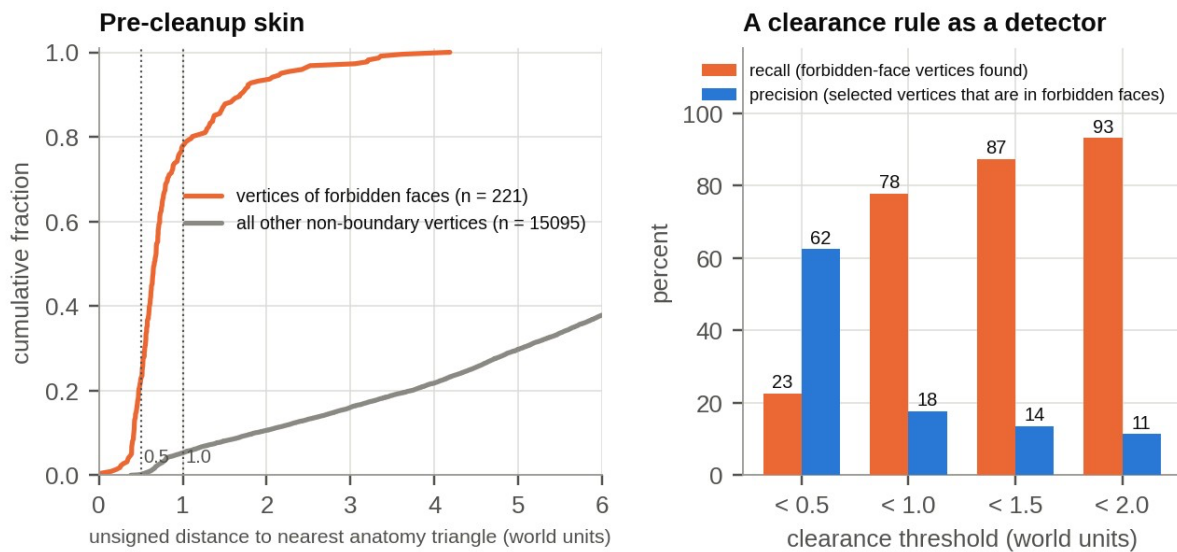
Figure 14. Journal screenshots of the true-intersection selection on the pre-cleanup skin: (a) front, (b) from below. The dotted brown patch at the chin, together with smaller patches on the jaw and cheeks, is the selection of intersecting skin *faces* in Maya; the small yellow marks are fat showing through the skin. Journal screenshots Region Intersection (front, bottom).

6.4 Face intersection versus vertex distance

The reason for a separate face-level test can be measured. In the pre-cleanup skin the detector reports 221 vertices that belong to forbidden faces. Their unsigned distance to the nearest anatomy triangle has a median of 0.66 world units, and one of them is 4.18 units away, which is consistent with a large skin face (edges reach 16 units) whose centre is pierced while its corner vertices are clear. If one instead flagged every vertex under a clearance threshold, the rule would have to be loose to find most of the problem and then produce mostly false alarms (Table 10, Figure 15): at 1 unit it finds 78% of the vertices of forbidden faces but only 18% of what it selects belongs to one.

Table 10. A clearance threshold used as a detector of forbidden-face vertices on the pre-cleanup skin (15,316 non-boundary vertices; 221 belong to forbidden faces). The face-level detector is the reference, not a human label.

Clearance threshold (units)	Vertices selected	of which in forbidden faces	Recall	Precision
< 0.25	3	3	1.4%	100.0%
< 0.5	80	50	22.6%	62.5%
< 1.0	968	172	77.8%	17.8%
< 1.5	1,419	193	87.3%	13.6%
< 2.0	1,808	206	93.2%	11.4%
< 3.0	2,634	214	96.8%	8.1%

**Figure 15.** Left: cumulative distribution of the unsigned clearance of the 221 vertices of forbidden faces and of all other non-boundary vertices in the pre-cleanup skin; the two distributions overlap heavily below 1 unit. Right: recall and precision of a clearance-threshold rule with respect to the face-level detector.

The same distinction holds at the other end of the research. In the saved final scene the three residual forbidden faces have ten vertices, none of which is closer than 0.5 units to anatomy: a threshold of 0.5 selects 30 other vertices and none of the ten, and only at 1.5 units does the rule reach all ten, by selecting 1,156 vertices. Conversely `_solver200` has 592 vertices under one unit and no forbidden face at all. Distance says that a vertex is close; only the face test says that the surface is crossed. This is why the whole later work is measured with faces, and why clearance appears only as a secondary indicator (Section 14).

The visual comparison that motivated this step was made by the author in Maya: selecting the intersecting skin faces highlighted the same places where yellow or red anatomy was visible through closed skin (Figure 14), whereas selections built from vertex-distance heuristics were reported not to line up as well. The journal does not contain a screenshot of a distance-heuristic selection, so that second observation is stated here as the author's account and not as figure evidence.

6.5 Limits of the detector

The detector answers a yes/no question per face. It does not measure how deep a crossing is (an intersection-depth output exists in the analysis files but its definition is not documented, so no depth statistics are reported), it does not measure skin thickness or clearance, and its counts inherit the quad-triangulation

tolerance described above. It also depends on the anatomy meshes being the intended geometry: intersections with an anatomy mesh that is itself wrongly placed would be reported as skin defects. None of these limits changes the qualitative conclusions, but they define what “zero forbidden faces” does and does not certify.

7. Repair Experiments

With a safety metric available, the first correction experiments tried to remove the intersections directly. Two repair versions are preserved in the repository for A/B comparison, selected by `repair_mode`; they are documented here together with the third screenshot in the journal series.

7.1 V1: per-vertex normal push

Objective and hypothesis. A crossing is caused by a few vertices that lie too far inside; moving exactly those vertices outward should remove it.

Method. Function `_legacy_normal_push_intersection_repair`. The core is the set of vertices of forbidden faces. In each of up to 20 iterations every core vertex steps along its own outward skin normal by 10% of its local mean edge length, with a binary search for the smallest step, and the detector is re-run. Neighbours of the core do not move.

Screenshot. Figure 16(a) is the journal screenshot “V1_Repair (Spikes)”.

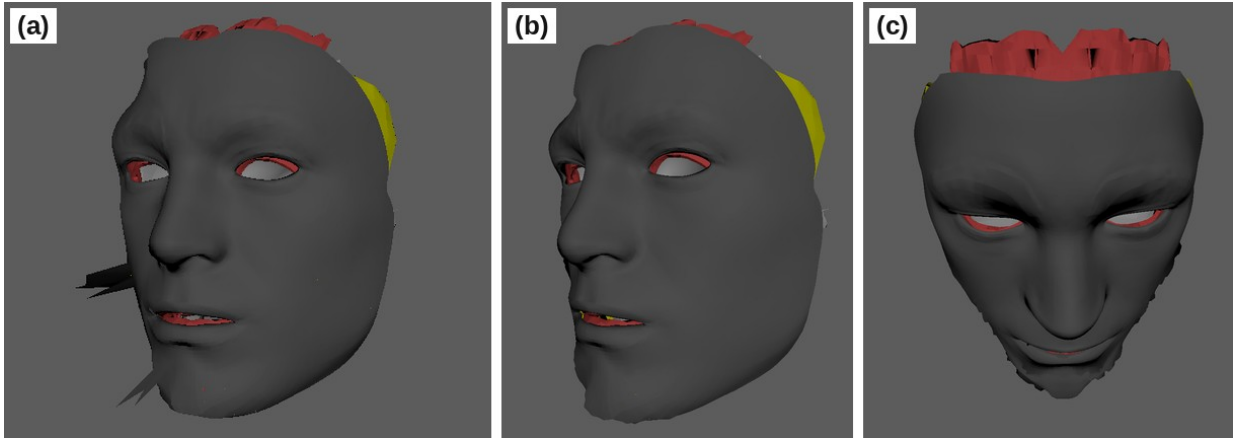


Figure 16. Repair experiments on the pre-cleanup skin. (a) V1 per-vertex normal push: dark spikes project from the jaw and the chin. (b) V2 anatomy-supported patch repair (journal name “V2_Repair (Rough)”): the spikes are gone but the jaw and chin surface is rough and mounded. (c) Journal image “V3”, whose stage is not identified in the notes; the chin contour is still serrated. Journal screenshots `ri_v1`, `ri_v2`, `ri_v3`.

Result. Isolated spikes several units long appeared at the chin and jaw. The intersections were reduced but the surface was unusable.

Why it failed technically. Moving a single vertex by a displacement δ while its neighbours stay fixed changes its umbrella vector by exactly δ ($L_i - L_i + \delta$), so the vertex acquires roughness of about $\|\delta\|$ regardless of how smooth it was: a 3-unit push produces a roughness of about 3, thirty times the skin-wide mean of 0.1. The displacement field of V1 is nonzero on the core and exactly zero one edge away, the roughest possible way to introduce a displacement. Repeating the push for up to 20 iterations at 10% of the local edge length can move a vertex by up to twice its local edge length, which on this mesh is several units.

Insight. A collision is not a vertex problem; the correction has to be a *patch* problem with a smooth displacement field. What V1 falsified was the hypothesis that a local, unconstrained normal push is a valid fix.

7.2 V2: anatomy-supported patch repair

Objective and hypothesis. Move the offending core to just outside the anatomy, then spread that displacement smoothly over a patch so that no spike is created.

Method. Function `_anatomy_supported_patch_repair`. (i) The intersecting vertices are split into topologically connected components. (ii) Each core vertex is moved along the oriented support normal of every offending anatomy triangle until it lies a small offset ε outside each in turn (never an average, because an average can re-cross one of the surfaces). (iii) The core displacement D_i is extended to a patch P by a discrete harmonic field with the core pinned and the outer ring pinned to zero, which is Jacobi iteration of

$$d_i^{(k+1)} = \frac{1}{|N(i) \cap P|} \sum_{j \in N(i) \cap P} d_j^{(k)} \quad (i \in P_{\text{int}}), \quad d_i = D_i \quad (i \in \mathcal{C}), \quad d_i = 0 \quad (i \in P_{\text{out}}) \quad (22)$$

for up to 50 sweeps or until the largest change is below 10^{-8} . (iv) The whole field is scaled by the smallest factor found by an eight-step binary search that leaves the patch valid under the unchanged detector, and (v) three Taubin-style relaxation steps of strength 0.1 follow. Two profiles exist (Table 11): a local one (two blend rings, pinned boundary) and the default broad one (six rings and a soft boundary in which the last two rings are capped at 12% and 25% of the mean core displacement).

Table 11. V2 repair profiles as implemented (`anatomy_constraint_apply_repair_profile` and defaults).

Parameter	legacy_v2_local	broad_fair (default)
Blend rings around the core	2	6
Displacement field	harmonic, 40 sweeps	harmonic, 50 sweeps
Outer boundary	fixed (all outer vertices pinned to 0)	soft (last ring pinned; last two rings capped)
Ring weights (envelope)	(1.0, 0.6, 0.3)	(1.0, 0.6, 0.3)
Binary-search steps / max passes	8 / 5	8 / 5
Post-repair relaxation	3 iterations, strength 0.1	3 iterations, strength 0.1

Screenshot. Figure 16(b) is the journal screenshot “V2_Repair (Rough)”. The image shown as (c) is the journal file V3.png; the notes do not say which parameters or which solver stage produced it, so no claim is attached to it beyond the visible serrated chin contour.

Result. The spikes disappeared and intersections could be removed, so the qualitative failure of V1 was fixed. The repaired surface, however, was rough and mounded.

What worked. Patch coherence: a smooth displacement field over a connected patch removes the discontinuity that made V1 unusable, and the intersection test can be used as the accept/reject criterion for the whole patch.

What failed and why. The default patch is large. The 221 vertices of the 93 pre-cleanup forbidden faces form 16 connected clusters, and their six-ring neighbourhood covers 1,804 vertices, 11.8% of the non-boundary skin, against 728 (4.8%) for two rings (Figure 17). A displacement field pinned to a positive value on the core and to zero at the rim is a mound: it raises a footprint-sized dome over anatomy that only needed to be cleared by a small amount, and it is applied to all clusters at once. Because the repair is re-run inside every solver iteration, a mound created in one iteration can be re-fed to the smoother in the next, and a repair triggered later can overwrite regions that smoothing had just improved.

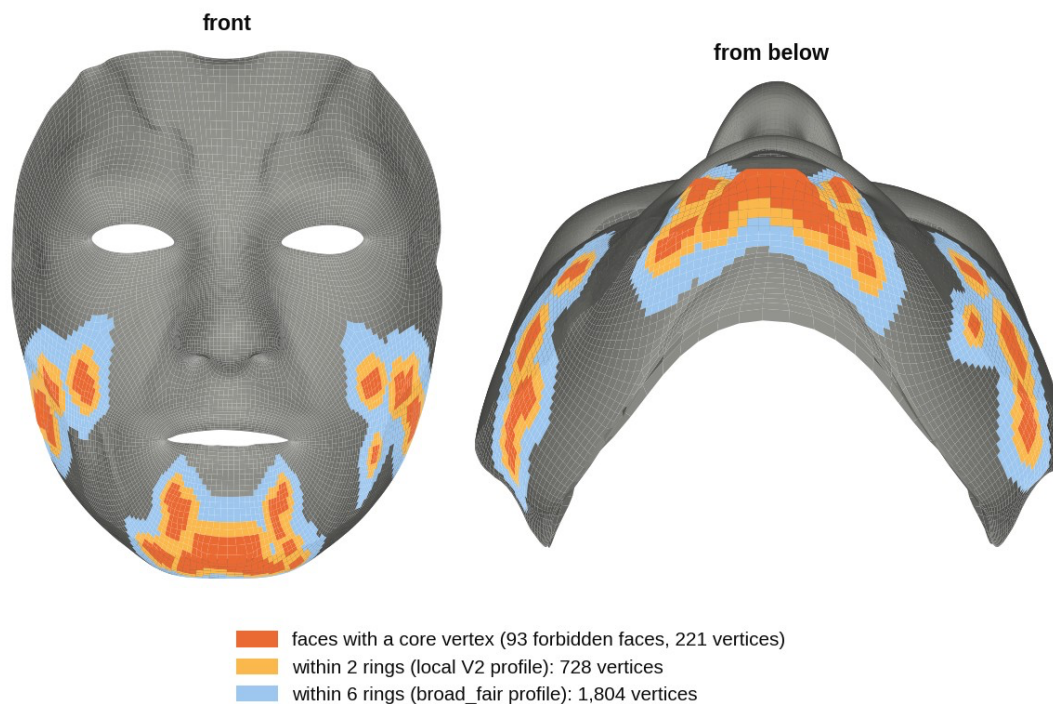


Figure 17. Repair footprint on the pre-cleanup skin. Orange: faces that contain a core vertex of a forbidden face; amber: vertices within two rings of the core (the local profile); blue: within six rings (the default broad profile), 1,804 vertices or 11.8% of the non-boundary skin. Topological growth from the offline detector output; the boundary buffer of the repair itself is not applied.

Insight. A repair must be as small as the defect. A broad harmonic patch is safe but visibly wrong, and it competes with any later smoothing of the same vertices.

Motivation for the next stage. Rather than repairing after the fact, put the safety criterion inside the smoother: only take smoothing steps that keep the surface valid (Section 8).

8. Constrained Smoothing and the Global-Step Problem

8.1 The strict constrained smoother

Objective and hypothesis. If every smoothing step is required to leave the surface intersection-free, a plain Laplacian smoother can be run inside the safe set and needs no repair at all.

Method. `smoothing_utils.constrained_smooth_mesh_region` in its strict mode. Each iteration builds a Laplacian proposal for the active region, applies the existing per-vertex clearance projection, and tests the *whole* proposal with the true detector. If the full step is unsafe, the search of Eq. (17) looks for a single scalar α that scales the displacement of *every* active vertex, by binary search over $[0,1]$ with eight steps and a floor $\alpha_{\min} = 10^{-3}$; a step is safe if it introduces no new forbidden faces and no more pairs than the baseline (function `_find_largest_safe_smoothing_step`). If no α works the iteration is recorded as stalled and nothing moves.

Logged result. In the early strict run the proposed mean displacement per iteration was about 0.701 world units, while the mean displacement actually applied was about 0.0587, that is 8.4% of the request (logged; working draft). The run repeatedly reported unsafe candidates and rejected steps, and the mesh hardly moved.

8.2 Why a theoretically safe algorithm did almost nothing

Three properties of the rule explain the result. First, *coupling*: one scalar controls all active vertices, so the safe step of the whole region is set by its tightest face. If each face k tolerates a fraction α_k of the step and safety is monotone in each, the global rule picks $\alpha^{\text{ast}} = \min_k \alpha_k$ and every other vertex moves only α^{ast} of what it could. Second, *resolution*: eight bisection steps restrict an accepted step to multiples of $2^{-8} = 1/256 \approx 0.0039$ (or the full step), so the floor $\alpha_{\min} = 10^{-3}$ can never bind; the smallest nonzero step the code can accept is 0.0039. The working draft reports accepted global steps of about 0.001; that value cannot be produced by this code with eight steps, so it is treated as unverified. Third, *stalling*: once a step has moved the surface up to the edge of contact, every further step is unsafe at every resolution and the run stops progressing.

The first and third properties can be reproduced exactly from the saved scene. Starting from the clean baseline `_solver200`, the pure Jacobi proposal of the later experiments ($\alpha = 0.35$, region and weights of Eq. (13)) was generated and passed through the strict rule with the repository detector (eight bisection steps as in the code). The proposal introduced 6 forbidden faces at full strength, formed from only 22 vertices, which is 0.24% of the 9,035 active vertices. The bisection accepted $\alpha^{\text{ast}} = 0.137$, so 86% of every vertex's requested motion was discarded because of those 6 faces (probes in Figure 18a). At the second iteration no step was safe at any of the eight resolutions, and the emulated run stalled for the remaining eight iterations. After ten iterations the rough-core mean roughness had fallen from 0.2762 to 0.2687, that is by 4.7% of the reduction (0.2762 to 0.1137) that the unconstrained run achieves.

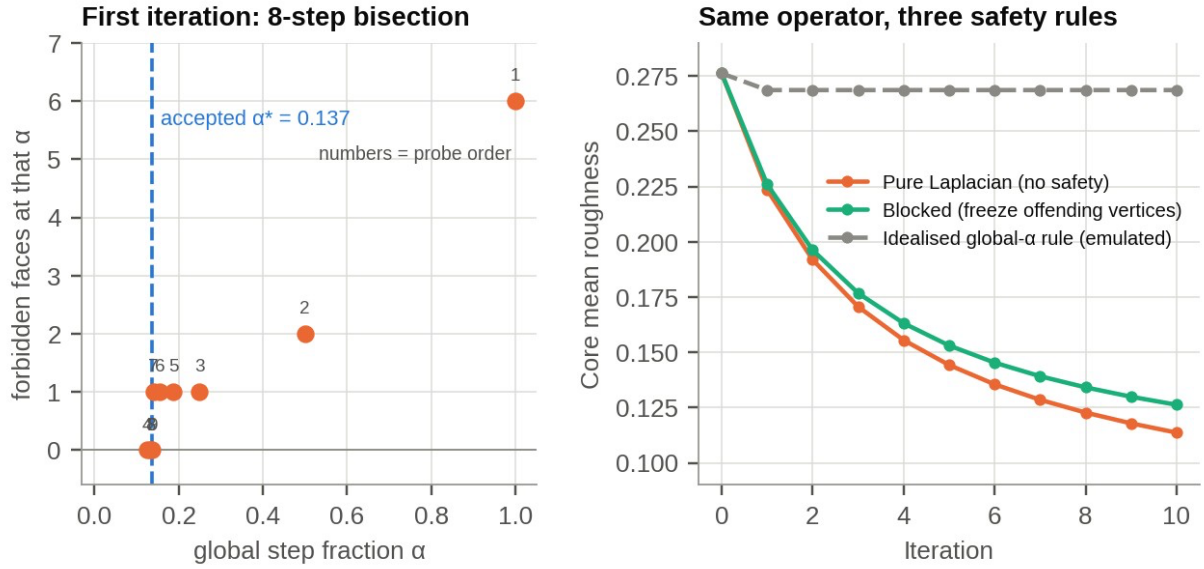


Figure 18. Left: the eight-step bisection of the global-step rule at the first iteration from `_solver200`; the probes (numbered in order) show that 6 forbidden faces at full strength force a step of 0.137 for all 9,035 active vertices. Right: core mean roughness per iteration for the same Jacobi operator under three safety rules; the idealised global-step rule stalls after one iteration. Offline emulation with the repository detector; the historical strict smoother had further terms (clearance projection) that are not emulated.

This emulation is idealised and is **not** the historical run: it uses the pure operator and the exact detector only, whereas the historical smoother also applied clearance projection. It is included because it isolates the mechanism with real data and because it makes the logged 0.701 versus 0.0587 comparison interpretable: a global scalar tied to a handful of faces reproduces the same order of loss.

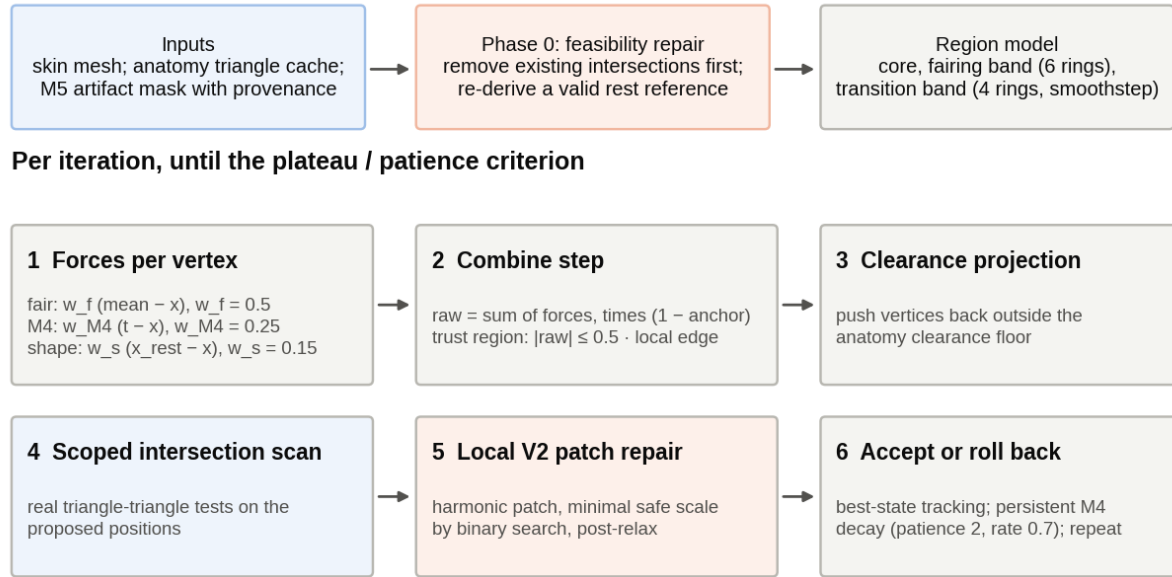
Key lesson

A single global safety scalar couples unrelated parts of the surface. A local constraint must stay local; otherwise the worst face decides how far every other vertex may move.

9. The Unified Cleanup Solver

9.1 Design

The unified solver (`final_cleanup_solver.run_cleanup_solver`) was written to remove the global-step bottleneck and to use the provenance kept by M5. It orchestrates existing, unchanged components (M5 detection, the SDF target of M4, exact clearance projection, the true intersection test and the V2 repair) and adds five things: a graded region, a per-vertex force combination, a per-vertex trust region, a feasibility phase and a set of convergence and oscillation controls. Figure 19 shows one iteration.



Logged outcome: 92 → 0 forbidden faces (Maya log). Offline re-scan of the saved states: 93 → 0. Roughness plateaus after ~200 iterations.

Failure mode: every force competes with the safety projection, so accepted steps shrink until the surface stops improving.

Figure 19. Architecture of the unified cleanup solver: a feasibility phase (Phase 0) and region model precede an iterative loop of per-vertex forces, clearance projection, scoped intersection scan, local repair and accept/roll-back. Generated for this report from the repository code.

The update of vertex i is Eq. (15): a fairing force toward the neighbour centroid, an M4 pull toward the anatomy-derived target, and a rest-shape term that anchors low-frequency form, all multiplied by $(1-a_i)$ where a_i is a smoothstep anchor that fades the update to zero across a transition band (Eq. (16)). The step of each vertex is clamped to a fraction $\rho = 0.5$ of *its own* local edge length, so one tight vertex limits only itself. M4-only vertices receive a reduced fairing scale (0.3) and overlap vertices a damped M4 pull (0.6), because both forces already act there. After the proposal the solver projects to the clearance floor, scans only the affected faces with the true detector, repairs whatever actually broke with the V2 repair, and accepts or rolls back. The best feasible state is tracked throughout.

Table 12. Main default parameters of the unified solver (`final_cleanup_solver.py`).

Parameter	Value	Role
Rings: fairing band R_f / transition R_t	6 / 4	Region grading around the M5 core.
w_f, w_{M4}, w_{sh}	0.5, 0.25, 0.15	Fairing, M4 pull, rest-shape weights.

Parameter	Value	Role
Core shape scale σ	0.3	Rest-shape weight kept in the core.
Trust ratio ρ	0.5	Per-vertex step cap (fraction of local edge).
Rest-reference low-pass	12 iterations, strength 0.3	Filters the feasible rest copy (see below).
Phase 0 passes	≤ 5	Repairs pre-existing intersections before any force is computed.
M4 decay (patience, rate)	2, 0.7	Persistent, monotone reduction of w_{M4} on plateau.
Per-vertex M4 freeze	5% of the target offset	Stops pulling a vertex already near its target.
Oscillation window / repeat / damping	5 / 3 / 0.2	Damps vertices that are repeatedly repaired.
Roughness ceiling	$1.5 \times$ original roughness	Roughness as an acceptance target.
Convergence patience; iteration cap	5; 100 (default)	Cap is a safety limit, not the definition of done.
Clearance policy; repair profile	preserve_valid_baseline; broad_fair	A vertex closer than the floor is never forced outward; V2 broad repair.

9.2 Research findings inside the solver

9.2.1 Invalid rest reference and the Phase-0 fix

The first version captured its rest position from the registered mesh *before* any repair. Where that mesh contained intersections, the rest-shape term pulled freshly repaired vertices straight back toward the position that had just been repaired away, so the loop of fair, repair, fair, repair never converged to a fair surface, only to a defended one. The logged early run illustrates the symptom: forbidden faces fell from 92 to 0 while roughness *rose* from 0.1473 to 0.2810 (Table 13). Phase 0 (function `_run_initial_feasibility`) repairs every pre-existing intersection over the whole mesh, in at most five passes, and captures the resulting valid mesh as the rest reference before any force is computed, so that “rest” and “valid” coincide. A second refinement low-pass filters only the rest copy (12 iterations, strength 0.3) so that bumps introduced by Phase-0 repair are not treated as shape to be defended; the filtered copy is never enforced as geometry.

9.2.2 The M4 decay state bug

M4 was designed as guidance that fades once its improvement plateaus. An early implementation recomputed the effective M4 weight each iteration from a counter that reset whenever M4 disagreement *changed*. When the first decay step visibly worked, the resulting rise in M4 disagreement, an expected and acceptable side effect once fairing is allowed to dominate, reset the counter and the weight snapped back to full strength; M4 decayed for one iteration and returned. The fix makes the effective weight persistent, monotone state (function `decay_m4_weight` ratchets it down by 0.7 on every second consecutive plateaued iteration and never up), and defines improvement as a signed relative decrease so that an increase caused by decay cannot reset the streak (`m4_is_improving`). A state-derived phase label (anatomy correction, balanced cleanup, fairing finish) replaced iteration-number thresholds.

Table 13. Representative development runs of the unified solver as logged in Maya (working draft). The vertex set over which roughness and M4 disagreement were averaged is not recorded, so these values are not comparable with the offline tables of Section 14.

Development run (logged)	Forbidden faces	Mean roughness	M4 disagreement	Interpretation
Initial unified solver (20 iterations)	92 – 0	0.1473 – 0.2810	7.2508 – 3.5289	Safety achieved, roughness worse; rest reference pulls toward the invalid surface.
After Phase-0 feasible reference	92 – 0	0.1469 – 0.2572	6.8828 – 2.2772	Better reference handling; still oscillatory and bumpy.

Development run (logged)	Forbidden faces	Mean roughness	M4 disagreement	Interpretation
Persistent M4 decay and fairing phase	92 – 0	0.1469 – 0.1966	6.8828 – 3.3422	Much better roughness and a safe final state; remaining bumps still visible.

9.2.3 The long-run plateau

Very long runs showed diminishing returns: a continuation of about 250 iterations took roughly 70 minutes for only minor visible change (logged; working draft). The saved states allow this to be checked offline (Table 14). Between `_solver50` and `_solver200`, 150 more iterations moved 5,381 vertices by 0.56 units on average, yet the all-vertex mean roughness improved by only 1.7% and the 99th percentile got slightly worse. The forbidden-face count was already zero at iteration 50. More iterations of the same objective were therefore not the answer.

Table 14. Solver progress from the saved scene states (offline). The rough core is defined from `_solver200`, which favours later states and penalises `_solver200` itself; the all-vertex columns carry no such selection.

Saved state	Forbidden faces / pairs	All-vertex mean	p50	p99	max	Rough-core mean
<code>_precleanupsolver</code>	93 / 497	0.1345	0.0909	0.681	2.22	0.2521
<code>_solver50</code>	0 / 0	0.1153	0.0787	0.658	5.80	0.2598
<code>_solver200</code>	0 / 0	0.1133	0.0741	0.666	4.91	0.2762

Two facts stand out. The solver did remove the safety problem: 93 forbidden faces at the start, none from iteration 50 onward. But the worst roughness *grew* from 2.22 to 4.91. That maximum, at a repair mound of the kind analysed in Section 7, is the largest sharp feature left in `_solver200` and is exactly the vertex that a pure Laplacian later flattens (maximum roughness 4.91 to 0.52, Section 11). The solver had traded intersections for a sharper worst case.

What the solver established

Intersection-free correction and broad registration cleanup can coexist, so the problem had moved from “make the mesh safe” to “make the mesh visibly fair”. It also exposed the architectural cost: every force competes with the safety projection, so accepted steps shrink until the surface stops improving.

10. Final Fairing Research

With safety solved, the last stage of the solver family was a dedicated fairing pass (`final_cleanup_solver.run_final_fairing`) that reuses the safety pipeline but replaces the force balance by a fairing operator, tested in three forms.

10.1 Taubin fairing

Objective and hypothesis. A shrinkage-resistant filter can fair the surface without pulling it into the anatomy ([23, 24]).

Method. The Taubin pair of Eq. (14) with the default $(\lambda, \mu) = (0.33, -0.34)$ (pass-band frequency $k_{PB} \approx 0.089$), fairing strength 0.5, at most 20 iterations, feasibility phase, anatomy projection, scoped scan, local repair, best-state tracking, and an aggressive preset (0.5, -0.51) for stronger runs. Figure 20 compares the transfer functions: after ten iterations the plain Laplacian at $\alpha = 0.35$ removes low graph frequencies as well, whereas the Taubin pair preserves them.

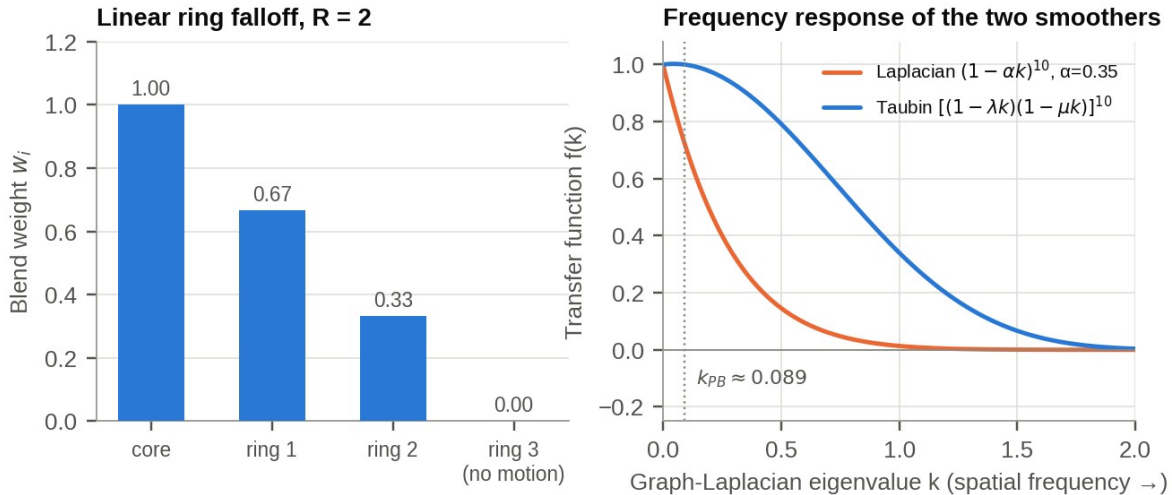


Figure 20. Left: the linear ring falloff used by the Laplacian smoothers ($R = 2$; weights 1, 2/3, 1/3, 0). Right: transfer function of ten iterations of the plain Laplacian ($\alpha = 0.35$) and of the Taubin pair ($\lambda = 0.33$, $\mu = -0.34$) over graph-Laplacian eigenvalue; the dotted line marks $k_{PB} \approx 0.089$. Computed from the two formulas in Section 4.

Result. A representative 20-iteration run reduced roughness in the fairing region from about 0.3903 to 0.2126 with no intersections before or after (logged; the working draft’s table lists 0.2246 for the core, and which value belongs to which region is unresolved). Stronger and longer runs typically reached their best state early, around iteration 13, and then entered a smooth–collide–repair cycle in which later iterations triggered repair displacements with maxima of two to five units.

What worked. It proved that fairness could improve independently of the M4 machinery, and the saved best-state node (`_fair_best17`) confirms an improvement of the tail: the all-vertex 99th percentile fell by 13.4% and the maximum by 14.8% relative to `_solver200`, with the forbidden-face count at 1 (one face, offline).

What failed and why. The gain was modest (all-vertex mean roughness 1.7% lower). Raising the global strength did not solve the ridges; it drove more vertices into the anatomy, and the broad repair then overwrote part of the progress. The limiting factor was not the filter but the surrounding stack: the safety projection, the shape-restraint force and the repair patch all acted against the fairing step.

10.2 High-tail roughness analysis

Mean roughness alone hid the interesting behaviour, so the percentiles of $\|L_i\|$ were tracked (Eq. (19)). One stronger global fairing run gave (logged) a mean change of -6.78% and changes of -14.89% , -17.01% , -8.98% and -8.71% at p_{90} , p_{95} , p_{99} and the maximum, while the median p_{50} became 10.57% worse (mean $0.09866 - 0.09197$). Selecting the 100 vertices of highest roughness in Maya showed that they coincided with the visually objectionable bumps; this observation is the author's (there is no screenshot of it in the journal), but it is the reason the percentile metric is trusted as a proxy for visible defects in the rest of this report. The offline saved states show the same pattern with less magnitude (Figure 21). Uniform fairing improves the worst regions while slightly degrading regions that were already smooth, since every vertex is smoothed by the same amount.

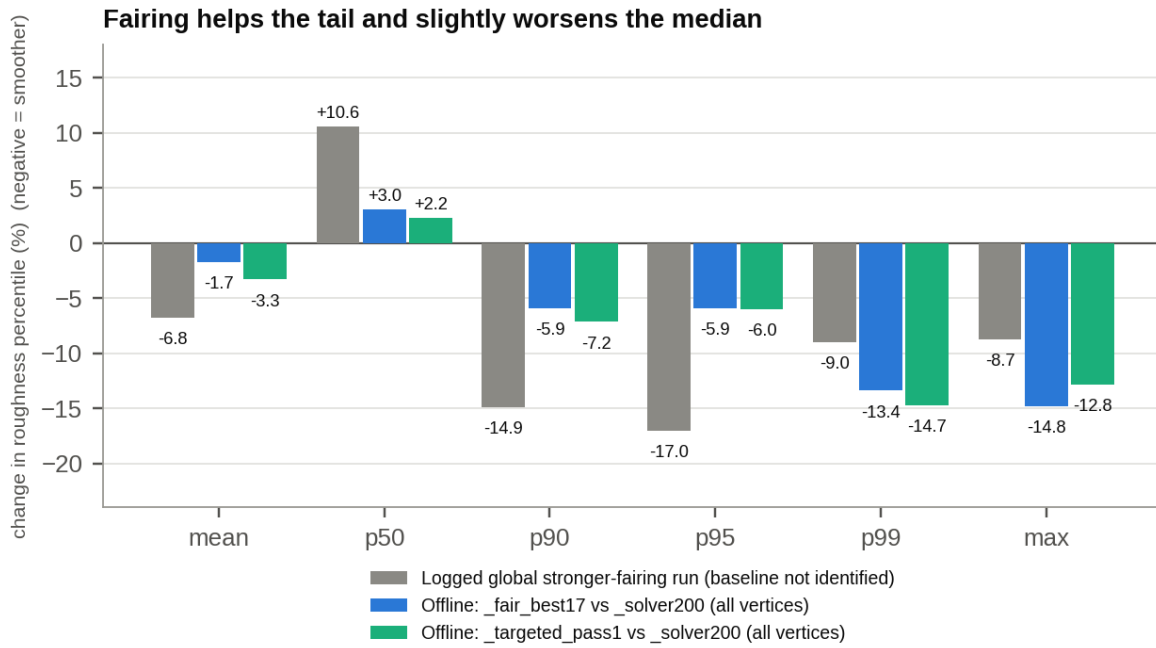


Figure 21. Change in roughness percentiles relative to the baseline: the logged stronger global fairing run (baseline not identified) and, from the saved scene, the Taubin best state and the roughness-targeted pass relative to `_solver200` over all non-boundary vertices. The tail improves, the median gets slightly worse.

10.3 Roughness-targeted fairing

Uniform weights spread fairing over already-smooth skin while the ridges barely moved. The targeted fairer therefore weights each vertex by its own roughness: a piecewise-linear map through percentile knots ($r_{\min} - 0.75$, $p_{75} - 1.0$, $p_{90} - 1.25$, $p_{95} - 1.5$, $p_{99} - 1.75$, $r_{\max} - 2.0$), diffused over three one-ring averaging passes so that a ridge is a coherent patch and not an isolated spike (`build_roughness_fairing_weights`). The saved state `_targeted_pass1` moved the right regions in the right direction: relative to `_solver200` the all-vertex mean fell by 3.3% , the 99th percentile by 14.7% and the maximum by 12.8% , with zero forbidden faces. The visual change nonetheless remained modest because the broad repair and the shape-restraint force were still active.

10.4 What the fairing research showed

Each additional layer produced a smaller visible gain than the one before, while the system grew more complex. This is the signature of a limit set by the architecture, not by the operator, and it raises the question

that the next experiment isolates: *if nothing resisted the smoother, could the visible bumps actually be removed?*

11. The Decisive Pure-Laplacian Experiment

The fairing research of Section 10 ended with a puzzle. Every added layer of machinery produced a smaller visible gain, yet the operator at the centre of it (a Laplacian step) is among the simplest in geometry processing. Two explanations were possible: either the bumps are intrinsically hard to smooth, or the safety architecture around the smoother prevents it from doing its job. They predict different outcomes for a smoother that has nothing around it, so the experiment was designed to remove everything else.

11.1 Objective, hypothesis and method

Objective. Determine whether the remaining bumps are difficult to smooth, or whether the safety architecture is what prevents smoothing.

Hypothesis. If the architecture is the limiting factor, a plain Laplacian smoother acting on the same region will flatten the bumps quickly, at the price of pushing the skin into anatomy. If the bumps are intrinsically hard to smooth, even a plain smoother will leave them.

Method. A separate module, `pure_laplacian_smoothing.run_pure_laplacian_smoothing`, was written with none of the existing solver machinery: no M4 pull, no anatomy constraint, no repair, no rest reference, no trust region, no rollback, no shape force and no safety check of any kind. It is explicitly labelled *experimental, no anatomy safety* in its own log banner. It selects the rough core as the vertices at or above a roughness percentile, grows it by a number of rings with linear falloff, and applies the Jacobi step of Eq. (13) for a fixed number of iterations, protecting boundary vertices. The experiment was deliberately allowed to expose anatomy: an unsafe result was an acceptable outcome because the purpose was to learn what the operator can do.

Table 15. Parameters of the pure-Laplacian experiment (module defaults differ: percentile 75, growth 3, strength 0.5, 20 iterations).

Parameter	Value used	Comment
Start mesh	_solver200	The clean baseline: zero forbidden faces (offline), all earlier safety work already done.
Roughness percentile	75	Selects the rough core: 3,564 non-boundary vertices.
Growth rings	2	Linear falloff $w = 1 - \rho/(R+1)$ with $R = 2$; 9,035 active vertices including falloff.
Strength α	0.35	Direct multiplier on the Jacobi step; not damped or clamped.
Iterations	10	The run took 0.38 s in Maya (logged).
Boundary	protected	Eye, mouth, nostril and cap-edge vertices are not moved.
Safety checks	none	The module's own log ends with: no intersection or clearance check was performed.

11.2 Logged result

The Maya Script Editor printed the run summary, reproduced in Figure 22. The numbers are the values in Table 16; the offline reproduction of the same run from the saved `_solver200` state matches every printed statistic (mean $0.27625 - 0.11366$).

```

PURE LAPLACIAN SMOOTHING -- EXPERIMENTAL, NO ANATOMY SAFETY
=====
selected: 3564 (auto_percentile)  active incl. falloff: 9035  boundary-protec
strength=0.35  growth_rings=2  iterations=10
roughness (mean): 0.27625 -> 0.11366
  before p50=0.20467 p90=0.44982 p95=0.59747 p99=1.35007 max=4.90895
  after  p50=0.10460 p90=0.18676 p95=0.21509 p99=0.29739 max=0.52002
displacement from STARTING mesh: mean=0.31878 max=5.82042 (vertex 12017)
NOTE: no intersection/clearance check was performed -- this run may have pushed
runtime: 0.38s
=====

```

Figure 22. Journal screenshot of the Maya Script Editor after the pure-Laplacian run: selected 3,564 vertices, 9,035 active, strength 0.35, 2 growth rings, 10 iterations; mean roughness 0.27625 – 0.11366; displacement from the starting mesh mean 0.31878, maximum 5.82042 at vertex 12017; runtime 0.38 s. The last line warns that no intersection or clearance check was performed. The image is cropped from the working draft (image 12).

Table 16. Logged roughness statistics of the pure-Laplacian run (world units). The table reproduces the values printed in the Script Editor; the offline reproduction agrees to the printed precision.

Roughness $\ L_i\ $ over the rough core (n = 3,564)	Before (<code>_solver200</code>)	After (pure $\alpha = 0.35 \times 10$)	Change
Mean	0.27625	0.11366	-58.9%
Median (p50)	0.20467	0.10460	-48.9%
90th percentile	0.44982	0.18676	-58.5%
95th percentile	0.59747	0.21509	-64.0%
99th percentile	1.35007	0.29739	-78.0%
Maximum	4.90895	0.52002	-89.4%

Every statistic fell, and the tail fell most: the maximum roughness dropped from 4.9 to 0.52 and the 99th percentile from 1.35 to 0.30. The median was roughly halved. The displacement needed to do this was modest for most vertices (mean 0.319, 95th percentile 0.84 world units over the active set, Figure 32) and large only for a few (maximum 5.82 at vertex 12017, the same vertex whose roughness of 4.91 is the maximum in `_solver200`).

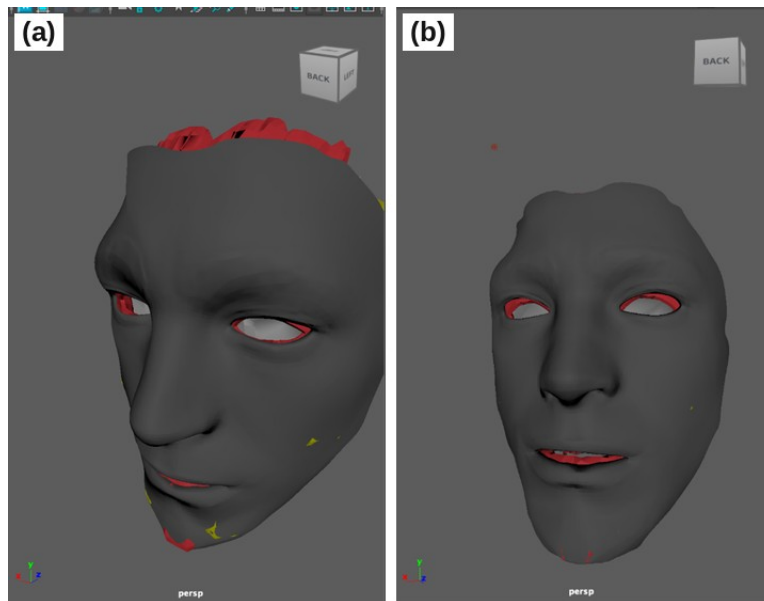


Figure 23. Journal screenshots. (a) Viewport after the logged pure-Laplacian run, seen from behind and the left: the surface is smooth, the open cap edge and the eye and lip openings show red anatomy (valid exposure), and small yellow

marks at the jaw and lower cheek are fat showing through closed skin, which is the unwanted case. (b) The scene with the node `_pre_blocked_laplacian` selected in the Outliner; this is the state that the blocked-Laplacian module started from (Section 12). The two images use different camera positions and different strengths and are not a matched before/after pair.

11.3 What the run showed, and what it did not show

Visual result. The bumps were, for the first time in the research, clearly gone. The eye-catching ridges along the cheeks and chin that the fairing experiments left in place flattened after ten cheap iterations (Figure 24). The viewport screenshot also shows the cost: yellow fat and red muscle became visible through skin that should be closed.

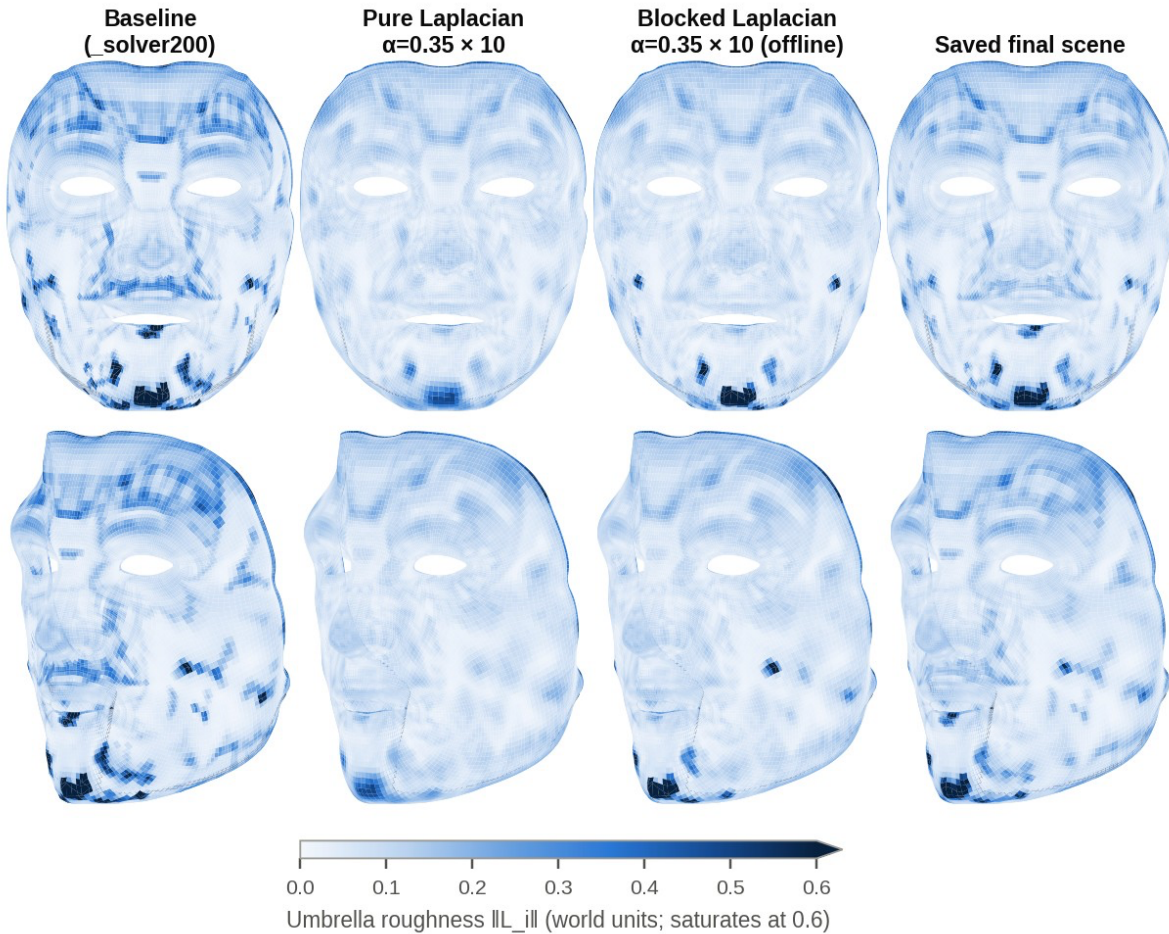


Figure 24. Umbrella roughness $\|L_i\|$ drawn on the skin (front view above, three-quarter view below) for the clean baseline `_solver200`, the pure Laplacian run ($\alpha = 0.35$, 10 iterations), the blocked Laplacian (offline reproduction, Section 12) and the saved final scene. Colour saturates at 0.6, so the darkest chin marks in the baseline are much rougher than they look. The pure run flattens nearly everything but also creates the anatomy exposure of Table 17; the blocked run keeps most of the flattening; the final scene keeps a few dark marks at the chin that are discussed in Section 13.

Price. The offline detector finds 143 forbidden faces (452 crossing pairs) after the run, against zero before it: 277 pairs with fat meshes, 85 with muscle meshes and 90 with the jaw bone. Unsigned clearance to anatomy fell below 0.5 world units for 186 of the 9,035 active vertices, against 1 in `_solver200` (Table 17).

Table 17. Smoothness gained and anatomy exposed by the pure-Laplacian run (offline detector and clearance, except the logged displacement).

Quantity	_solver200	Pure $\alpha = 0.35 \times 10$
Rough-core mean roughness	0.2762	0.1137
All-vertex mean roughness (n = 15,316)	0.1133	0.0822
Forbidden skin faces (crossing pairs)	0 (0)	143 (452)
Active vertices under 0.5 units of anatomy	1	186
Active vertices under 1.0 unit of anatomy	537	657
Minimum clearance (world units)	0.445	0.0013
Mean / maximum displacement from _solver200 (active set)	—	0.319 / 5.82

Insight. The surface was easily smoothable. The bumps are not intrinsically hard to smooth: a plain operator, applied for less than half a second, removes most of them. What the previous stages had been doing, for weeks, was suppressing or undoing that useful smoothing with competing forces (the M4 pull, the rest-shape term, the repair mounds) and with a global safety rule. The experiment therefore isolated the root cause, and it changed the question from *how do we smooth this surface?* to *where exactly does smoothing have to stop?*

What it does not show. It does not show that unconstrained smoothing is acceptable: 143 forbidden faces are a real defect, and the smoother also shrinks features slightly, as every Laplacian smoother does ([8, 24]). Its 0.38 s runtime and its result are a diagnostic, not a deliverable. It also does not identify *which* bumps are legitimate: the roughest vertices are the repair mounds left by the earlier solver, and flattening them is what re-creates the intersections that the mounds had been placed to avoid (Section 12 returns to this).

Motivation for the next stage. If the operator is right and only a few local regions conflict with anatomy, the constraint should act only there. The next module keeps the operator of this experiment unchanged and adds the smallest mechanism that stops a region once it really crosses into anatomy (Section 12).

Central finding

Pure Laplacian smoothing proved that the surface could be made smooth immediately. The remaining problem was not smoothing but where to stop smoothing, and that is a *local* question.

12. The Blocked-Laplacian Strategy

12.1 Design

Objective and hypothesis. Keep the pure Laplacian exactly as it is, and add the smallest mechanism that prevents the smoothing from creating a *real* anatomy crossing. If constraints are localised to the vertices that actually cause a crossing, most of the smoothing effect of Section 11 should survive with no forbidden faces.

Method. `blocked_laplacian_smoothing.run_blocked_laplacian_smoothing` imports the Jacobi step, region selection and falloff from the pure module, so the operator is identical, not a re-implementation. Each iteration proposes a Jacobi step for all movable active vertices, scans the *proposed* positions with the true triangle–triangle detector, and if any forbidden face appears it (1) restores the vertices of the offending faces and, with the default `freeze_rings = 1`, their one-ring neighbours to their last safe positions, (2) marks them frozen for the rest of the run, and (3) re-scans. If a forbidden face remains, a second tier rolls back one further ring and re-scans; if that also fails the proposal is discarded and the run stops with the last safe state. Nothing else resists the smoothing: no repair, no anatomy attraction, no clearance floor, no global step reduction. Frozen vertices stay in the neighbour averages of the vertices around them, so the free vertices treat them as fixed anchors, which is the formal statement of Eq. (18).

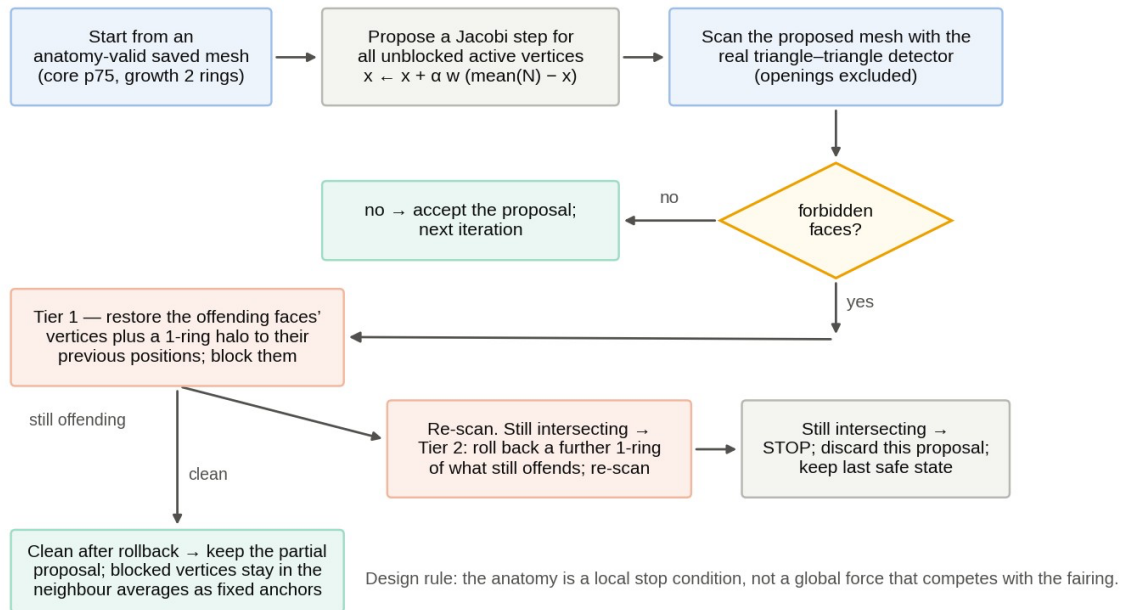


Figure 25. Flow of one blocked-Laplacian iteration: a Jacobi proposal is scanned with the true detector; if forbidden faces appear, the offending vertices plus a one-ring halo are restored and frozen (tier 1), a further ring is rolled back if the re-scan still fails (tier 2), and only if that fails the proposal is discarded. Generated for this report from the repository code.

Table 18. Parameters of the blocked-Laplacian module (`blocked_laplacian_smoothing.py`).

Parameter	Default	Meaning
<code>roughness_percentile / growth_rings</code>	75 / 2	Same region as the pure module (3,564 core, 9,035 active).
<code>strength α</code>	0.35	Direct multiplier for free vertices; frozen vertices have an effective step of exactly 0.
<code>iterations</code>	10	A cap; the run may stop earlier if a proposal cannot be made safe.

Parameter	Default	Meaning
freeze_rings	1	Offending vertices plus this many rings are frozen; 0 freezes exactly the offending vertices.
boundary_buffer_rings; tolerance	1; 10^{-6}	Passed unchanged to the detector; eye, mouth and cap openings are excluded as everywhere else.
Precondition	valid start	The start mesh must contain no forbidden face in the active region; the module does not repair a baseline (it prints a warning by design).

12.2 Why one freeze ring

Freezing exactly the offending vertices (*freeze_rings* = 0) leaves them as isolated fixed points inside an otherwise moving neighbourhood: their neighbours keep moving toward the average and can create a fresh crossing next to the frozen vertex, which then has to be frozen in the next iteration. Freezing one ring around them (*freeze_rings* = 1) gives the fixed set a soft margin. It does not become another broad repair patch because a frozen vertex is never moved: the halo is a set of vertices that *stop* smoothing at their last safe position, and the footprint stays small (342 vertices, 3.8% of the active region, after ten iterations, against 11.8% of the non-boundary skin that the six-ring repair of Section 7 would have displaced for the same 93 faces). The three settings were compared offline with the repository module and the same start mesh (Table 19).

Table 19. Effect of the freeze halo on the blocked Laplacian (offline reproduction with the repository module, start *_solver200*, $\alpha = 0.35 \times 10$). All three settings finish with zero forbidden faces in the whole mesh. Baseline core mean 0.2762.

freeze_rings	Vertices frozen	Share of active	Offending faces per iteration (first four)	Core mean	Core p99	All-vertex p99	Forbidden faces
0	148	1.6%	6 9 6 8 ...	0.1244	0.595	0.250	0
1 (default)	342	3.8%	6 5 5 6 ...	0.1263	0.536	0.262	0
2	507	5.6%	6 5 4 1 ...	0.1274	0.537	0.263	0

The differences in roughness are small (core mean 0.1244, 0.1263 and 0.1274) and no setting is uniformly best: zero rings freezes the fewest vertices and gives the best all-vertex tail, whereas one ring leaves fewer offending faces per iteration to roll back (five instead of nine in iteration 2) and a better core tail. The metrics here do not measure seams, which is the motivation of the module’s author for the default; the table shows that the choice is not costly. Following the design rationale, one ring is the default used in this report.

12.3 Result

Starting from *_solver200* with module defaults, the run completed all ten iterations with zero forbidden faces at every iteration after rollback (Figure 26). It proposed 6, 5, 5, 6, 3, 3, 4 and 3 offending faces in iterations 1–8 and none in iteration 9, and each time the rollback cleared them; 342 vertices (3.8% of the active region) were frozen by the end. The rough-core mean roughness fell from 0.2762 to 0.1263, which is 92% of the reduction that the unsafe pure run achieved (0.1137). Table 20 sets the three rules that share the same operator side by side.

Table 20. Three ways of combining one smoothing operator with the same anatomy safety requirement. The idealised global-step rule stalls after one iteration; the blocked rule keeps 92% of the pure reduction and ends without forbidden faces.

Rule (same Jacobi operator, $\alpha = 0.35$, 10 iterations, start_solver200)	Forbidden faces	Core mean	Share of the pure reduction	Core p99	Core max
No safety (pure Laplacian, Section 11)	143	0.1137	100%	0.297	0.52
Global step bisection (idealised, Section 8)	0	0.2687	4.7%	—	—
Blocked Laplacian (freeze on real intersection)	0	0.1263	92%	0.536	4.75

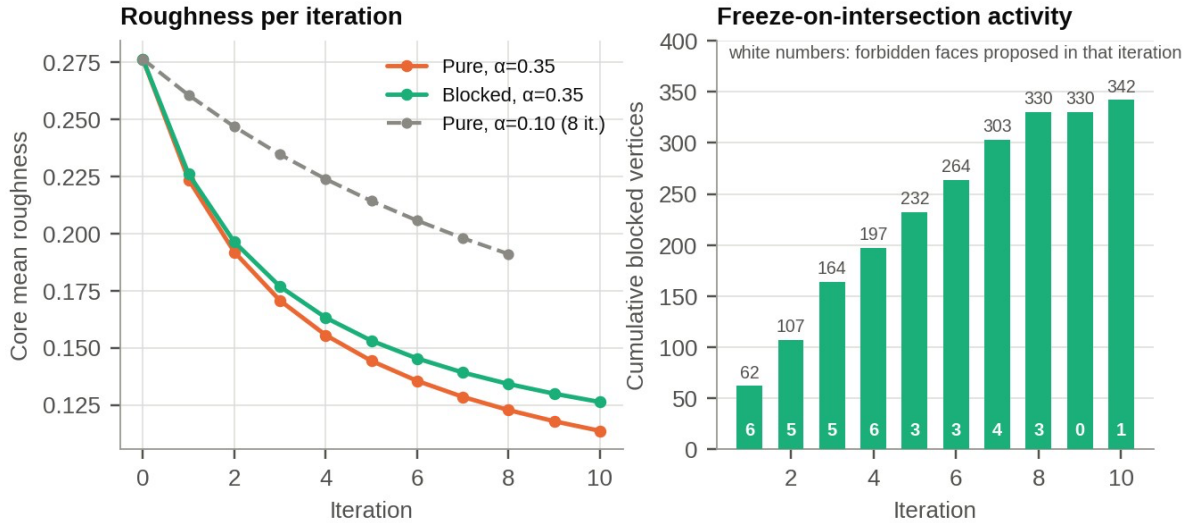


Figure 26. Left: rough-core mean roughness per iteration for the pure Laplacian ($\alpha = 0.35$), the blocked Laplacian ($\alpha = 0.35$, offline) and the pure $\alpha = 0.10 \times 8$ run whose result is the saved pre-manual state. Right: cumulative number of frozen vertices in the blocked run; the white numbers give the forbidden faces proposed in each iteration and cleared by rollback.

The blocked curve follows the pure curve closely for two iterations and then separates slowly, since frozen vertices are anchors that the free vertices average toward. The freeze activity shrinks with iteration: 62 vertices are frozen at the first iteration and 12 at the last, and no proposal in iteration 9 needs rollback.

12.4 What the blocked strategy does not do

The tail differs from the pure result: the core maximum of the blocked run is 4.75 against 0.52 for the pure run, and the 99th percentile is 0.536 against 0.297. The sharpest bumps of `_solver200` are the vertices 12017, 15012 and 1924 (roughness 4.91, 4.75 and 3.33), and they are also the places where flattening pushes the skin into anatomy. In the blocked run vertices 15012 and 1924 are frozen at their initial positions from the first iteration onward and vertex 12017 stops after a displacement of 2.8 units. These are the same mounds that the broad repair of Section 7 created in order to clear the anatomy: they are sharp because they are *where* geometry and anatomy conflict, so a smooth surface is not available there without moving the anatomy or accepting a broader bulge. That is a limit of the constraint, not of the freezing strategy, and it is the residual that Section 13 addresses by hand.

Runtime. The blocked module needs a detector scan of every proposal, which makes it roughly fifty times slower than the pure run (21.5 s offline for ten iterations against 0.38 s logged in Maya for the pure run; the two timings come from different hardware and are indicative only).

12.5 A valid start is a precondition

The design assumes a valid start. The saved state that the blocked module actually began from in Maya, `_pre_blocked_laplacian`, is not valid: it has 21 forbidden faces (94 pairs) because it is geometrically a pure Laplacian with $\alpha = 0.10$ and eight iterations applied to `_solver200` (Appendix C). Running the repository module from that state offline stops in the first iteration, because the rollback and the one-ring fallback cannot clear the 21 pre-existing forbidden faces, so it moves no vertex. The module prints a warning in this situation and, by design, does not repair the baseline.

12.6 Did the saved final scene use this output?

No. The evidence does not support saying that the saved final mesh is the output of the blocked run above. The saved final scene differs from `_pre_blocked_laplacian` in only 130 vertices, whereas a completed blocked run moves several thousand (9,035 active vertices, mean displacement 0.30, Table 29); if a blocked run had been kept, the live skin would differ from its own pre-run backup in the thousands, not in 130. The saved final is therefore best described as the pure $\alpha = 0.10 \times 8$ state plus a documented manual finish (Section 13). The blocked Laplacian is presented here as the design that the research arrived at and as an offline reproduction with the repository module, not as the origin of the final geometry. The journal contains no screenshot of a blocked run's *result*; Figure 23(b) shows only its start state.

12.7 The micro-clearance push

A third module, `micro_clearance_push.run_micro_clearance_push`, was written after the blocked module for exactly the situation that remained: a finished, smooth mesh with a very small number of closed-skin regions where anatomy is still visible. It detects forbidden faces with the same detector, pushes only their vertices outward to a small clearance (default 0.15 world units) with the existing closest-point machinery, blends the correction one ring outward at half weight, re-scans once, and stops. A backup node with its default suffix (`_pre_micro_clearance_push`) exists in the scene and is geometrically identical to `_pre_blocked_laplacian`, and the first stage of the manual finish has the signature of an outward push (Section 13); the notes do not record whether the module was used for it, so the module is documented but not credited with any result.

13. Final Manual Finishing

This stage is part of the result and is reported as such. After the automated pipeline had removed the large-scale problem, a small residual set of vertices remained. The author's account is that the finish consisted of selecting specific skin vertices, pushing them slightly outward where fat or muscle was still visible through the skin, and locally smoothing selected vertices where needed. The saved scene lets that account be checked against geometry.

13.1 What the saved geometry shows

Three saved states bracket the finish: `_pre_blocked_laplacian` (the pure $\alpha = 0.10 \times 8$ state, 21 forbidden faces), the scene node `skin_cloth_copy_v5_pre_final` (a saved intermediate whose place in the sequence is established by geometry, Appendix C) and the live skin. Between the first and the last, exactly 130 vertices differ (mean displacement 0.099, 95th percentile 0.48, maximum 0.68 at vertex 1924; Figure 27). They fall into two stages with different characters (Table 21).

Table 21. The two stages of the manual finish, recovered from the saved scene states (offline). Roughness is the umbrella-Laplacian norm at the moved vertices only.

	Stage 1: outward push	Stage 2: local smoothing
Saved states	<code>_pre_blocked_laplacian</code> – <code>pre_final</code>	<code>pre_final</code> – live skin
Vertices moved	46	87
Mean / 95th percentile / maximum displacement	0.254 / 0.51 / 0.68	0.021 / 0.085 / 0.147
Vertices moved outward (positive along the normal)	100%	11%
Share of displacement along the surface normal	99%	45%
Vertices moved toward their neighbour centroid	0%	100%
Roughness at these vertices, before – after	0.547 – 0.632	0.0839 – 0.0768
Forbidden faces, before – after	21 – 3	3 – 3

Stage 1. 46 vertices move by a mean of 0.25 units, all outward, with 99% of the displacement along the surface normal; the forbidden-face count falls from 21 to 3. This is the signature of the hand or module push that the author describes: a small, targeted outward correction at exactly the places where anatomy was still visible. It has a cost, visible in the table: the pushed vertices become locally *rougher* (0.547 – 0.632), because an isolated outward step is the operation that Section 7 showed to create local bumps, here kept to a few tenths of a unit.

Stage 2. 87 vertices move by a mean of only 0.021 units, all toward the centroid of their neighbours, and their roughness falls from 0.0839 to 0.0768. This is a local smoothing pass over the vertices around the stage-1 edits. It does not change the number of forbidden faces (3 before and after).

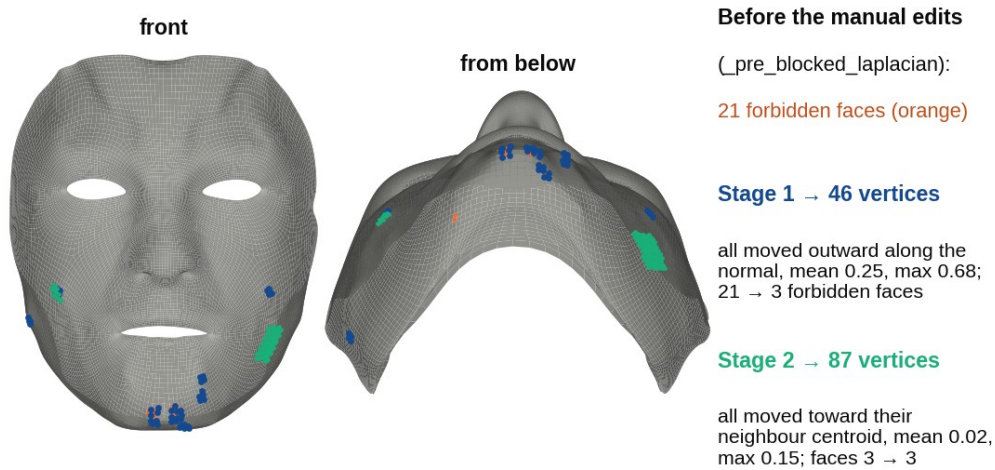


Figure 27. Manual finishing recovered from the saved scene states. Blue: the 46 vertices of stage 1 (outward pushes); green: the 87 vertices of stage 2 (local smoothing), drawn on the pre-manual skin (front and from below); orange: the 21 forbidden faces before the finish. Stage 1 reduces the forbidden faces from 21 to 3. The edits sit at the chin and along the jaw line, where the forbidden faces are.

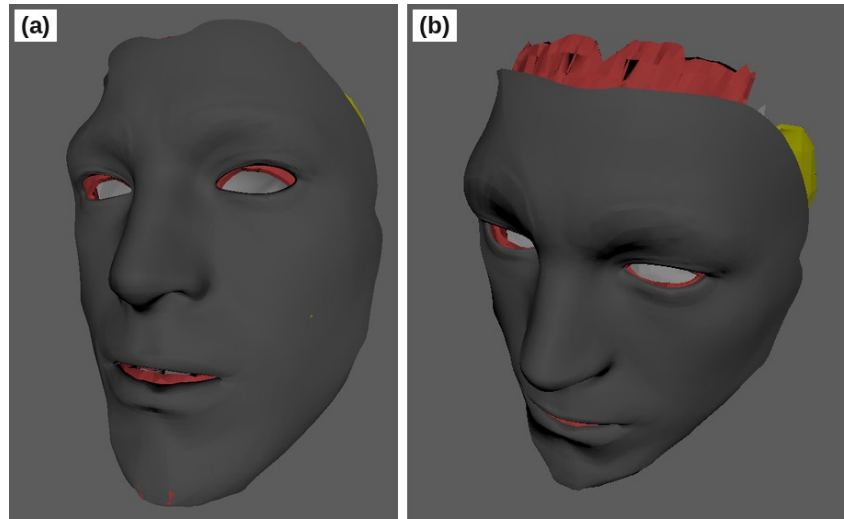


Figure 28. Journal screenshots of the skin (a) before the manual finishing (Pre_Manual_Cleanup) and (b) in the saved final state (Final). The cap edge at the top, and the eye and lip openings, show red muscle by design; the face is smooth at this scale in both images. The two views use different cameras, so they cannot show the effect of 130 sub-unit vertex edits; that effect is measured in Table 27 and the roughness maps of Figure 24.

13.2 Why this was an appropriate stopping point

Three reasons support finishing by hand. First, the residual was tiny: 21 forbidden faces in 8 small clusters (5.4% of the non-boundary skin within six rings, against 11.8% for the original 93 faces, Figure 17), and about 130 vertices touched. Second, every global attempt so far had spread its side effects over good areas, and a further automated pass would have had to be designed for a problem of that size. Third, the goal of the project was a usable final mesh; automation for its own sake was not the objective. These are judgements, and the report treats them as such rather than as a proof.

13.3 Status of the result

Table 22. Which parts of the saved final mesh are automated and which are not.

Component of the saved final mesh	Automated?	Evidence
Detection of the artifact regions (M1–M5)	Yes	Repository code; screenshots of each milestone.
Safety metric (true intersection detector)	Yes	Repository code; offline and Maya logs.
Removal of the pre-cleanup intersections (unified solver, 200 iterations)	Yes	Saved state _solver200; Maya logs.
Smoothing of the rough region (pure Laplacian, $\alpha = 0.10 \times 8$)	Yes	Saved state _pre_blocked_laplacian, reproduced offline.
Outward push of 46 vertices (stage 1)	Manual or module (unconfirmed)	Saved states; author's description.
Local smoothing of 87 vertices (stage 2)	Manual	Saved states; author's description.
Blocked Laplacian	Implemented; not part of the saved final	Offline reproduction; Section 12.

Limitation stated plainly

The saved final mesh is a *hybrid* result: automated detection, correction and smoothing followed by a documented manual pass of about 130 vertices. It still contains 3 forbidden faces by the offline detector (Table in Section 14). It must not be described as the output of a fully automatic pipeline.

14. Quantitative Evaluation

The supervisor asked whether the mesh had been measured before and after the algorithm with common mesh-quality metrics. This section does that. The topology of the skin never changes (15,905 vertices, 31,543 edges, 15,636 faces in every saved state), so every metric below responds to vertex-position changes only, and states can be compared vertex by vertex. All numbers are recomputed offline from the saved Maya scene t34.mb by the methods of Section 3.3; the run-time values quoted in earlier sections are labelled as logged.

14.1 Baselines and metrics

Two baselines are used. The *pre-cleanup skin* (`_precleanupsolver`) is the registered mesh at the start of the cleanup research and is the honest "before". The *clean baseline* `_solver200` is the mesh after the unified solver, from which all later smoothing started; comparing against it isolates the contribution of the smoothing and finishing stages. The *exact final mesh* is the live skin of the saved scene. Metrics were chosen because they respond to what was done to the mesh; common metrics that would not were left out and are listed at the end of the section.

Table 23. Metrics used in the evaluation and what each means for this problem.

Metric	What it measures for this problem	Reference
Umbrella-Laplacian roughness $\ L_i\$ (mean, p50, p90, p95, p99, max)	Local deviation of each vertex from the centroid of its one-ring. Large at bumps and ridges; the quantity the smoothers minimise. Reported for all 15,316 non-boundary vertices and for the rough core.	Eq. Eq. (3); [8, 24]
Forbidden skin faces and crossing pairs, by anatomy mesh	Anatomy validity: skin triangles exactly crossed by anatomy triangles, excluding legitimate openings.	Eq. (12); [18, 26]
Displacement from a baseline (mean, RMS, p95, max, vertex of maximum)	How far the smoothing moved the vertices, in world units.	—
Point-to-surface distance and Hausdorff distance from the baseline	How far the <i>surface</i> moved, which is smaller than the vertex displacement when vertices slide within the surface.	[7]
Signed normal displacement	Whether smoothing shrinks the surface (inward) or inflates it; inward is toward the anatomy.	[24, 8]
Face interior angles (minimum, maximum) and edge-length ratio per face	Element quality of the quad-dominant mesh: sliver-like faces and stretched faces.	[14, 5]
Edge-length coefficient of variation	Uniformity of the sampling; only changes if vertices slide along the surface.	[5]
Adjacent-face normal angle (dihedral variation)	Faceting: how sharply neighbouring faces turn relative to each other.	[5]
Angle defect (discrete Gaussian curvature) and umbrella mean-curvature magnitude	Discrete curvature distribution at interior vertices; tests whether smoothing removes curvature outliers rather than shape.	[16]
Unsigned clearance to the nearest anatomy triangle	Secondary risk indicator only. It does not detect crossings (Section 6).	[12]

14.2 Headline before/after

Table 24. Quantitative before/after summary (offline detector and metrics; world units). Roughness is the umbrella-Laplacian norm. The rough core is defined from `_solver200` (Section 3.5), which is by construction roughest on its own core; the all-vertex rows carry no such selection. The forbidden-face row counts skin faces (crossing pairs in the next row).

Metric	Pre-cleanup	<code>_solver200</code> (clean baseline)	Saved final	Change vs pre-cleanup	Change vs <code>_solver200</code>
Forbidden skin faces	93	0	3	-97%	+3 faces
Crossing pairs	497	0	15	-97%	+15 pairs
Roughness, all vertices, mean	0.1345	0.1133	0.0965	-28%	-15%
Roughness, all vertices, p50	0.0909	0.0741	0.0767	-16%	+3%
Roughness, all vertices, p90	0.2922	0.2223	0.1751	-40%	-21%
Roughness, all vertices, p95	0.4052	0.3229	0.2313	-43%	-28%
Roughness, all vertices, p99	0.6815	0.6658	0.4021	-41%	-40%
Roughness, all vertices, maximum	2.2235	4.9089	2.2614	+2%	-54%
Roughness, rough core (n = 3,564), mean	0.2521	0.2762	0.1915	-24%	-31%
Roughness, rough core (n = 3,564), p95	0.5885	0.5975	0.3693	-37%	-38%
Roughness, rough core (n = 3,564), p99	1.0897	1.3501	0.8058	-26%	-40%
Roughness, rough core (n = 3,564), maximum	2.2235	4.9089	2.2614	+2%	-54%

Read against the pre-cleanup skin, the final mesh has 90 fewer forbidden faces ($93 - 3$), a 28% lower all-vertex mean roughness, a 43% lower 95th percentile and a 41% lower 99th percentile. Read against the clean baseline, the smoothing and finishing stages lowered mean roughness by 15% and the 99th percentile by 40% without re-introducing more than three forbidden faces. Two entries need candour. The maximum roughness (2.26) is close to the pre-cleanup maximum (2.22), but it belongs to a different vertex: the pre-cleanup maximum was at vertex 6262, whereas the final maximum is the repair mound at vertex 12017, whose roughness was 0.39 before the cleanup, 4.91 after the unified solver and 2.26 in the final mesh. And the 3 residual forbidden faces are not zero.

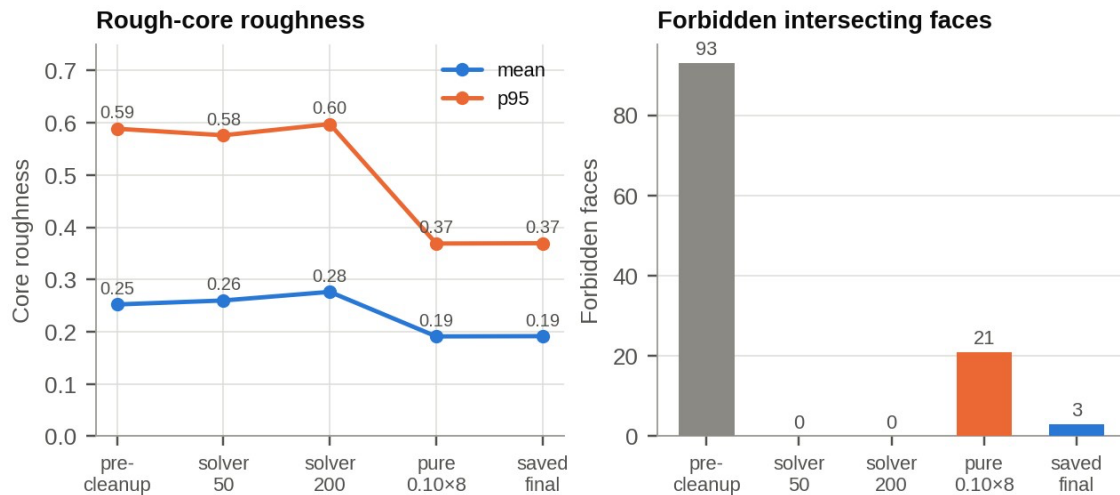


Figure 29. Progression across the saved states. Left: rough-core mean and 95th-percentile roughness. Right: forbidden skin faces. The unified solver (`_solver50`, `_solver200`) removes all forbidden faces without lowering the core roughness; the pure $\alpha = 0.10 \times 8$ smoothing lowers it; the saved final scene keeps that roughness with 3 forbidden faces.

14.3 Anatomy validity

Table 25. Every saved or reproduced state, in order of the research. The two offline rows were generated from `_solver200` with the repository modules (Maya I/O replaced by in-memory stand-ins). Node names carry the prefix `skin_cloth_copy_v5_pull_back`.

State	Scene node	Forbidden faces / pairs	All-vertex mean	All-vertex p99	All-vertex max	Core mean
Pre-cleanup	<code>_precleanupsolver</code>	93 / 497	0.1345	0.681	2.22	0.2521
Unified solver, 50 it.	<code>_solver50</code>	0 / 0	0.1153	0.658	5.80	0.2598
Unified solver, 200 it.	<code>_solver200</code>	0 / 0	0.1133	0.666	4.91	0.2762
Taubin fairing best state	<code>_fair_best17</code>	1 / 12	0.1113	0.577	4.18	0.2528
Roughness-targeted pass	<code>_targeted_pass1</code>	0 / 0	0.1096	0.568	4.28	0.2442
Pure $\alpha = 0.35 \times 10$ from <code>_solver200</code>	(offline)	143 / 452	0.0822	0.237	0.55	0.1137
Blocked $\alpha = 0.35 \times 10$ from <code>_solver200</code>	(offline)	0 / 0	0.0843	0.262	4.75	0.1263
Pure $\alpha = 0.10 \times 8$ (pre-manual)	<code>_pre_blocked_laplacian</code>	21 / 94	0.0965	0.398	2.26	0.1910
Saved final	(live skin)	3 / 15	0.0965	0.402	2.26	0.1915

Table 26. Crossing pairs by anatomy mesh for four states (offline detector). The unified solver states and the blocked reproduction have none. The saved final retains 15 pairs with only the jaw and the mentalis.

Crossing pairs with	Pre-cleanup	Pure 0.35×10	Pure 0.10×8 (pre-manual)	Saved final
Mentalis	115	76	49	8
fat_8	63	26	3	0
Jaw	60	90	21	7
fat_4	54	63	5	0
fat_mirrored3	48	71	0	0
fat_11	47	55	9	0
fat_12	45	50	7	0
fat_7	35	12	0	0
Depr. labii inf. R	23	7	0	0
Depr. labii inf. L	7	0	0	0
Orbicularis oris	0	2	0	0
All anatomy meshes	497	452	94	15

The residual set in the saved final scene is small and specific: 3 faces (ids 4185, 10931, 10934) made of 10 vertices, crossing the jaw bone in 7 pairs and the mentalis muscle in 8 pairs. All three lie at the chin and jaw (Figure 13). None of the ten vertices is closer than 0.5 units to anatomy, which is why a clearance floor would not have found them (Section 6). The offline detector reproduces Maya’s count for the pre-cleanup skin to within one face, so the residual count carries the same tolerance; a Maya-side scan of the final scene has not been logged and is listed as an open item (Appendix E).

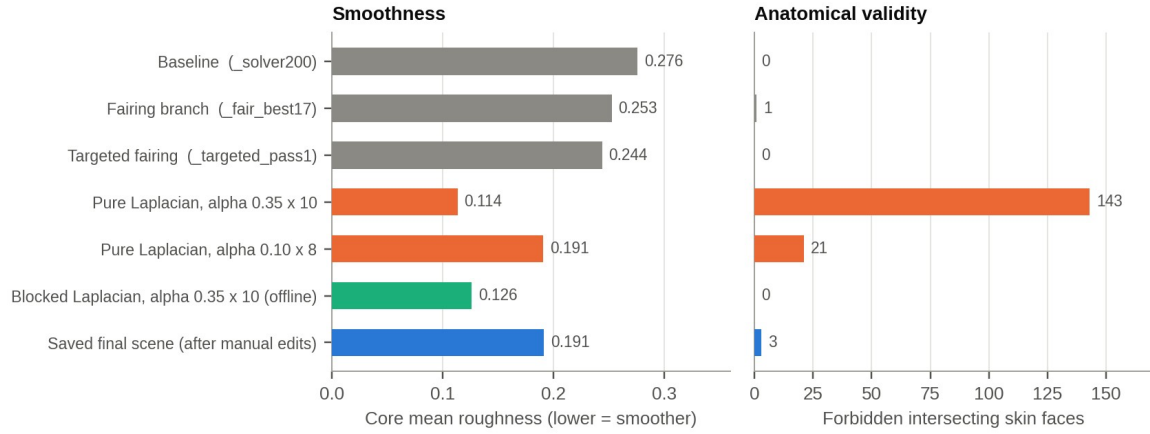


Figure 30. The trade-off that organises the research: smoothness (left, rough-core mean roughness, lower is smoother) against anatomical validity (right, forbidden faces). The unified-solver and fairing states are safe but not much smoother; the pure runs are smoother but unsafe; the blocked reproduction is smooth and safe; the saved final scene is smooth with three residual faces.

14.4 Roughness distribution

Table 27. Roughness $\|L_i\|$ over all non-boundary vertices, for the two baselines, the two Laplacian experiments and the final mesh (offline).

All non-boundary vertices (n = 15,316)	Mean	p50	p90	p95	p99	Max
Pre-cleanup	0.1345	0.0909	0.2922	0.4052	0.6815	2.224
_solver200	0.1133	0.0741	0.2223	0.3229	0.6658	4.909
Pure 0.35 × 10	0.0822	0.0722	0.1446	0.1744	0.2375	0.547
Blocked 0.35 × 10	0.0843	0.0710	0.1446	0.1765	0.2616	4.746
Pure 0.10 × 8	0.0965	0.0767	0.1754	0.2323	0.3978	2.261
Saved final	0.0965	0.0767	0.1751	0.2313	0.4021	2.261

Table 28. Roughness over the rough core. The core is the roughest 25% of _solver200, so _solver200 is by construction the worst state on it; use with the all-vertex table.

Rough core (n = 3,564; defined from _solver200)	Mean	p50	p90	p95	p99	Max
Pre-cleanup	0.2521	0.1908	0.4650	0.5885	1.0897	2.224
_solver200	0.2762	0.2047	0.4498	0.5975	1.3501	4.909
Pure 0.35 × 10	0.1137	0.1046	0.1868	0.2151	0.2974	0.520
Blocked 0.35 × 10	0.1263	0.1038	0.1968	0.2349	0.5358	4.746
Pure 0.10 × 8	0.1910	0.1562	0.3007	0.3688	0.7789	2.261
Saved final	0.1915	0.1562	0.3009	0.3693	0.8058	2.261

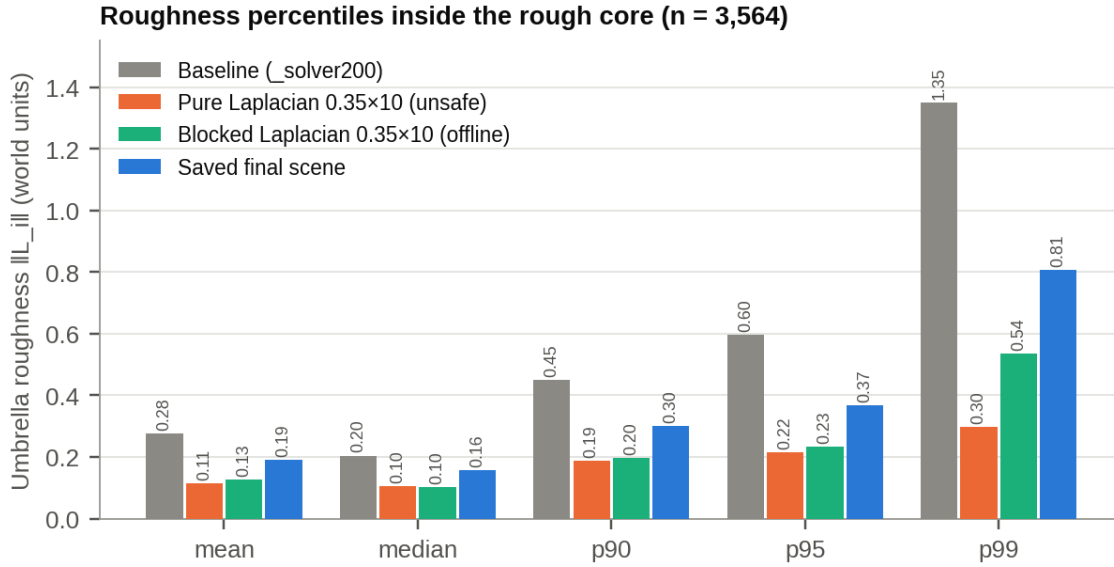


Figure 31. Roughness percentiles inside the rough core for the baseline, the unsafe pure run, the blocked reproduction and the saved final scene. The blocked run retains most of the tail reduction (p99 1.35 $\rightarrow$ 0.54); the saved final keeps the pure 0.10 $\times$ 8 reduction (p99 $\rightarrow$ 0.81) with the residual chin mounds still in the tail.

The distribution tables show a consistent picture. Every smoothing state lowers all percentiles, the tail more than the centre, and the pure run is smoothest. The saved final scene is smoother than `_solver200` at every percentile but not as smooth as the pure run, because $\alpha = 0.10$ with eight iterations is a gentler operator than $\alpha = 0.35$ with ten. It also keeps the chin mounds: its core maximum (2.26) and 99th percentile (0.81) show that what remains rough is a small set of vertices, the same ones that the blocked strategy freezes (Section 12).

14.5 Displacement and fidelity

Table 29. Vertex displacement between states over all 15,905 vertices (world units; mean edge length 2.5). The last column gives the mean over the active set that the smoothers may move, which is comparable with the logged displacement of the pure run (0.319).

Pair of states	Mean	RMS	p95	Max	Max at vertex	Mean over the 9,035 active vertices
<code>_precleanupsolver</code> $\rightarrow$ <code>_solver200</code> (unified solver)	0.458	0.886	2.04	10.66	12017	—
<code>_precleanupsolver</code> $\rightarrow$ saved final	0.496	0.882	2.04	7.87	12017	—
<code>_solver200</code> $\rightarrow$ pure 0.35 $\times$ 10	0.181	0.336	0.70	5.82	12017	0.319
<code>_solver200</code> $\rightarrow$ blocked 0.35 $\times$ 10	0.169	0.303	0.67	2.97	6262	0.298
<code>_solver200</code> $\rightarrow$ pure 0.10 $\times$ 8 (pre-manual)	0.053	0.116	0.21	2.82	12017	0.093
<code>_solver200</code> $\rightarrow$ saved final	0.053	0.114	0.21	2.82	12017	0.093

The saved final differs from `_solver200` by a mean of 0.053 world units over the whole skin (RMS 0.114; 0.21 at the 95th percentile), about 2% of an edge, and by 0.093 on the active set. The largest displacement, 2.82 units at vertex 12017, is again the repair mound. The unsafe pure run moved the skin about 3.4 times as far (0.181), and its displacement distribution has a long tail (Figure 32); the saved final matches the gentle pure 0.10 $\times$ 8 curve, which is what one would expect since it is that state plus 130 small edits.

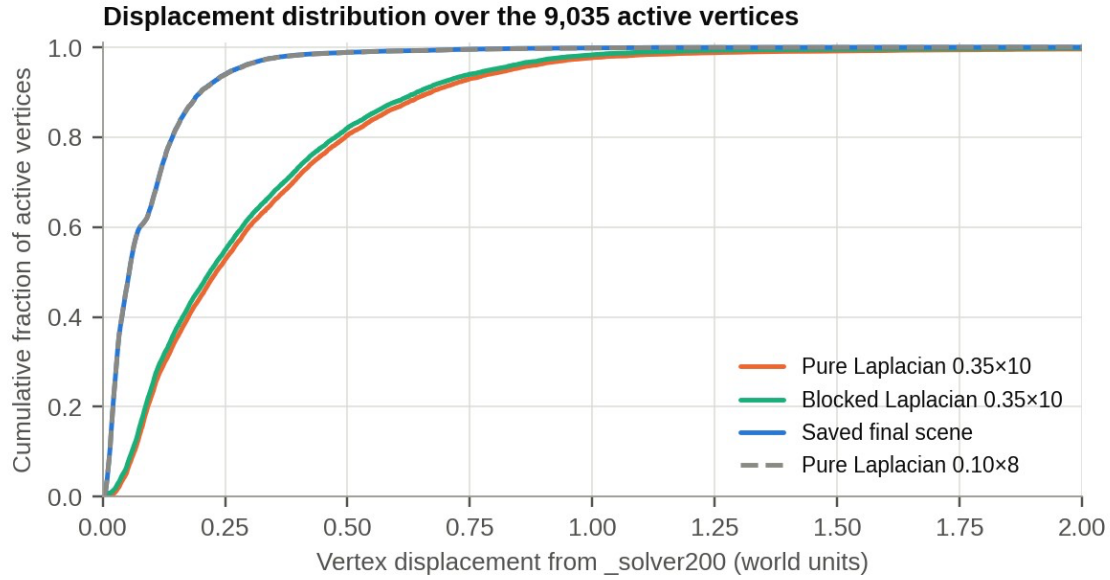


Figure 32. Cumulative distribution of the vertex displacement from `_solver200` over the 9,035 active vertices for the four smoothing results. The saved final scene and the pure 0.10×8 state coincide; the two $\alpha = 0.35$ runs move vertices several times further.

Table 30. Point-to-surface (p2s) distance from the moved vertices of the first state to the surface of the second, world units. Hausdorff is the maximum. Sampled points are a subset of the moved vertices for the large sets; the manual edits are all 130 vertices.

Pair of states	Vertices moved	Points sampled	Hausdorff	Mean p2s	p95	RMS
<code>_precleanupsolver</code> $\rightarrow$ <code>_solver200</code>	6,516	2,172	3.99	0.560	1.708	0.797
<code>_solver200</code> $\rightarrow$ pure 0.10×8	9,035	2,259	0.95	0.043	0.134	0.071
<code>_solver200</code> $\rightarrow$ saved final	9,074	2,269	0.95	0.042	0.137	0.072
pure 0.10×8 $\rightarrow$ saved final (manual edits)	130	130	0.68	0.091	0.460	0.170

The surface moved less than the vertices did. Over all 9,074 vertices that the smoothing moved, the mean point-to-surface distance from `_solver200` to the saved final is 0.043 world units (95th percentile 0.136, RMS 0.070, Hausdorff 0.98), while the vertex displacement is more than twice as large: on average 46% of the displacement is out of the surface, and the remainder slides within it. The Hausdorff distance in the opposite direction, from the saved final to `_solver200`, is 0.56, so no part of the final surface is further than about one unit from the clean baseline. Facial form is preserved by this measure, and the 130 manual edits moved the surface by a mean of 0.091 units.

Table 31. Displacement along the outward vertex normal of `_solver200` over the 9,035 active vertices (negative = inward). The normal is the area-weighted average of the incident faces; outward orientation was fixed by the mean direction from the vertex centroid.

From <code>_solver200</code> to	Mean signed normal displacement	Mean absolute	Moved vertices inward	Mean signed, % of mean edge length
Pure 0.35×10	-0.0927	0.1683	71%	3.7%
Blocked 0.35×10	-0.0760	0.1516	71%	3.0%
Pure 0.10×8	-0.0227	0.0438	71%	0.9%
Saved final	-0.0215	0.0429	71%	0.9%

Every smoothing result moves the active vertices inward on average, as a Laplacian smoother does at convex features ([24, 8]): 0.021 world units for the saved final and 0.093 for the unsafe pure run, about 1% and 4% of an edge length. Inward is toward the anatomy. This is a second, independent reason why an unconstrained smoother exposes fat and muscle, and why the constraint has to be spatial: the operator has a systematic bias into the very tissue that the safety test protects.

14.6 Mesh quality

The mesh is quad-dominant, so angle metrics are computed on face interior angles (quads and triangles alike) and the aspect measure is the ratio of the longest to the shortest edge of a face. Table 32 reports them for the two baselines, the two smoothing experiments and the final mesh. Because the operators move vertices by a small fraction of an edge, element quality is expected to change little; the measurement is there to show that it did not degrade.

Table 32. Geometric quality of the skin mesh in five states (topology identical). Face angles are interior angles of quads and triangles; the angle defect at an interior vertex is 2π minus the sum of the incident face angles.

Metric	Pre-cleanup	_solver200	Pure 0.35×10	Blocked 0.35×10	Saved final
Minimum face angle, mean (°)	75.78	75.79	76.42	76.32	76.13
Minimum face angle, median (°)	78.52	78.67	79.27	79.20	78.98
Faces with a minimum angle below 10°	24	34	18	18	29
Maximum face angle, 99th percentile (°)	149.5	150.8	148.3	148.4	149.2
Faces with a maximum angle above 170°	19	25	11	11	16
Edge-length ratio per face, 95th percentile	3.047	3.057	2.726	2.731	2.892
Edge-length ratio per face, maximum	20.2	58.2	20.3	20.3	15.4
Edge-length coefficient of variation	0.4887	0.4808	0.4687	0.4704	0.4751
Adjacent-face normal angle, mean (°)	4.54	4.33	3.71	3.76	3.98
Adjacent-face normal angle, 95th percentile (°)	12.46	12.00	9.74	10.05	10.92
Adjacent-face normal angle, 99th percentile (°)	23.61	22.61	16.98	17.65	19.22
Angle defect, mean of $ \cdot $ (rad)	0.00543	0.00556	0.00365	0.00393	0.00402
Angle defect, 99th percentile (rad)	0.0727	0.0680	0.0432	0.0473	0.0484
Umbrella mean-curvature magnitude, 99th percentile	0.487	0.455	0.341	0.341	0.401
Zero-length edges / degenerate faces	0 / 0	2 / 3	0 / 0	0 / 0	0 / 0

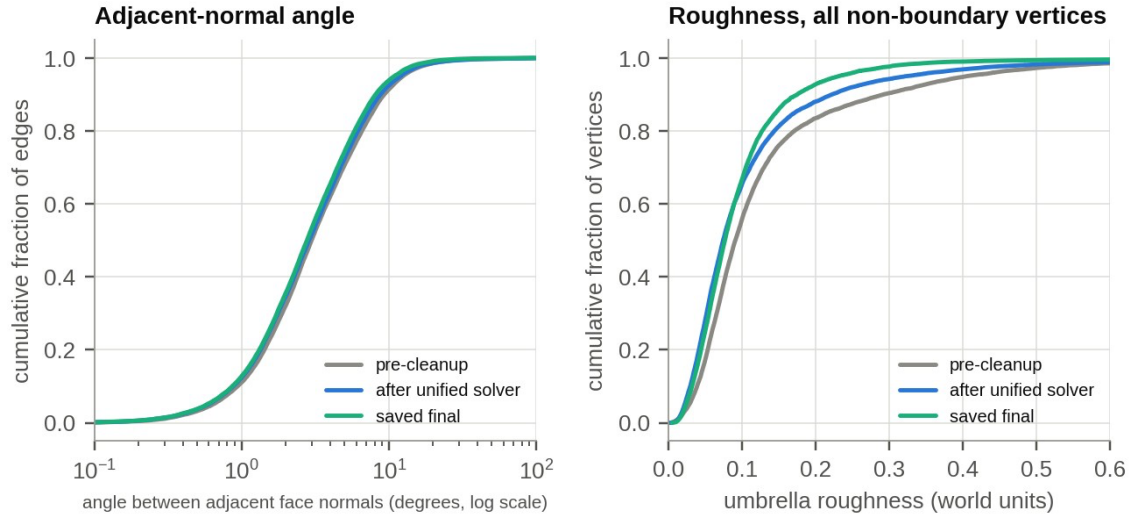


Figure 33. Left: cumulative distribution of the adjacent-face normal angle for the pre-cleanup skin, the unified-solver result and the saved final. Right: cumulative distribution of umbrella roughness over all non-boundary vertices. Smoothing shifts the roughness distribution left and leaves the dihedral distribution almost unchanged.

The result is uneventful in the intended way. The distribution of the face minimum angle is unchanged to within 0.35° in the mean, no degenerate face or zero-length edge appears, and the counts of very stretched faces do not grow: 16 faces exceed 170° in the saved final against 19 before the unified solver and 25 in `_solver200`. The 99th percentile of the adjacent-face normal angle falls from 23.6° to 19.2° and the mean angle defect from 0.00543 to 0.00402, which is the expected signature of removing curvature outliers without changing the overall shape. The maximum edge ratio (15.4) is a property of the registration mesh, not of the cleanup; it is 20.2 in the pre-cleanup skin and 58.2 in `_solver200` (a repair mound stretches faces around it).

14.7 Clearance to anatomy

Table 33. Unsigned distance from the 9,035 active skin vertices to the nearest anatomy triangle (world units, exact closest-point query; offline).

State	Min	p1	p5	Median	Vertices < 0.5	Vertices < 1.0
<code>_solver200</code>	0.445	0.713	0.905	7.48	1	537
Pure 0.35×10	0.001	0.313	0.770	7.47	186	657
Blocked 0.35×10	0.016	0.509	0.826	7.47	85	620
Pure 0.10×8	0.155	0.585	0.874	7.48	46	573
Saved final	0.228	0.598	0.882	7.48	30	566

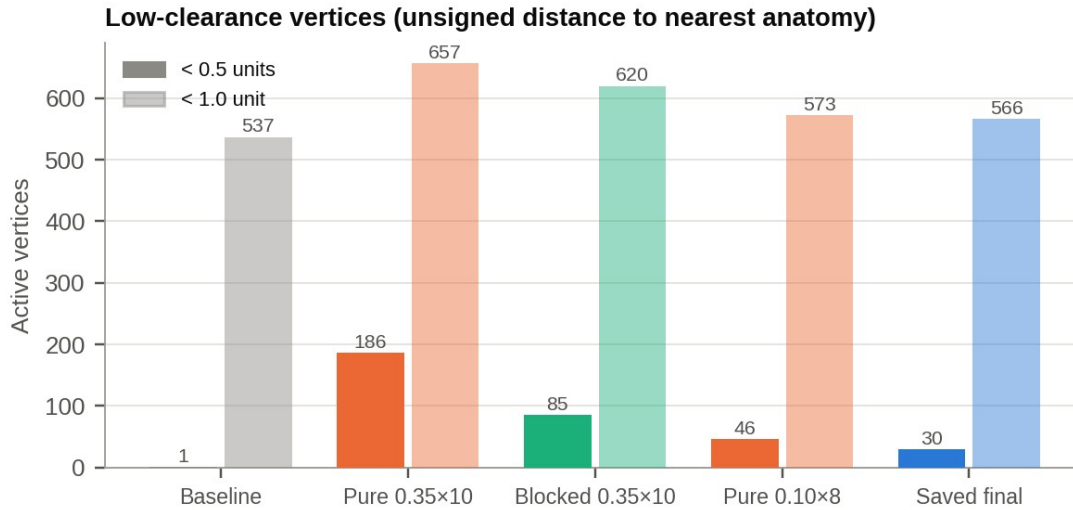


Figure 34. Number of active vertices closer than 0.5 and 1.0 world units to anatomy in five states. Every smoothing result brings some vertices closer; the pure runs bring many more. Clearance is a risk indicator; the forbidden-face count is the safety measure.

Clearance behaves as the earlier analysis predicts (Section 6). The clean baseline has one active vertex under 0.5 units and 537 under 1.0; the saved final has 30 and 566, and the unsafe pure run 186 and 657. The saved final therefore has a few dozen vertices within half a unit of the anatomy without crossing it, and the three residual faces are not among them. The number is reported and not optimised.

14.8 What could not be measured

- Detector precision and recall for M1–M5 cannot be computed: there is no labelled ground truth for artifacts, and none was invented. M5 is judged visually and against the intersection detector, not scored.
- Roughness and M4 disagreement values printed by the Maya solver runs are averaged over a vertex set that the logs do not record, so they are not comparable with the offline tables above (Table 13).
- Intersection depth is not reported: an intersection-depth output exists in the analysis files, but its definition is not documented in the evidence, so no statistic is quoted.
- Visual quality at normal inspection distance is judged from journal screenshots taken with different cameras. No matched before/after camera pair exists for the final stage; the roughness maps (Figure 24) substitute for it.
- Volume or area change is not reported: the skin is an open surface and no closed volume is defined. The signed normal displacement is the shrinkage indicator used instead.
- Common metrics that would not respond to this work were not computed: cotangent-weighted curvature on a skewed quad mesh, genus and manifoldness (unchanged topology), and texture-space distortion (UVs were not part of the study).

15. Why the Project Is Complete

The project is not declared complete because time ran out. It is declared complete because the original problem, as stated in Section 2, has been answered by a chain of results, each of which is supported in this report by a screenshot, a log, a saved state, a measurement, or all four. The argument has twelve steps; where a step is only partly supported, that is stated.

Table 34. The completion argument: twelve claims, the evidence in this report and their status.

#	Claim	Supporting evidence	Status
1	The main visual failure modes were defined.	Sharp local artifact, broad smooth artifact, closed-skin anatomy protrusion, valid exposure and low clearance are separated (Section 2, Section 1.5); the starting state is documented with screenshots (Figure 1).	Supported
2	Detection progressed through multiple reproducible hypotheses.	Five milestones with commits, screenshots and stated reasons for each failure (Section 5).	Supported
3	M5 gave a stable, interpretable detection result.	Fused mask with provenance; two recorded runs with unrecorded parameter difference (Section 5.6).	Partly: no ground truth; M5 written notes missing
4	True surface intersections were separated from low-clearance cases.	Exact triangle test; 93 faces against 968 vertices under 1 unit, 82% of which belong to no forbidden face (Table 10).	Supported (measured)
5	Failure modes of naive and sophisticated repair were understood.	V1 spikes and V2 mounds, with footprint measurement; global-step stall reproduced offline (Section 7, Section 8).	Supported
6	The long-running solver plateau was diagnosed, not just observed.	Invalid rest reference, M4 decay state bug, and the 50 – 200 iteration comparison (Section 9).	Supported
7	Pure Laplacian proved the remaining surface errors were removable.	Core mean 0.2762 – 0.1137 in 0.38 s; the logged run (Section 11).	Supported (logged and offline)
8	Blocked Laplacian localised safety instead of suppressing smoothing globally.	92% of the pure reduction, zero forbidden faces, 3.8% of active vertices frozen (Section 12).	Supported offline; not the origin of the saved final mesh
9	The final mesh visually meets the project objective.	Journal screenshots, roughness maps (Figure 28, Figure 24); 28% lower mean roughness than the pre-cleanup skin.	Partly: three residual forbidden faces; largest mound remains
10	Residual corrections are small and documented.	About 130 vertices, mean 0.099 world units (Section 13).	Supported; provenance of stage 1 to be confirmed
11	Further global iterations gave negligible visible gain at rising cost and risk.	_solver50 – _solver200: 1.7% lower mean roughness for 150 further iterations; fairing runs: modest gains and repair cycles (Section 9, Section 10).	Supported
12	The code and report preserve the research trail, including failed hypotheses.	Milestone commits, branch history, state catalogue, reproducibility appendices (Section 19).	Supported

15.1 Tie to the original success criteria

The success criteria of Section 2 can now be checked one by one. (i) Closed skin covers the anatomy while eye and mouth exposure remains: met in the journal screenshots except for three residual faces at the chin and jaw, which the offline detector reports. (ii) The major bumps, ridges and faceted patches are reduced: skin-wide the roughness percentiles fell by 16% (median) to 43% (95th percentile) and the mean by 28% relative to the pre-cleanup skin, and at the rough core the mean fell by 24% and the 95th percentile by 37%; the pure and blocked results show that more is available at the price of the constraint. (iii) Correction adds no large spikes or repair mounds: met for the smoothing stages, but the largest repair mound of the unified solver (vertex 12017) is halved and not removed, and the manual finish makes its 46 pushed vertices locally rougher. (iv) The broad form is preserved: the Hausdorff distance to the clean baseline is below one unit and the mean

point-to-surface change is 0.043. (v) The result is reproducible from documented code and saved stages with the manual finishing stated: met, with the provenance of manual stage 1 to be confirmed. (vi) Further automated iteration shows diminishing returns: shown for the unified solver and the fairing runs; the blocked strategy is the exception, since it made a further step available, and it is not part of the saved final.

15.2 Research completion versus a production system

Two statements must not be confused. The first is that the *research question is answered and an acceptable final mesh exists*. The second, which this report does not claim, is that a *completely general, fully automatic production system exists*. The table separates what is complete from what remains.

Table 35. Research completion versus the future goal of a general fully automatic production system.

Complete: the research and the acceptable final mesh	Not complete: a general fully automatic system
The failure modes are defined, detected and measured on one character.	Generalisation to other characters, registrations and anatomies has not been tested.
The cause of the plateau is identified and a principled strategy exists (blocked Laplacian).	The blocked strategy was not run in Maya on a valid baseline for the saved final; a Maya-side run and log are missing.
The final mesh has 3 forbidden faces (offline) against 93, and is smoother by every skin-wide roughness measure.	A fully automatic pipeline that yields zero forbidden faces without manual edits has not been demonstrated.
Residual manual edits are small (about 130 vertices) and documented.	The manual finish cannot be scripted from the evidence: stage-1 provenance is not recorded.
The research trail, negative results included, is preserved.	The mounds that the repair creates are not solved: a smooth surface at those vertices needs an anatomy-aware solution.

Summary of the narrative

The research progressed from increasingly sophisticated artifact detection to increasingly sophisticated safety-constrained correction. The decisive result came from simplifying the problem: pure Laplacian smoothing showed that the surface could be made smooth immediately, and the final solution therefore focused on localising the anatomical constraint rather than continuing to weaken the smoothing globally.

15.3 Research progression at a glance

Table 36. Milestone progression: what each stage asked, what it established and what limited it. Dates are commit dates on the branch feature/m5-final-artifact-detection.

Stage (commit, date)	Question	Outcome	What limited it
M1 (7b2627a, 10 Jul)	Can a local score find bumps automatically?	Upper-tail umbrella-Laplacian mask selects sharp features.	Legitimate high curvature scores as high as bumps; boundaries; no broad error found.
M2 (d207c61, 17 Jul)	Can a target-referenced Laplacian cancel legitimate curvature?	The residual highlights differences from the supplied target.	The target is a different surface; disagreement is not artifact probability.
M3 V1 (8a38590, 17 Jul)	Does persistence over ring scales separate defects from valid features?	Multi-scale persistence with boundary and small-component removal.	Broad smooth errors (the cheeks) are missed.
M3 V2 (ae11d62, 27 Jul)	Can geometry, normals and anatomy be combined?	Robust-z hybrid of Laplacian, normal inconsistency and anatomy deviation.	New false positives at folds; hand-set weights.
M4 (f2aca65, 6 Aug)	Can the anatomy define the reference?	SDF iso-surface reference exposes broad disagreement.	One global offset ignores thickness variation; a detection signal, not a correction rule.
M5 (7a8faa2, 270bac2, 13 Sep)	Can M3 and M4 be combined interpretably?	Union with provenance; the detector is frozen.	No ground truth; detection is not correction.
Intersection detector (b655e80, 14 Sep)	What is actually wrong?	Exact triangle test; 93 forbidden faces; clearance separated from crossing.	Yes/no per face; no depth.

Stage (commit, date)	Question	Outcome	What limited it
Repair V1 / V2 (03fcfa4, 14 Sep)	Can the crossings be removed directly?	V2 patch repair removes them.	V1 spikes; V2 broad mounds.
Constrained smoothing (6ac97a3, 3c5c0eb, 14 Sep)	Can smoothing stay inside the safe set?	Safe but nearly immobile: a global scalar couples the whole region.	Worst face sets every vertex's step.
Unified solver (30aeba4, 461989d, bf3022f, 14 Sep)	Can forces, trust region and feasibility do both?	93 → 0 forbidden faces.	Roughness plateau; sharper worst case (max 2.22 → 4.91).
Final fairing (5cfed80, 635c00e, 14–15 Sep)	Can a dedicated fairing pass finish the job?	Taubin and targeted passes improve the tail modestly.	The surrounding stack acts against the fairing step.
Pure Laplacian (a17462c, 15 Sep)	Is the surface hard to smooth?	No: core mean 0.276 → 0.114 in 0.38 s.	143 forbidden faces: unsafe.
Blocked Laplacian (f687b5c, 16 Sep)	Can the constraint be local?	Offline: 92% of the reduction, 0 forbidden faces.	Needs a valid start; leaves the sharpest mounds frozen.
Micro push and manual finish (ce07f08, 16 Sep; scene)	How to remove the last faces?	21 → 3 forbidden faces with about 130 edits.	Manual; 3 residual faces.

16. Screenshot and Evidence Map

The following table follows the order of the research and lists, for each stage, the figures that document it, what the reader should notice in them, and how far they support the claim made in the text. Where a journal screenshot does not support a claim, or does not exist, the entry says so instead of stretching an image to fit. The complete list of figures with their captions is given at the front of the report.

Table 37. Figure sequence and evidence map.

#	Figure	Stage	What the reader should notice	Support for the written claim
1	Figure 1, Figure 2	Initial registered skin	Fine bumps on chin, jaw and cheeks; small yellow (fat) and red (muscle) marks through closed skin near the jaw; large yellow and red areas at the top and back belong to the open cap and are not defects.	Supports the problem statement.
2	Figure 4	M1 umbrella Laplacian	Selected vertices (yellow dots) trace the brow, nostrils, lip corners and jaw line as well as true bumps; the selection is dominated by legitimate curvature.	Supports the failure claim.
3	Figure 5, Figure 6	M2 reference Laplacian	The selection follows differences from the target, concentrating at the openings and the outer edge.	Supports the failure claim.
4	Figure 7	M3 V1 multi-scale	Cleaner masks than M1, but the cheeks show solid yellow fat through the skin with no selected vertices there.	Supports the blind-spot claim.
5	Figure 8	M3 V2 hybrid	More coverage of broad regions and new selections around folds and corners.	Supports, qualitatively.
6	Figure 9, Figure 10	M4 SDF reference and detection	The reference surface is smoother than the skin and inherits anatomy facets; the detector selects broad regions of disagreement, many near openings.	Supports the M4 claims. No written notes.
7	Figure 11	M5 unified detector	Two recorded runs (V1, V2) that differ visibly; the mask combines M3 and M4 evidence.	Supports the mask description; the V1/V2 setting difference is not recorded.
8	Figure 14	True intersection selection	Selected faces at the chin and along the jaw line coincide with the yellow and red marks through closed skin.	Supports the claim that faces, not vertices, mark the defect.
9	Figure 16	V1 spikes; V2 rough patch; V3	(a) Dark spikes at the chin and jaw. (b) Spikes gone, mounded surface. (c) A serrated chin contour.	(a), (b) support the claims; (c) stage unknown; no claim is attached.
10	Figure 19, Figure 18, Figure 17	Constrained solver and unified solver	Diagrammatic and offline evidence: the global-step stall, the repair footprint, the force architecture.	No journal screenshot of a solver result exists; flagged as missing.
11	Figure 22, Figure 23(a), Figure 24	Pure Laplacian	The log shows the statistics; the viewport shows a smooth surface with small yellow marks at the jaw and cheek; the roughness maps show the bumps gone.	Supports the smoothing and exposure claims; the viewport uses one camera.
12	Figure 23(b), Figure 25, Figure 26	Blocked Laplacian	The start state of the blocked module (pure 0.10×8); the mechanism; the offline run.	No journal screenshot of a blocked run's <i>result</i> exists; flagged as missing.
13	Figure 27, Figure 28	Manual finishing and final mesh	Vertices touched by each stage; the pre-manual and final journal views.	Camera angles differ; the edit effect is measured, not shown by the screenshots.
14	Figure 29, Figure 30, Figure 31, Figure 32, Figure 34, Figure 33	Quantitative plots	Roughness and validity across states; the trade-off; tail statistics; displacement; clearance; mesh quality.	Offline measurements from the saved scene.

Four screenshots would strengthen the report and are requested rather than forced into it: a viewport screenshot of the unified-solver result (`_solver200`) with the same camera as Figure 1; a viewport screenshot of an actual blocked-Laplacian run on `_solver200`; a matched-camera before/after pair for the manual finishing (`_pre_blocked_laplacian` and the live skin); and a screenshot of the vertex-distance heuristic selection described in Section 6.

17. Related Work and Literature

This section places the methods used in the project in the literature and separates what is taken from the literature from what was developed here. Sources are primary papers, textbooks and vendor documentation; each was checked against publisher metadata as described in the note to the bibliography.

17.1 Discrete Laplacians and mesh smoothing

The umbrella (uniform) Laplacian, the average of the neighbours minus the vertex, is the oldest smoothing operator on meshes and is the operator of this project [9, 5]. It is cheap and needs no geometry, but on irregular meshes it mixes normal and tangential movement and is not scale-invariant; improved variants and shrinkage corrections were proposed early [27]. Cotangent-weighted operators, which discretise the Laplace–Beltrami operator and its mean-curvature normal, remove much of the dependence on sampling [20, 16]. The skin of this project is a quad-dominant mesh with a high edge-length coefficient of variation (0.475) and a few triangles, so the cotangent formulas for triangles cannot be applied unchanged; the umbrella form was retained because it is defined identically for every vertex and gives comparable roughness values across the whole skin. Section 14.8 lists cotangent-based curvature as a metric that was not computed for that reason, and a cotangent variant of the roughness score is a natural extension.

17.2 Fairing, shrinkage and the frequency view

Taubin’s signal-processing view treats iterated Laplacian smoothing as a low-pass filter on the graph frequencies of the mesh, explains the shrinkage that it causes, and removes it with a second, negative step of the same form [23, 24]. Implicit fairing with diffusion and curvature flow allows larger, unconditionally stable steps [8], and multiresolution approaches such as interactive fairing separate detail from a smooth base [15]. The measurements of Section 14.5 reproduce the shrinkage in the project’s own data: every smoothing result moved the active vertices inward on average. The project’s Taubin experiment used the published pair and confirmed only a modest gain (Section 10); the decisive result was obtained with the plain operator, because the limit in this problem was the constraint and not the shrinkage.

17.3 Feature-preserving denoising

Feature-preserving methods, such as bilateral mesh denoising and non-iterative feature-preserving smoothing, smooth noise while retaining sharp features by weighting neighbours by geometric similarity [10, 13]. They address a problem that the project met in M1: legitimate high-curvature features (lip corners, nostrils, the brow) score as roughly as high as artifacts under a local roughness score, and a uniform smoother flattens them along with the bumps. The project handled this by detection (M3, M4, M5) and by restricting smoothing to a selected region; it did not test a feature-preserving filter, and the comparison is listed as future work (Section 20).

17.4 Laplacian editing, rest-shape terms and variational deformation

Laplacian surface editing preserves differential coordinates while constraining handles [22], and variational deformation methods generalise this to a family of energies [4]. The rest-shape term of the unified solver (Eq. (15)) plays the role of a differential-coordinate preservation term: it defends the registered shape. The research found that when the reference itself contains defects, defending it works against fairing (Section 9); this is not a criticism of the published methods, whose reference is assumed to be a good shape, but it explains why the shape term had to be removed in the final pipeline.

17.5 Distance fields, closest points and signed distance

Distance fields on meshes are surveyed in [12]. Signed distance requires a reliable inside/outside decision; the angle-weighted pseudonormal gives one for closed, consistently oriented meshes [2], and earlier point-set methods estimated signed distance from oriented tangent planes [11]. The anatomy meshes of this project are a mixture of closed and open surfaces and the Maya distance backend used (MFnMesh closest-point queries [1]) returns an unsigned distance. This is why the report never interprets the M4 distance as inside or outside, and why vertex penetration is treated as a separate, less reliable condition than face intersection.

17.6 Intersection tests and collision-aware processing

The triangle–triangle test of this project follows the Möller–Trumbore line-triangle formulation applied to finite edge segments, with a separating-axis test for coplanar pairs [17, 18]; broad-phase bounding boxes and the general structure of deformable-object collision detection follow the survey of [26]. Cloth simulation practice separates detection from response and treats intersections as states to avoid rather than repair [3, 6]. The blocked Laplacian is in that spirit: the detector is an oracle for accepting a step, and the response is only to withdraw the step locally. Its relation to constraint-based projection methods was not analysed here, and no claim of novelty is made for the freeze-on-intersection mechanism without a systematic literature search.

17.7 Mesh quality and error measurement

Element-quality measures are collected in [14] and in the mesh-processing textbook [5]. The Hausdorff-based point-to-surface error used in Section 14.5 follows the practice of the Metro tool for measuring the error between a reference and a modified surface [7]. Angle defect and the umbrella mean-curvature vector are discrete curvature quantities on meshes [16]. Only measures that respond to vertex-position changes on a fixed-topology mesh were used.

17.8 Anatomically based facial modelling and skin registration

Physically based facial animation with layered muscle, fat and skin began with the layered model of [25], and modern systems personalise the anatomy and drive it from motion [21]. The framework of this project [19] registers a template skin to personalised internal tissues as a one-time step and then animates through muscle dynamics; the reference list of that manuscript was not part of the material supplied for this report, so the coverage of related anatomical face models here is limited to the sources above. This research adds a post-registration quality stage that the manuscript does not describe.

17.9 What is taken from the literature and what is new here

Table 38. Literature and project contributions by element.

Element	From the literature	Developed in this project
Umbrella Laplacian score, Jacobi smoothing	Standard operators [9, 5].	Percentile region selection with linear ring falloff; roughness percentiles as a tail metric.
Taubin smoothing	Published filter [23, 24].	Roughness-targeted weight map and its use inside the safety pipeline.
Triangle intersection	Möller tests [17, 18].	Exact skin–anatomy detector with opening exclusions and per-mesh reporting; face-versus-vertex analysis.
Distance-field reference	Unsigned distance fields [12].	Offset iso-surface reference, damped-Newton projection, $M3 \cup M4$ fusion with provenance.
Repair and constrained smoothing	Harmonic fields and step-size search are standard tools.	V2 anatomy-supported patch repair; global-step bisection; unified per-vertex trust-region solver, with negative results.

Element	From the literature	Developed in this project
Blocked Laplacian	Detector-as-oracle from collision handling [6].	Freeze-on-intersection with a one-ring halo for a smoother; not claimed to be new without a systematic search.
Evaluation	Metro-style distance; element-quality metrics [7, 14].	Offline pipeline that reproduces Maya states from the saved scene.

17.10 Bibliography

References are listed alphabetically by first author. DOI-bearing entries were checked against Crossref metadata for title, authors, year, venue, pages and DOI. The Beneath the Skin manuscript is cited as supplied with the project (no venue or DOI exists to verify), and the Maya documentation was located by web search; its version is not recorded.

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18. Limitations and Future Work

The report's main limits are stated where they arise; this section collects them so that the strength of the conclusions can be judged in one place.

One character, one registration. Everything was developed and evaluated on a single skin and a single anatomy. The thresholds and the roughness metrics are relative to this mesh, and no statement is made about other characters or registrations.

No ground truth for detection. Detector precision and recall were not measured, because no labelled artifacts exist; the detectors were compared visually and against the intersection detector. The strongest quantitative results concern correction, not detection.

The final mesh is a hybrid. It combines automated stages with about 130 manual vertex edits (Table Table 22) and retains three forbidden faces by the offline detector. It also retains the largest repair mound of the unified solver in a reduced form. The blocked strategy is shown offline and is not the origin of the final geometry.

Offline versus Maya. The offline pipeline reproduces the logged pure-Laplacian statistics exactly and the pre-cleanup face count to within one face, but the quad triangulation of Maya is not recorded and absolute face counts inherit that tolerance. A Maya-side scan of the final scene has not been logged.

Roughness is a proxy. Umbrella-Laplacian roughness responds to bumps but also to legitimate curvature and to sampling irregularity; a cotangent or feature-aware measure may rank vertices differently. The report treats it as one measure among several, and reports quality metrics that are independent of it.

Future work. The clearest technical gaps are (i) an anatomy-aware solution for the remaining mounds, where geometry and anatomy conflict, for example a small, local, energy-based solve on the frozen set instead of a manual push; (ii) a Maya-side run of the blocked Laplacian on a valid baseline with its log and screenshots; (iii) comparison with feature-preserving smoothers [10, 13] and cotangent-weighted operators [16]; and (iv) generalisation tests. Section 20 turns these into an experimental plan.

19. Reproducibility

19.1 Code, scene and environment

Table 39. Environment and code state.

Item	Value
Repository / branch / HEAD	skin-to-muscle-registration; feature/m5-final-artifact-detection; ce07f08 (16 Sep 2026)
Scene	t34.mb (derived from the t32 converged result of 1,600 registration iterations)
Live skin node; SDF parameters	skin_cloth_copy_v5_pull_back; skin_offset 1.5, sdf_blend_k 1.0, 6 SDF smoothing passes, K = 5
Maya / Python version	Not recorded in the evidence; the Maya SDK documentation version is likewise not recorded.
Pipeline entry	exec(open("src/d98-target_shrinkwrap_registration.py").read()) in the Maya Script Editor (repository README).
Offline pipeline	Python 3 with NumPy; the repository modules are imported with Maya I/O replaced by stand-ins (analysis_scripts/blocked_emu.py, emulate.py).

19.2 Baseline names

The states used as baselines and the operation that produced each are catalogued in Appendix C. The two baselines of the evaluation are `skin_cloth_copy_v5_pull_back_precleanupsolver` (pre-cleanup) and `skin_cloth_copy_v5_pull_back_solver200` (clean baseline); the exact final mesh is the live node `skin_cloth_copy_v5_pull_back`.

19.3 Commands in Maya

The wrappers below are the repository entry points that delegate to the modules of Appendix B. Parameter values marked *logged* were recorded in the Maya log or the working draft; values marked *inferred* were recovered from the saved geometry and were not recorded as commands.

```
# detection milestones (logged parameters: percentile 97.5; scales (1, 2, 3); persistence >= 2)
detect_skin_artifacts(percentile=97.5) # M1
detect_reference_skin_artifacts(...) # M2
detect_multiscale_skin_artifacts(scales=(1, 2, 3)) # M3 V1
detect_hybrid_skin_artifacts(...) # M3 V2
detect_sdf_reference_skin_artifacts(...) # M4 (d0 = 7.3998 logged)
detect_m5_skin_artifacts(fusion_mode="union", m3_percentile=97.5, m4_percentile=97.5) # M5

# correction (unified solver, produces _precleanupsolver -> _solver50 -> _solver200)
run_final_skin_cleanup(max_iterations=200)

# pure Laplacian, logged run from _solver200
run_pure_skin_smoothing(roughness_percentile=75, growth_rings=2, strength=0.35, iterations=10)

# pure Laplacian that produced _pre_blocked_laplacian (inferred from geometry: alpha 0.10, 8
iterations)
run_pure_skin_smoothing(roughness_percentile=75, growth_rings=2, strength=0.10, iterations=8)

# blocked Laplacian (module defaults; requires a valid start such as _solver200)
run_blocked_skin_smoothing(roughness_percentile=75, growth_rings=2, strength=0.35, iterations=10,
freeze_rings=1)

# one-shot residual push (implemented; use in the saved final unconfirmed)
run_micro_clearance_push(target_clearance=0.15, blend_rings=1)
```

19.4 Offline reproduction of the numbers

The numbers of this report were generated by scripts in the supplementary folder in the following order: `scene.py` and `mbmesh.py` read the vertex positions of every saved node and of the 43 anatomy meshes from `t34.mb`; `emulate.py` and `blocked_emu.py` run the repository pure and blocked modules with Maya I/O mocked; `isect.py`, `isect_all.py`, `clear.py`, `haus.py`, `haus_full.py` and `final_metrics.py` compute the detector, clearance, Hausdorff and roughness/quality statistics; `clear_vs_face.py`, `global_alpha.py`, `global_alpha10.py` and `freeze_rings_cmp.py` compute the analyses of Section 6, Section 8 and Section 12; and the figure scripts draw the plots. The scripts contain absolute paths of the analysis session that must be edited on another machine. Deterministic operators (Jacobi smoothing, the detector, closest-point queries) were used throughout; no random seeds are involved in the reported numbers.

Table 40. Values that a reproduction should obtain.

Check value	Expected result
Pre-cleanup forbidden faces / pairs	93 / 497 (Maya logged 92 faces; alternative quad diagonal: 99)
Pure 0.35×10 from <code>_solver200</code>, rough-core mean	0.27625 – 0.11366 (matches the Maya log: 3,564 selected, 9,035 active)
Pure 0.35×10, forbidden faces	143 (452 pairs)
Blocked 0.35×10 from <code>_solver200</code> (defaults)	10 iterations, 342 vertices frozen, 0 forbidden faces, core mean 0.1263
Blocked from <code>_pre_blocked_laplacian</code>	Stops in iteration 1; 21 forbidden faces remain
Saved final vs <code>_pre_blocked_laplacian</code>	130 vertices differ; forbidden faces 21 – 3

20. Toward a Workshop Submission

The work as reported is a research-software report on one character. Turning it into a workshop paper needs additional experiments that address the limits of Section 18. A defensible contribution statement would be that a detector-as-oracle, freeze-on-intersection smoother recovers most of the fairing of an unconstrained Laplacian while keeping a skin free of anatomy intersections, and that the constraint must be spatially local. The experiments needed are these.

Table 41. Experiments required to turn the report into a workshop submission.

Experiment	Purpose	Output
Multiple characters and registrations (at least five)	Test generalisation of thresholds, region selection and blocked smoothing.	Per-subject before/after tables with the metrics of Section 14; distributions and confidence intervals.
Labelled artifacts and blinded visual rating	Give M1–M5 and the final mesh a ground truth; measure precision and recall; check that roughness tracks perceived quality.	Detector precision/recall; rating study.
Ablation of the unified solver	Show which of the M4 pull, rest-shape term, trust region and repair profile causes the plateau.	Table of variants under identical metrics.
Blocked Laplacian in Maya on a valid baseline, with sweeps	Replace the offline reproduction; vary α , iterations, percentile, growth and freeze rings; time it.	Logs, screenshots, sensitivity plots.
Comparison with published smoothers	Compare uniform, cotangent, Taubin, implicit and feature-preserving smoothers [24, 8, 10, 13] under the same safety oracle.	Trade-off curves of roughness against forbidden faces.
Automated treatment of the residual mounds	Replace the manual finish, for instance with a local energy-based solve on the frozen set.	Zero forbidden faces without manual edits.
Downstream animation test	Check that the cleaned skin behaves like the registered skin under muscle-driven deformation.	Deformation differences under standard expressions.
Sensitivity to quad triangulation and detector tolerance	Bound the absolute face counts.	Counts under both diagonals and several tolerances.
Formal statement of the blocked smoother	State the invariant (accepted states are intersection-free) and the termination behaviour.	Short proof or counterexamples.

Appendix A. Formula Appendix

Table 42 indexes the equations of Section 4 and of the solver sections by the quantity they define and the function in which they are implemented. Table 43 gives the notation.

Table 42. Index of the equations.

Equation	Quantity	Where defined	Implemented in
Eq. (3)	Umbrella Laplacian and roughness $\ L_i\ $	Section 4	artifact_detection.compute_laplacian_scores
Eq. (4)	Reference (target) Laplacian residual, M2	Section 5.2	artifact_detection (reference detector)
Eq. (5), Eq. (6)	Multi-scale score and persistence, M3 V1	Section 5.3	artifact_detection (multi-scale detector)
Eq. (7)	Robust-z hybrid score, M3 V2	Section 5.4	artifact_detection (hybrid detector)
Eq. (8), Eq. (9)	Soft-minimum distance and damped-Newton projection to the iso-surface, M4	Section 5.5	artifact_detection (SDF reference)
Eq. (10)	M5 fusion by set union with provenance	Section 5.6	artifact_detection.detect_m5
Eq. (11), Eq. (12)	Finite-segment triangle test and forbidden-face set	Section 4.4	anatomy_constraint.analyze_skin_anatomy_intersections
Eq. (22)	Harmonic patch displacement, V2 repair	Section 7.2	anatomy_constraint_anatomy_supported_patch_repair
Eq. (17)	Global safe step by bisection	Section 8.2	smoothing_utils.find_largest_safe_ssmoothing_step
Eq. (15), Eq. (16)	Unified per-vertex force balance and smoothstep anchor	Section 9	final_cleanup_solver.run_cleanup_solver
Eq. (14)	Taubin pair (λ, μ)	Section 10	final_cleanup_solver.run_final_fairing
Eq. (13)	Jacobi Laplacian step with ring falloff	Section 4	pure_laplacian_smoothing.jacobi_laplacian_step
Eq. (18)	Blocked update (frozen vertices as anchors)	Section 4	blocked_laplacian_smoothing.run_blocked_laplacian_smoothing
Eq. (19), Eq. (20), Eq. (21)	Percentile statistics and evaluation metrics	Section 4	metrics_utils; offline analysis scripts

Table 43. Notation.

Symbol	Meaning
$\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n), n = 15,905$	Skin vertex positions; fixed connectivity.
$\mathbf{N}(i), \mathbf{B}$	One-ring neighbours of vertex i ; boundary vertex set (589 vertices).
$\mathbf{L}_i = (1/ \mathbf{N}(i)) \sum_j \mathbf{x}_j - \mathbf{x}_i$	Umbrella Laplacian; roughness is $\ \mathbf{L}_i\ $.
$\mathbf{A}_1, \dots, \mathbf{A}_{43}$	Anatomy triangle meshes (bone, muscle, fat, other).
$\phi(\mathbf{x}), d_o$	Soft-minimum unsigned distance over $K = 5$ nearest anatomy meshes; iso-value (7.3998 logged).
$\mathcal{F}(\mathbf{X})$	Set of forbidden skin faces: crossing or coplanar-overlap with anatomy, excluding faces near the buffered boundary.
α, ρ	Smoothing strength; solver trust ratio (0.5).
$\mathcal{C}, \mathcal{A}$	Rough core (3,564 vertices) and active set (9,035 vertices).

Appendix B. Code and Module Architecture

The module sizes are given in Table 5. Table 44 lists the functions that carry the research and what they depend on. Detection was frozen at M5; all later modules take the detector output or the true intersection detector as input and do not change it.

Table 44. Functions that carry the research (as read from the repository).

Module	Key functions	Depends on
artifact_detection.py	compute_laplacian_scores; detect_irregular_region; reference, multi-scale, hybrid and SDF-reference detectors; detect_m5	mesh_utils, region_selection
anatomy_constraint.py	analyze_skin_anatomy_intersections; intersection_vertices_from_report; enforce_anatomy_clearance; _anatomy_supported_patch_repair; _legacy_normal_push_intersection_repair; analyze_skin_anatomy_penetration; MayaAnatomyBackend	mesh_utils
smoothing_utils.py	constrained_smooth_mesh_region (strict mode); _find_largest_safe_smoothing_step; Laplacian and Taubin smoothers	anatomy_constraint, mesh_utils
final_cleanup_solver.py	run_cleanup_solver; _run_initial_feasibility; decay_m4_weight; m4_is_improving; run_final_fairing; build_roughness_fairing_weights	artifact_detection, anatomy_constraint, smoothing_utils
pure_laplacian_smoothing.py	run_pure_laplacian_smoothing; jacobi_laplacian_step; build_falloff_weights; select_rough_vertices; ring_distances	mesh_utils, artifact_detection (scores)
blocked_laplacian_smoothing.py	run_blocked_laplacian_smoothing; freeze_footprint; _scan_intersections	pure_laplacian_smoothing (imported), anatomy_constraint, artifact_detection
micro_clearance_push.py	run_micro_clearance_push; blend_corrections	mesh_utils, anatomy_constraint
d98-target_shrinkwrap_registration.py	Wrappers: detect_*_skin_artifacts, run_final_skin_cleanup, run_final_skin_fairing, run_pure_skin_smoothing, run_blocked_skin_smoothing, run_micro_clearance_push	all of the above; Maya
cleanup_pipeline.py; metrics_utils.py; maya_io.py	Orchestration and reports; displacement and diagnostic metrics; Maya read/write and duplicate_mesh backups	Maya API

The pure, blocked and micro-push modules are deliberately independent of the unified solver: they do not import final_cleanup_solver, so the results of Section 11 and Section 12 cannot be affected by its state. The blocked module imports the Jacobi step and the region logic of the pure module, so its operator is identical, not a copy.

Appendix C. Catalogue of Saved States

All skin nodes in the scene t34.mb, their geometric relations as measured from the saved vertex positions, and what is known of their origin. Geometric relations are exact; provenance statements are marked *recorded* (in a log or the repository) or *inferred* (from geometry).

Table 45. Skin nodes in the saved scene and their relations.

Node (prefix skin_cloth_copy_v5_pull_back unless stated)	Role and geometric relation	Forbidden faces / pairs
_target	Target mesh for the M2 reference; geometrically far from every cleanup state (tens of world units).	n/a
_backup	Earlier backup; far from every cleanup state; role not recorded.	n/a
_precleanupsolver	Registered skin before the unified solver (pre-cleanup baseline; recorded).	93 / 497
_solver50	Unified solver after 50 iterations (recorded).	0 / 0
_solver200	Unified solver after 200 iterations; clean baseline (recorded).	0 / 0
_prefairing	Identical to _solver200 (no vertex differs); backup taken before the fairing stage (inferred).	0 / 0
_fair_best17	Taubin fairing, best state at iteration 17 (recorded name).	1 / 12
_targeted_pass1	Roughness-targeted fairing, first pass (recorded name).	0 / 0
_pre_pure_laplacian	Backup written by the pure module. Not identical to _solver200 : it differs in 1,678 vertices (maximum 3.5 scene units); its provenance is not recorded. The logged 0.35×10 run reproduces exactly from _solver200 .	2 / 16
_pre_blocked_laplacian	Backup written by the blocked module. Geometrically identical to a pure Laplacian (p75, growth 2, $\alpha = 0.10$, 8 iterations) of _solver200 (inferred).	21 / 94
_pre_micro_clearance_push	Backup written by the micro-push module; identical to _pre_blocked_laplacian (no vertex differs).	21 / 94
skin_cloth_copy_v5_pre_final	Scene node between the two manual stages: differs from _pre_blocked_laplacian in 46 vertices and from the live skin in 87.	3 / 15
(live skin) skin_cloth_copy_v5_pull_back	Saved final scene: differs from _pre_blocked_laplacian in 130 vertices.	3 / 15

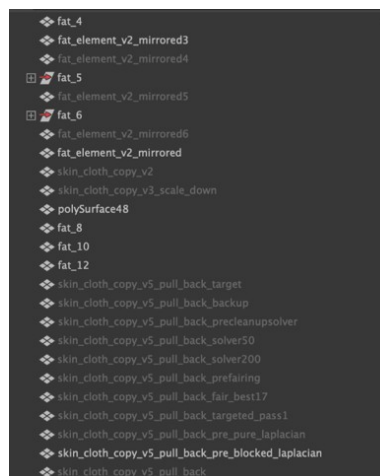


Figure 35. Journal screenshot of the Maya Outliner listing the skin backup nodes of the cleanup research, cropped from the working draft (image 13).

Appendix D. Data and Script Inventory

All statistics of the report can be regenerated from the following files in the supplementary folder (Research Journal / Final Report / supplementary). Data files are JSON or NumPy archives; the scripts are the analysis scripts named in Section 19.4.

Table 46. Data files behind the report.

File	Content
states.npz, viz_data.npz	Vertex positions (world space) of every saved skin state; anatomy vertices and triangles for the figures.
final_metrics.json	Roughness statistics (all, core, active), displacement and mesh-quality metrics for all states.
isect_all.json, forbidden_face_vertices.json, flagged_face_loops.json	Forbidden faces, pairs and per-mesh counts per state; face and vertex ids of the forbidden faces; face loops used for drawings.
clear_json, clearance_.npz, clearance_vs_face.json	Clearance of the active vertices per state; per-vertex clearance of all non-boundary vertices; clearance-versus-face analysis.
haus.json, haus_full.json, disp_pairs.json, normal_disp.json	Point-to-surface and Hausdorff distances; vertex displacement between states; signed normal displacement.
blocked_default.json, blocked_from_pre_blocked.json, blocked_freeze_rings.json	Offline blocked-Laplacian runs: defaults from _solver200; from _pre_blocked_laplacian (stops in iteration 1); freeze_rings 0, 1, 2.
global_alpha.json, global_alpha10.json	Emulation of the global-step rule from _solver200.
repair_footprint.json	Core, two-ring and six-ring footprints of the forbidden faces.
manual_edits.json, manual_stages.json	Statistics of the 130 manual vertex edits and of the two stages.
isect_depth.json	Intersection-depth output whose definition is not documented (no statistics are reported).

Appendix E. Missing Evidence and Open Items

Nothing below was filled in with a guess. Each item names what is missing, why it matters and what would resolve it.

Table 47. Open items and missing evidence.

#	Item	Why it matters	What would resolve it
1	Written notes for M4 and M5.	Milestone descriptions are derived from code, commits and screenshots.	The journal notes, if they exist.
2	Parameter difference between the two M5 runs (V1 and V2).	The two screenshots differ visibly and cannot be explained.	The Maya call or log for each run.
3	Stage that produced the journal image V3.	No claim is attached to it.	The Maya log or the author's note.
4	A screenshot of the vertex-distance heuristic selection; a screenshot of the worst-100-vertices check.	Both observations are stated as the author's account.	The screenshots.
5	Screenshots of the unified-solver result and of a blocked-Laplacian run result; a matched-camera before/after pair for the manual finish.	Figure evidence for stages 10, 12 and 13 of the evidence map.	New screenshots with a fixed camera.
6	Provenance of manual stage 1 (46 outward pushes): by hand or by <code>run_micro_clearance_push</code> .	Determines whether the finish is fully manual.	The author's confirmation or the log of the push.
7	Whether a blocked run was executed on the saved scene.	The saved final is not the blocked output; the offline reproduction stands in.	The Maya log of the run.
8	The accepted step size “about 0.001” in the working draft.	Cannot be produced by eight bisection steps (smallest nonzero step 0.0039).	The original log line.
9	Taubin run values 0.2126 (text) and 0.2246 (table) in the draft.	Region and run are ambiguous.	The log of that run.
10	Maya-side scan of the saved final scene (three residual faces by the offline detector); Maya's quad triangulation (92 against 93 faces).	Absolute face counts carry a tolerance.	A Maya log of <code>analyze_skin_anatomy_intersections</code> on the final scene.
11	Provenance of <code>_pre_pure_laplacian</code> (differs from <code>_solver200</code> in 1,678 vertices), of <code>_backup</code> and of <code>_target</code> .	State catalogue is incomplete for these nodes.	The author's notes.
12	Exact commands for pure 0.10×8 .	Recovered from geometry, not from a log.	The command history.
13	Ground truth for artifacts.	Detector precision and recall cannot be measured.	Labelled artifacts.
14	Vertex set over which the solver logs averaged roughness and M4 disagreement.	Logged values are not comparable with the offline tables.	The solver log format or code.
15	Definition of the intersection-depth output.	No depth statistics are reported.	The code or the author.
16	Venue and DOI of the Beneath the Skin manuscript and its reference list; Maya and Autodesk documentation versions.	Citation completeness.	The final publication; the Maya release number.
17	Some anatomical-face and collision-aware fairing references considered were not verified against publisher metadata and are not cited.	Related-work coverage is limited to verified sources.	A further literature pass.